

normal distribution of unknown variance. It is used to test a hypothesis.

Its formula is given by $f(t) = \frac{1}{\sqrt{\pi} \Gamma\left(\frac{1}{2}v\right)} \left(1 + \frac{t^2}{v}\right)^{-\frac{1}{2}(v+1)}$, $-\infty < t < \infty$,

where Γ is the gamma function and v is a positive parameter (usually an integer), known as the number of degrees of freedom. The distribution is symmetrical about its mode at 0, which is therefore (for $v > 1$), also its mean.

T-FORMULAE: A set of trigonometric identities, that are useful as a change of variables in integration that express functions in terms of

$t = \tan(\theta/2)$, in particular, $\sin \theta = \frac{2t}{1+t^2}$, $\cos \theta = \frac{1-t^2}{1+t^2}$, $\tan \theta = \frac{2t}{1-t^2}$.

T INVARIANCE (T-INVARIANCE): A symmetry of the fundamental (microscopic) equations of motion of a system. If it holds, the time reversal of any motion of the system is also a motion of the system. With one exception (K_L meson decay), all observations are consistent with time reversal invariance (T invariance).

T TAURI STAR: An unstable young variable star with 0.5 to 2.5 solar masses in its pre-main sequence phase. The instability, brought about by the beginning of nuclear fusion in the core of the star, causes pulsations and stellar winds, possibly with bipolar outflows.

T-TEST: In statistics, an examination of two population means to determine if they came from the same population; especially a test of the hypothesis that the normal distribution from which a given random sample is drawn has a given mean. There are three kinds of t-tests: independent-samples (measurements made on two different sets of items), paired-samples or dependent samples (samples in which natural or matched couplings occur) and one-sample (used to determine whether a sample comes from a population with a specific mean).

T VIOLATION: The study of thermodynamics and the symmetry of certain physical laws, where the concept of time is reversed; the mirror or reverse transformation is applied to a forward direction time state.

T1: A leased data communications line capable of transmitting 1,544,000 bits per second. T1 lines are often used for transferring a company's network data to the Internet or for an ISP's connection to the Internet. T2 Lines can transmit 6.3 megabits per second; and T3 Lines can transmit digital signal at 44.746 megabits per second and often used for large companies requiring large amounts of bandwidth and are also often used as backbones.

TABEBUIA: A genus made up of about 67 species of mostly flowering plants in the family, Bignoniaceae. The species are mostly found in American tropics, sub-tropics of Mexico, the Caribbean to northern Argentina. They are mostly trees but a few species are large shrubs. The trees or shrubs are evergreen or dry season deciduous. Some of the species can grow up to 50 metres high. The leaves are usually palmately compound with 3-7 leaflets. It bears inflorescences with few-flowered panicles which branch profusely without a well-developed central rachis. Some species have petal colours of yellow, white or pink. A few species are used for timber and their roots are used for floats, inner soles and razor straps. Some species of *Tabebuia* have been cultivated as honey plants for beekeepers. Examples of species include *Tabebuia aurea* with the common name Caribbean Trumpet Tree, a small dry season-deciduous tree growing to 8 metres tall; *Tabebuia rosea* a neotropical tree that grows up to 30 metres tall; and *Tabebuia serratifolia*, with the common name Yellow poui, grows up to 37 metres in height and has large buttresses at the base of the tree.

TABLE: 1. A list of facts or figures systematically arranged especially in columns. It is also defined as data mostly in numeric form in science, set out as two-dimensional pattern of rows and columns. Anything arranged or displayed in a table or list is described as **tabular**.

Tabulate is to arrange information systematically in columns and

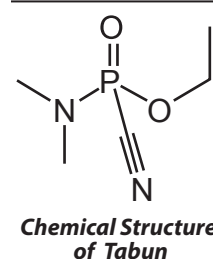
rows or in a table. 2. A flat layer of bone, especially either one of the inner or outer surfaces of the skull that separated by a more spongy bone tissue found between the harder inside and outside bone layers of the cranium.

TABLET: 1. A solid small pill containing a measured medicinal dose, usually intended to be taken orally. 2. A thin small portable personal computer that relies on a pen or touchscreen instead of a keyboard to input information. Many tablets now support multi-touch input, which allows performance with multiple fingers, such as pinching an image to zoom out or spreading fingers apart to zoom in. Tablets without physical keyboards allow texts to be entered using a pop-up keyboard that appears on the screen.

TABULAR DIFFERENCES: The differences between consecutive function values when tabulated or recorded in a table. Also, it is the difference between two consecutive numbers in a table.

TABULATE: To organise something such as data or information into a table or chart, with rows and columns.

TABUN: A colourless or brown liquid organophosphorus compound, ethyl N,N-dimethylphosphoramido-cyanidate, which is highly toxic. Its chemical formula is $C_5H_{11}N_2O_2P$, relative density 1.09, melting point -50°C (223 K) and boiling point 247.5°C (520.65 K). It belongs to the G-series of nerve agents or gas. It was used by Iraq in the Iran-Iraq war (1980-88).



TACHOGRAPH: An instrument that produces a record of the use and readings of a tachometer, especially one in a commercial vehicle or bus, recording speeds and distance travelled.

TACHOMETER: An instrument for measuring angular speed, especially the number of revolutions made by a rotating shaft in unit time. It could be an electrical, electronic or a mechanical device. An example is an electrical generator attached to a rotating shaft that drives it. The voltage generated is read as rate of rotation of the shaft that drives it.

TACHYCARDIA: An increase in heart rate over 100 and as high as 240 beats per minute. The heart is controlled by the exposing actions of the sympathetic nervous system, which accelerates the heartbeat and the parasympathetic nervous system which slows it down. The heart rate is approximately 70 beats per minute. During exercise it goes up to 110 beats per minute and drops to 60 per minute immediately after the exercise due to an increase in the parasympathetic activity. When it is a normal response to exercise or stress, it is not dangerous to healthy people, otherwise it is an arrhythmia. Symptoms include fatigue, faintness, shortness of breath and feeling the heart thumping. It may subside within minutes or hours with no lasting ill effects, but in a serious heart, lung or circulatory disease it can precede heart attack and demands immediate medical attention. Tachycardias can be treated by an electric shock to the heart, by antiarrhythmic drugs and by pacemakers.

TACHYMETABOLISM: A type of metabolism that is sustained at a very high rate to maintain heat in the organism, occurring in homoiothermic organisms. It generates the heat required to maintain the body temperature of such animals at a constant high level. A tachymetabolic animal typically has a metabolic rate at least five times that of an ectothermic animal of equivalent size.

TACHYON: A hypothetical or theoretical subatomic particle that has a speed in excess of the speed of light. In electromagnetic theory, a charged particle travelling through a medium at a speed in excess of the speed of light in that medium emits Cerenkov radiation (electromagnetic radiation, usually a bluish light emitted by a beam of high-energy charged particles passing through a transparent medium at a speed greater than the speed of light in a medium). No

such particle has yet been detected. According to the special theory of relativity, it is impossible to accelerate a particle up to the speed of light as its energy would have to become infinite.

TACNODE (OSCULATION): In mathematics, a point at which two osculating circles to the curve at that point have a common tangent, when both branches extend on both sides of the point.

TACPOINT: A point at which two similar curves intersect with a common tangent.

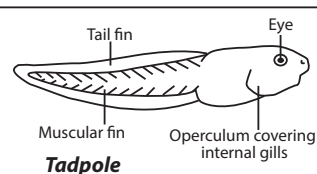
TACTIC MACROMOLECULE: A regular macromolecule in which essentially all the configurational (repeating) units are identical.

TACTIC MOVEMENT (TAXIS; TAXIC RESPONSE): 1. The movement of a cell or microorganism in response to an external stimulus towards or away from the source of the stimulus. Examples are chemotaxis and phototaxis. 2. The manipulating of a displaced body part to return it to its normal position such as the case of hernia.

TACTICITY: The orderliness of the succession of configurational repeating units in the main chain of a regular macromolecule, a regular oligomer molecule, a regular block or a regular chain.

TACTOSOL: A colloidal sol containing nonspherical particles capable of orientating themselves. When a tactosol is placed in a magnetic field its particles arrange themselves along the lines of force of the field. An example is an aged vanadium pentoxide sol, in which the particles are rod-shaped.

TADPOLE: The limbless aquatic larva of a frog or toad, having gills and a long flat tail. A tadpole has an oval body, gills for breathing and a long, muscular tail with fins along the upper and lower edges for swimming, two small eyes, one on each side of the head and a horny beak and rows of tiny, comb-like teeth that it uses to scrape algae from underwater plants and bite off bits of plant material. As the tadpole approaches the adult stage, legs and lungs develop and the tail gradually disappears.



TAFELPLOT: The graph of the logarithm of the current density j against the overpotential η in electrochemistry in the high overpotential limit. This limit can occur in electrolysis either when the overpotential is large and positive, the electrode being the anode or when the overpotential is large and negative, the electrode being the cathode. In the case of positive overpotential, the second exponential in the Butler-Volmer equation can be ignored, as it is much smaller than the first exponential and thus $j = j_0 \exp[(1 - \alpha)\eta]$. In the case of a negative overpotential, the first exponential can be ignored, which gives $j = j_0 \exp(-\alpha\eta)$, so that $\ln(-j) = \ln j_0 - \alpha\eta$. Thus, in the Tafel plot the value of α can be obtained from the slope and the value of j_0 can be obtained from the intercept at $\eta = 0$.

TAG: In computer science, it is a formatting command that is inserted in a document to show how it should be formatted, allowing it to be found again by browsing or searching. In Web pages, tags indicate what should be displayed on the screen when the page loads. Tags are the basic formatting tool used in HTML (hypertext markup language) and other markup languages, such as XML.

TAGGED IMAGE FILE FORMAT (TIFF): In computing, it is a raster graphics file format in which graphical images are stored and transmitted; the images are stored as a bit maps (arrangement of dots).

TAGMA: A section of the body of an arthropod that is formed by the fusion of mesodermal somites and has a distinct functional structure. The basic tagmata (plural of tagma) are the head, thorax and abdomen but the form of the tagmata varies between arthropod groups. For

example, many crustaceans have a cephalothorax and abdomen, while arachnids have a prosoma and opisthosoma.

TAIGA (BOREAL FOREST): A terrestrial subpolar biome occurring across North America and Eurasia. It is bounded by the colder tundra to the north and the warmer temperate zone to the south. It consists mainly of evergreen coniferous forests growing on swampy ground that is covered with lichens. The predominant trees are pine, fir and spruce. Soil organisms are protozoans, nematodes and rotifers; larger invertebrates such as insects that decompose plant litter are lacking, so humus accumulates very slowly. The taiga is rich in fur-bearing animals such as sable, fox and ermine and is home to elks, bears and wolves.

TAIL: 1. The movable part or extension of the vertebral column beyond the trunk part at the end of the body of a bird, a mammal, a fish or reptile. In fishes and other animals living completely or partly in water, it is very important to movement through water. Many tree-dwelling animals such as squirrels use the tail for balance and as a rudder when leaping. In some animals such as certain monkeys, it is adapted for grasping. Birds' tail feathers aid in flight manoeuvrability. Other animals such as porcupines use their tails for defence. 2. The set of points in a directed set that are greater than some given point; for example, if the natural numbers are the directed set, then the set of natural numbers greater than a googol is a (very distant) tail. 3. In astronomy, it is the luminous stream of gas and dust particles driven by the solar wind from a comet as it approaches and then recedes from the Sun. 4. The rear part of an aircraft together with horizontal and vertical stabilising surfaces attached to it. 5. The rear part of a missile or bomb, including structures for controlling the angle of the trajectory or any slender projection resembling such a structure.

TAIL EVENT: In statistics, an event that does not depend on any finite initial segment of a sequence of independent random variables.

TAIL POINTER: An abstract reference data type used to indicate the end of a queue.

TAIL SLAPPING: The forceful slapping of tails on the surface of water by dolphins and whales.

TAILING: Asymmetry of a peak such that, relative to the baseline, the front is steeper than the rear. In paper chromatography and in thin-layer chromatography, the distortion of a zone showing a diffuse region behind the zone in the direction of travel.

TAKE-ALL: A disease of grasses and cereals caused by the fungus *Gaeumannomyces graminis*. The pathogen survives in the soil on crop debris and can infect the crop at any stage. The dense mycelium on the roots and stem bases gives them a typical blackened appearance. If the plants survive until ear emergence, the ears have little grain and appear bleached and are, therefore, called **whiteheads**.

TALC: Soft white or grey-green mineral with the chemical formula $\text{H}_2\text{Mg}_3(\text{SiO}_3)_4$ or $\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$. It is used as the defining mineral for the value of 1 on Mohs scale hardness. Its purified form is a white powder used in talcum powder, in medicine and in ceramic materials. It forms as a secondary mineral by alternating of magnesium-rich olivines, pyroxenes and amphiboles of ultrabasic rocks. Talc is not soluble in water, but it is slightly soluble in dilute mineral acids.

TALINUM: A variety of plant species, belonging to the family Talinaceae, whose plants are herbaceous and succulent with edible leaves. The genus contains about 50 species of annuals, biennials and perennials that grow in tropical and subtropical regions of Africa, North America and Central America. Examples include *Talinum fruticosum* with the common name waterleaf, grown as a leaf vegetable; *Talinum aurantiacum*, known commonly as flame flower, is a plant with fleshy roots and erect stems with narrow succulent leaves; and *Talinum paniculatum*, commonly known as jewels of opar or fame flower is grown as an ornamental plant.

TALL OIL: A complex mixture of mainly resin and fatty acids, a by-product of wood-pulp manufacture by the Kraft process. It is used in soaps, paints and varnishes.

TALLY CHART: A table with tally marks to show a data set, or to keep track of data such as events, frequencies, etc. Tally marks are usually made by making lines in groups of five; the first four lines are vertical and the fifth one is represented by a diagonal to help in counting of the frequency for each value. Tally marks are frequencies, occurrences, or total numbers being measured for a specific category in a data set.

TALUS: 1. A pile of rocks at the bottom of a hill that forms a slope. 2. The large bone in the ankle, which together with the tibia and fibula forms the ankle joint.

TAMARIND (*Tamarindus indica*): A large long-lived leguminous tree belonging to the family, Fabaceae which can grow to a height of 24-30 metres and about 7.5 metres in thickness. Its crown can extend to 12 metres across. It is native to tropical Africa, where it grows wild, but also found in other tropical regions of the world. It thrives very well in sandy, clay, loamy and acidic soils. It is bushy with bright green elliptically ovular, pinnately compound leaves, each having 10 to 20 pairs of oblong leaflets, which fold at night. The flowers are borne in small racemes, have 5 petals with yellow colour and orange or red streaks. The flower buds are distinctly pink due to the outer colour of the 4 sepals, which are shed when the flower opens. The tree produces edible curved pod-like fruits with a fleshy, juicy and acidulous pulp having sweet or sour taste. The seeds in the fruits are flat and glossy brown. The ripened fruits are used as snacks and to sweeten drinks. It has antiseptic and anthelmintic properties and its various parts such as leaves, bark and flowers have been used medicinally to treat several ailments such as conjunctivitis, dysentery, jaundice, haemorrhoids, etc.

TAMOXIFEN: A synthetic drug with the molecular formula $C_{26}H_{29}NO$ used to treat breast cancer by blocking the oestrogen receptors on the cancer cells that require oestrogen to grow. It is also used to help prevent the disease in women at high risk of getting it.

TAMPERING (DATA TAMPERING): In information technology, it is a term used to describe any type of unauthorised modification of data.

TAN: 1. To convert an animal hide or skin into leather by treating it with a chemical called tannin. 2. Abbreviation for **Total Ammonia Nitrogen**. It is the two forms represented chemically as NH_4^+ and NH_3 in which ammonia occurs in water. NH_4^+ is called **ionised ammonia** because it has a positive electrical charge and NH_3 is called **unionised ammonia** because it has no charge. 3. Abbreviation and symbol for the tangent function. In mathematics, tangent is one of the six trigonometric functions that represent ratios of sides of right-angled triangles. The others are sine, cosine, cotangent, secant and cosecant. $\tan^{-1}$ is the symbol for the inverse tangent function, arctan. 4. Short form of **tannin**, a group of pale yellow to light brown amorphous substances widely distributed in plants and used in tanning leather, dyeing fabric and making ink. 5. Short form of **suntan** which is the brownish colour that the skin takes on after being exposed to ultraviolet light, especially from the Sun.

TANDEM ARRAY: A series of copies of genes arranged in tandem along a chromosome. Nucleolar organisers, for example, can contain up to 250 copies of a single ribosomal RNA gene in tandem. Genes for histone proteins also occur in tandem arrays. Such arrays ensure that large amounts of the gene product are synthesised by the cell.

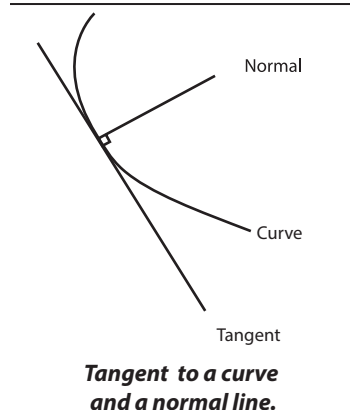
TANDEM GENERATOR: A type of particle generator, essentially consisting of a Van de Graff generator that maintains one electrode at a high positive potential; this electrode is placed between two earthed electrodes. Negative ions are accelerated from earth potential to the positively charged electrode, where surplus electrons are stripped from the ions to produce positive ions, which are accelerated again from the positive electrode back to earth. The ions are thus accelerated twice over by a single high potential. The arrangement of this tandem enables up to 30 MeV to be generated.

TANDEM MASS SPECTROMETER: An arrangement in which ions are subjected to two or more sequential stages of analysis (which may

be separated spatially or temporally) according to the quotient mass/charge. A hybrid mass spectrometer is an instrument which combines analysers of different types, for example, magnetic plus electric sector combined with quadrupole. The study of ions involving two stages of mass analysis has been termed **mass spectrometry**.

TANGENCY POINT (POINT OF CONTACT): The point at which a tangent line or plane is tangent to a given curve or surface.

TANGENT: 1. A line that touches a curve or surface at exactly one point but does not cross or intersect it. The radius to the point of tangency is always perpendicular to the tangent line. In geometry, tangent is one of the six trigonometric functions that represent ratios of sides of right-angled triangles. The others are sine, cosine, cotangent, secant and cosecant. The abbreviation and symbol for the tangent function is $\tan$. $\tan^{-1}$ is the symbol for the inverse tangent function, arctan. **Tanh** is the symbol for the hyperbolic function, hyperbolic tangent. It is the ratio the hyperbolic sine function $\sinh$ to the hyperbolic cosine function $\cosh$, given by



$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$. Its derivative is $\text{sech}^2 x$ and an antiderivative or

indefinite integral of $\ln(\cosh x)$. It has asymptotes of $y = \pm 1$. **Tanh** is the symbol for the inverse hyperbolic tangent function, arc-tanh.

2. For a given angle in a right-angled triangle, it is a trigonometric function which is equal to the length of the side opposite the angle divided by the length of the side adjacent to the angle. The tangent

is also given by $\tan x = \frac{\sin x}{\cos x}$. 3. The part of a survey line that is

straight. 4. A sudden digression or change of course. 5. An upright pin in a keyboard instrument, especially in a clavichord, that rises to sound a string when a key is depressed and stops the string at a preset length to set the pitch.

TANGENT GALVANOMETER: A type of galvanometer in which a small magnetic needle is pivoted horizontally at the centre of a vertical coil that is adjusted to be parallel to the Earth's magnetic field. When a current is passed through the coil, the needle is deflected so that it makes an angle with its equilibrium position parallel to the Earth's field.

TANGENT PLANE: A plane with every line in it that passes through a point is tangent to the surface. The direction numbers of the normal lines to this plane are the partial derivatives of the equation of the surface evaluated at the tangency point.

TANGENT RULE: The rule that is used in solving plane triangles. It is of the form $\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$, where a, b, c are respectively the sides opposite the vertices A, B, C of the triangle.

TANGENT VECTOR: A vector that is tangent to a curve or surface at a given point.

TANGENTIAL: 1. A term used in biology to describe a longitudinal section obtained down a cylindrical organ that does not pass through the centre of the organ. A tangential section through a woody stem is at a tangent to the growth rings and the growth-ring pattern appears paraboloid. 2. Of, relating to or in the direction of a tangent.

TANGENTIAL ACCELERATION (TANGENTIAL VELOCITY): The component of angular acceleration directed along a tangent to the path of a body travelling in angular motion that indicates change in linear speed.

TANGENTIAL COMPONENT: In mathematics, it is a component of a given vector acting at right angles to a given radius of a given circle.

TANGENTIAL COORDINATES: For a surface, a set of four coordinates, three of which are the direction cosines of the normal to the surface, while the fourth is the algebraic distance from the origin to the plane tangent to the surface.

TANGENTIAL CURVATURE (GEODESIC CURVATURE): In mathematics, for a point on a curve lying on a surface, the curvature of the orthogonal projection of the curve onto the tangent plane to the surface at the point. It measures the departure of the curve from a geodesic.

TANGENTIAL FLOW FILTRATION (TFF): A separation technique that uses membrane systems to purify proteins. It recirculates the retentate across the membrane surface, which minimises the fouling of the membrane. This permits longer membrane use, resulting in higher product yields.

TANGENTIAL FOCUS (PRIMARY FOCUS; MERIDIONAL FOCUS): In an astigmatic system, a line at which some of the bundle of rays from an off-axis point meet. This line is perpendicular to a plane which contains the point and the optical axis and has a smaller image distance than the secondary focus.

TANGENTIAL FORCE: A force which acts on a moving body in the direction of a tangent to the path of the body, its effect being to increase or diminish the velocity. It is distinguished from a normal force, which acts at right angles to the tangent and changes the direction of the motion without changing the velocity.

TANGERINE: A small, thin-skinned variety of the mandarin orange species (*Citrus reticulata deliciosa*) of the rue family (citrus family). The tree is smaller than other orange trees, with slender twigs and lance-shaped leaves. The fruit is slightly flattened at each end and has a loose, reddish orange peel. Easily separated segments of tender, juicy, richly flavoured pulp are abundant in vitamin C. Oil from the fragrant skin is a characteristic ingredient in several flavourings and liqueurs. The tangerine crossed with the grapefruit produced the hybrid known as the **tangelo**.

TANGRAM: A Chinese puzzle in which a square cuts into five triangles, one square and one parallelogram. It is used to make different figures.

TANKAGE: Dried animal residue after slaughter, made from heat-digested animal abattoir residues without gut contents, hide, horn and hoof. It is concentrated and dried and possessing a high biological value protein content of 60%. It is used as a feed supplement and fertiliser.

TANNERY: An industrial area where animal hides (raw hides) are converted into leather by soaking them in a liquid containing tannic acid or by the use of mineral salts.

TANNIC ACID: 1. A white or yellowish astringent powder with the chemical formula $C_{76}H_{52}O_{46}$. It is derived from nutgalls and used as a denaturant and in tanning and textiles. 2. Also known as **acidum tannicum** or **gallotannic acid**, it is a lustrous yellowish to light brown amorphous, powdered, flaked or spongy mass having the approximate composition $C_{76}H_{52}O_{46}$. It is derived from the bark and fruit of many plants and used in photography, as a mordant and to clarify wine and beer.

TANNIN: A member of a group of complex organic chemicals commonly found in leaves, unripe fruits and the bark of trees. Tannin is also present in tea, coffee and walnuts. A solution of tannic acid is obtained from one of these natural sources by extraction with hot water; in particular, gallotannic acid is obtained from plant gall. Some tannins have commercial uses especially in the tanning of leather and making ink. Tannin varies somewhat in composition, having the approximate empirical formula $C_{76}H_{52}O_{46}$. Tannin is used in tanning animal skins to make leather; it transforms certain proteins of animal tissue into compounds that resist decomposition. It is also used in manufacturing inks, as a mordant in dyeing and in medicine as an astringent and for treatment of burns.

TANTALUM: A heavy, hard, blue-grey metallic element in Group 5 (VB) of the Periodic Table (a transition element). Its chemical symbol is Ta, melting point 2,996 °C (3269 K), atomic number 73 and relative atomic mass 180.948. It is formed with niobium in the ore

columbite-tantalite and extracted by dissolving in hydrofluoric acid, separating the tantalum and niobium fluorides to give K_2TaF_7 , followed by reduction of this with sodium. The element contains the stable isotope tantalum-181 and the long-lived radio isotope tantalum-180 which has a half-life greater than 10^7 years. It contains several other short-lived isotopes. It is used in electronic and chemical equipment and for making surgical instruments.

TAPE: 1. A roll of plastic coated magnetically and wound on a wheel used for storing data. This technology has been superseded by disk drives, etc. 2. A narrow strip of material, that is sticky and used to hold or put something in place.

TAPE MEASURE: A long roll or strip of fabric, plastic, paper or thin metal that is marked in centimetres or inches or both and used for measuring length.

TAPE READER: An input device that feeds data off punched paper tape into a computer.

TAPETUM: A layer of cells in the wall of the eye of nocturnal and deep-sea animals that reflects light back onto the retina, enhancing visual sensitivity in dim light. It contains crystals of guanine in the choroid of the eye. Light reflected by this layer is responsible for the shining eyes of cats, seen when they are illuminated at night.

TAPEWORM: A ribbon-like flatworm belonging to the class Cestoda of the phylum

Platyhelminthes that lives as a parasite in the intestines of man and other animals. There are two types of tapeworm, which infest humans especially by eating undercooked meat containing tapeworm larvae.

Taenia solium commonly known as **pork tapeworm**

is a flattened ribbon-like tapeworm that is white in colour, belonging to cyclophyllid cestodes in the family Taeniidae that infects pigs and humans. It is an intestinal zoonotic parasite found, especially in areas, where pork is eaten and sanitation is poor. The adult tapeworm grows to about 6 mm in width and an average length of 2-3, although some individuals can grow much longer. Its body is called **strobila**, which has about 800 segments called **proglottids**. Its head is called **scolex** and contains 4 large suckers with a double row of hooks, which serve as organs of attachment. It infests humans, when they eat infected raw or undercooked pork. The most common disease caused by the pork tapeworm is known as **cysticercosis**. **Taenia saginata**, commonly known as **beef tapeworm** is a long flattened ribbon-like zoonotic tapeworm belonging to the order Cyclophyllida. It is white in

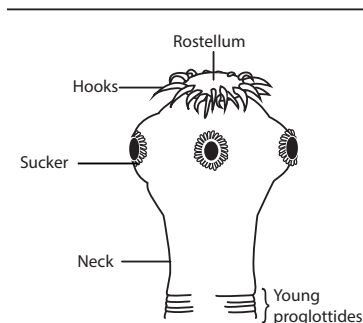
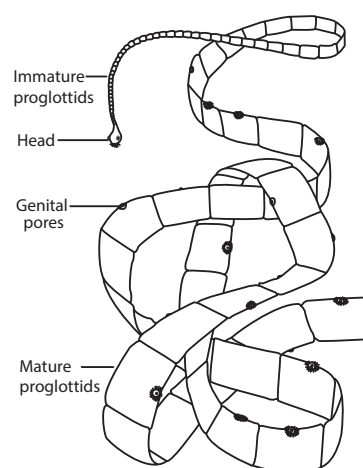


Fig. I: Head of *Taenia solium*



Fig. II: Head of *Taenia saginata*



Tapeworm

colour, with about 1000 segments called proglottids. It is an intestinal parasite of cattle and humans, causing taeniasis in humans. It is about 6 to 7 millimetres in width and usually grows to about 4 to 10 metres in length. Cattle is the intermediate host, ingesting the eggs from contaminated vegetation. Humans are the only known definitive hosts for *T. saginata* and get infected by the eating raw or undercooked beef containing the larvae. The larvae are digested from the beef in the human intestinal system. The worm then attaches on the intestinal mucosa of the upper small intestine.

TAPHONOMY: The study of biological, chemical and physical natural environmental processes that leads to preservation of dead remains of organisms as fossils.

TAPROOT: A single robust main root that is derived from the radicle, which is the primary or embryonic root of a plant. A taproot is formed when the embryonic root continues to elongate downwards, often to considerable depth, with fine lateral roots, making the plants difficult to transplant. The typical shapes may be conical, widest at the top and tapering steadily towards the bottom such as carrot; a fusiform root is widest in the middle and tapers towards the top and the bottom as seen, for example, in radish; and a napiform root is very broad at the top and tapers suddenly like a tail at the bottom, for example, turnip.

TAR: 1. A sticky black or brown liquid or semisolid mixture of hydrocarbons and free carbons used to make road surfaces. It is residue from the fractional distillation of crude oil or from the destructive distillation of peat, wood, coal, bones and other materials of organic origin. Typical tars are rich in organic compounds that are related to benzene. 2. The particulate residue from tobacco smoke that enters the lungs. Sticky brown tar leaves yellow stains on fingers and teeth. Some inhaled tar is absorbed by lung cells, causing them to die. Tar also damages the cilia in the upper airways that protect against infection.

TAR SAND: A mixture of sand, water, clay and bitumen lying at or near the surface of the Earth or sandstone bearing high molecular weight hydrocarbons (highly viscous bitumen) that can be recovered to yield oil products. Oil recovered from tar sands, commonly referred to as synthetic crude, is a potentially significant form of fossil fuel. The largest known deposits of tar sands occur in Athabasca River valley in Canada.

TARGET CELL: A cell that a particular hormone effects by its direct action either passing through the membrane or binding to a surface receptor.

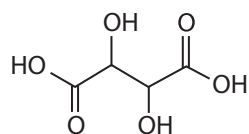
TARPAULIN: A heavy waterproof material such as canvas or polyester coated with urethane or made of plastics such as polyethylene, used to protect things from moisture and as a covering.

TARSAL: 1. One of the angular bones that make up the ankle of many terrestrial vertebrates, including humans. 2. Relating to the small section of connective tissue tarsus along the edge of the eyelid.

TARSUS: The ankle or corresponding part of the hind limb in land vertebrates, consisting of a number of small bones called tarsals, which varies from species to species. Humans for example, have seven. The top of the tarsus articulates with the tibia and fibula and the front articulates with the metatarsals.

TARTAR EMETIC: Potassium antimonyl tartrate, a poisonous, odourless compound with metallic sweetish taste. Its chemical formula is $K_2Sb_2(C_4H_4O_6)_2 \cdot 3H_2O$. It can be produced by dissolving antimony oxide in tartaric acid. It is used as an insecticide and as mordant in dyeing.

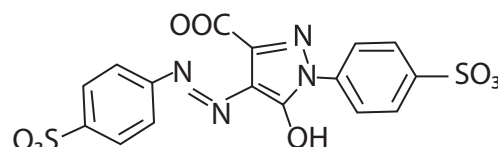
TARTARIC ACID (2,3-DIHYDROXY-BUTANEDIOIC ACID; RACEMIC ACID; PARATARTARIC ACID): A white crystalline naturally occurring dicarboxylic or hydroxycarboxylic acid with the chemical formula $C_4H_6O_6$. It occurs in grapes and other fruits. It is used in baking powders and as food additives and in dyeing and printing. Its salts, the tartrates, are frequently used in medicine.



Chemical Structure of Tartaric Acid

TARTRATE: A salt or ester of tartaric acid with the chemical formula $C_4H_4O_6^{2-}$ with a molar mass of 148.07 g/mol. Examples of tartrates are **sodium tartrates** which include monosodium tartrate, disodium tartrate and sodium ammonium tartrate; and **potassium tartrates** including monopotassium tartrate also known as cream of tartar, dipotassium tartrate and potassium sodium tartrate. They are used as food additives, antioxidants, acidity regulators and emulsifiers.

TARTRAZINE: A synthetic lemon yellow azo dye with the chemical formula $C_{16}H_9N_4O_9S_2$. It is used as food colouring that gives food a yellow colour. It can cause toxic response in the immune system and is banned in certain countries.



Chemical Structure of Tartrazine

TASK: In computer programming, it is a basic unit of programming that an operating system controls. Most operating systems support multitasking, which allows multiple tasks to run concurrently, taking turns using the resources of the computer. When a program is executed, the operating system creates a new task for it.

TASK MANAGER: A software utility found in Microsoft Windows that enables a user to view each of the tasks currently running on the computer, each of the processes and the general status of the computer. It is accessed by pressing Ctrl + Alt + Del on the keyboard.

TASKBAR: In computing, it is a bar located at the bottom of the screen and contains the Start button on the left side and the systray (system tray) on the right. The Taskbar allows the user to locate and launch programs through the Start button, see any program that is currently open, display the time or date, items running in the background through the Systray and with subsequent versions of the Windows operating system, the Quick Launch.

TASSEL: The staminate inflorescence in corn (*Zea*).

TASTE: 1. The ability of an animal to detect flavour or the sense that detects some of the chemical constituents of food. In humans, the receptor cells of taste are taste buds, which are sensitive to chemical stimuli and are stimulated in the mouth. The human tongue can distinguish only four basic tastes: sweet, sour, bitter and salty. 2. The flavour of a substance.

TASTE BUD: A group of taste-sensitive cells on the tongue or a small sense organ in most vertebrates specialised in the detection of taste. In land animals, they are concentrated on the upper surface of the tongue and are sensitive to four types of taste: sweetness, saltiness, bitterness or sourness. It transmits information about a particular type of taste to the brain via nerve fibres. In fishes, they are distributed over the entire surface of the body and provide information about the surrounding water.

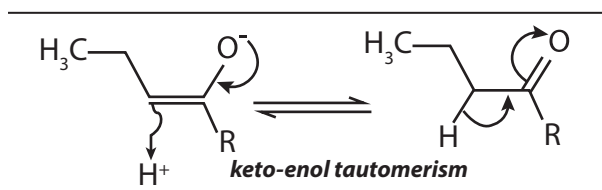
TATA BOX (HOGNESS BOX): A sequence of nucleotides rich in adenine (A) and thymidine (T) residues, located about 25-30 nucleotides upstream of the transcriptional initiation site. It serves as the main recognition site for the attachment of RNA polymerase in the promoter region of eukaryotic genes.

TATA-BOX BINDING PROTEIN (TBP): A protein that binds to the TATA box in the promoter region of eukaryotic genes and is necessary for initiation of transcription. It bends the DNA molecule and associates with many other transcription factors to assemble a pre-initiation complex, enabling binding of RNA polymerase.

TAU-PARTICLE: A subatomic particle or a lepton and its associated anti-lepton that have unusually high mass observed as resonance in electron-positron collisions. It interacts only weakly with other particles such as the electron, muon, neutrino and their anti particles.

TAUTOLOGY (LOGICAL TRUTH): In logic, a statement in which the conclusion is equivalent to the premise that is always true. In other words, it is a statement that is always true whatever the truth value of the individual statements. It is of the form: (P and (if P then Q)) implies Q, which can be written symbolically as $(P \wedge (P \rightarrow Q)) \Rightarrow Q$.

TAUTOMERISM: A type of isomerism in which the two or more isomers are in equilibrium or converts easily from one to the other. It occurs between two or more isomeric organic compounds. It usually involves a shift in the point of attachment of a mobile hydrogen atom and a shift in the position of a double bond in a molecule. For example, in keto-enol tautomerism, the hydrogen atom bonded to the carbon atom in a carbonyl (keto) group (CHCO functional group) moves to the oxygen atom, making it an enol group (CCOH). The keto form predominates in many aldehydes and ketones, the enol form in phenols. Sugars such as glucose exhibit tautomerism between open (chain) forms and closed (ring) forms.



TAUTONYM: A specific or infraspecific name used in zoology, in which the generic name and the specific name are the same, for example, *Magnolia magnolia*. This is not allowed in botanical nomenclature.

TAXALES: An order of gymnosperms containing five extant genera (including *Taxus*, the yews) and about 20 species all in the family Taxaceae. They are often included within the order Coniferales but differ from conifers in having solitary ovules borne on short terminal shoots. The Taxales also have a distinct fossil record going back to the Triassic.

TAXIS (TACTIC MOVEMENT; TAXIC RESPONSE): 1. The movement of a cell or microorganism in response to an external stimulus towards or away from the source of the stimulus. Examples are chemotaxis and phototaxis. 2. The manipulating of a displaced body part to return it to its normal position such as the case of hernia.

TAXOMETRICS (NUMERICAL TAXONOMY): A classification method based on the numerical analysis of the variation of a large number of characters in a group of organisms.

TAXON: A unit or rank of classification based on seven basic levels: kingdom, phylum (division), class, order, family, genus and species in hierarchical sequence with kingdom as the first and largest taxon and species as the last and smallest taxon. Papilionidae, Lepidoptera, Hexapoda and Uniramia are named examples of a family, order, class and phylum respectively.

TAXONOMIC CHARACTERS: Features, such as form, physiology, structure and behaviour that are assessed in isolation from the rest of the plant by taxonomists, in order to make comparisons and interpretations. It is important to distinguish between characters and character states. For example, leaf width may be a character, while leaves 4 mm wide are an expression of that character. Characters are often referred to as 'good' or 'bad' but this is strictly relative. Thus a good diagnostic character, such as compound leaves in a group of plants that mainly have entire leaves, would be bad for separating taxa in a group in which leaf divisions were either variable or often compound.

TAXONOMY: The study of the theory, practice and rules of classification of living and extinct organisms or the science of classifying plants, animals and micro organisms into increasingly broader categories based on shared features that reflect evolutionary and usually morphological relationships.

TAY-SACHS DISEASE: A hereditary metabolic disorder that mostly affect children due to a defective gene, causing an enzyme deficiency resulting in fatty substances to build up the brain that affect the nerve cells. This leads to loss of body function, blindness, deafness,

paralysis, retardation and death. It is most common in people of Eastern European Jewish descent.

TAYLOR POLYNOMIAL: A finite initial segment that is a partial sum of a Taylor series that approximates to the value of a function in a small interval around a given argument.

TAYLOR SERIES: 1. A representation of a function as an infinite sum of terms calculated from the values of its derivatives at a single point or the basic theorem of calculus relating an approximation value of a continuous function at a point to the successive derivatives of the function evaluated at that point. Taylor's series is given

$$\text{by } f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots, \text{ where } f(x) \text{ is a}$$

function which is to be approximated where "a" is a point close to x. 2. Resistance charts based upon model tests of a series of ships derived by altering the proportions of a single parent form. It is used to study the effects of these alterations on resistance to the ship's motion and to predict the powering requirements for new ships.

TAYLOR'S THEOREM: The theorem that if $x > a$, $f(n)$ is continuous on the interval $[a, x]$ and $f(n+1)$ exists on (a, x) , there is a point $c \in (a, x)$ with $f(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$. A similar statement also

holds if $x < a$.

TCP/IP (TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL): Two protocols, TCP and IP that are used together to allow computers to communicate over the public Internet, many wide area networks (WANs) and most local area networks (LANs). TCP ensures the reliability of data transmission across Internet connected networks and takes care of keeping track of the individual units of data, called packets that a message is divided into for efficient routing through the Internet. For example, when an HTML file is sent from a Web server, the Transmission Control Protocol (TCP) program layer in that server divides the file into one or more packets, numbers the packets and then forwards them individually to the IP program layer. IP handles the actual delivery of the data by moving data packets between nodes using an address such as 166.70.10.23 of a computer or other network device on a network using IP or TCP/IP. TCP/IP software is built into all major operating systems, such as Unix, Windows and the Mac OS.

TEAK: A large deciduous tree, *Tectona grandis*, of the verbena family and its wood, one of the most valuable and durable timbers. It has a yellowish brown colour that has a straight stem, often thickened at the base, a spreading crown and four-sided branchlets. The rough leaves are opposite or sometimes whorled and the branches end in many small white flowers. The timber is resistant to the effects of water and is used for shipbuilding, fine furniture, door and window frames, wharves, bridges, cooling-tower louvres, flooring and panelling.

TEAR: A salty fluid exuded by the lachrymal (lacrimal) glands in the eyes. It is a continuous and largely involuntary process stimulated by the autonomic nervous system. Fluid is secreted into the lacrimal lake, the area between the eyeball and the upper eyelid and spread across the surface of the eye by blinking. Tears serve to bathe and lubricate the cornea, the sensitive outer covering of the eyeball. Typically, the fluid either evaporates or is drained off through tiny canals at the inner corner of the eye, but in times of excessive tearing the apparatus is overwhelmed and tears overflow the eyes.

TEAR GAS: Any of various volatile gases that produce weeping and irritation of the eyes or any of a group of substances, most often synthetic organic halogen compounds that irritate the mucous membranes of the eyes, causing a stinging sensation and tears. They may also irritate the upper respiratory tract, causing coughing, choking and general debility. It is used by police against crowds and also in chemical warfare.

TEAT: The nipple of a mammary gland. In humans, it is referred to as breast, from which milk is discharged. Similarly, in animals such

as cows and goats, teats are the projections from the udder through which milk is discharged.

TEAT PIPETTE (PASTEUR PIPETTE; EYE DROPPERS): A laboratory equipment that is used to transfer small amounts of measured volumes of liquid. It is a thin tube made of glass with a fine tip and rubber teat fitted to the other. It is not graduated and therefore needs a scale for precise measurements.

TEBIBYTE: A unit of data storage that is equals to 1,099,511,627,776 bytes.

TECHNETIUM: Artificial radioactive metallic element in Group 7 (VIIB) of the Periodic Table (a transition element), which occurs among the fission products of uranium. Its chemical symbol is Tc, atomic number 43 and relative atomic mass 99. Technetium resembles platinum in appearance and manganese and rhenium in chemical behaviour. It occurs naturally in minute amounts in the earth's crust and is a by-product of the nuclear industry and is a product of uranium decay. It can be made by the bombardment of molybdenum targets with deuterium nuclei. It has several isotopes, with half lives of up to 2.6×10^6 years. Tc-99m, the most stable isotope of technetium, is used extensively in nuclear medicine due to its short half-life and being a gamma emitter. It is also used as a metallurgical tracer and in corrosion-resistant products.

TECHNOCOLOUR THEORY: A model of physics beyond the standard model that addresses electroweak symmetry breaking, the mechanism through which elementary particles acquire masses. Early technicolour theories were modelled on quantum chromodynamics (QCD), the colour theory of the strong nuclear force, which inspired their name.

TECHNOLOGY: Application of scientific knowledge to the practical aims of human life or to changing and manipulating the human environment. Technology includes the use of materials, tools, techniques and sources of power to make life easier or more pleasant and work more productive. Science is concerned with how and why things happen and technology focuses on making things happen.

TECHNOLOGY SERVICES: Services involving technology, which include information technology (IT), technical support, computer networking, systems administration and other services. Common Internet services, such as Web hosting, e-mail and social networking websites also fall under the scope of technology services.

TECHNOLOGY TRANSFER: The process of imparting scientific knowledge from the places of origin to wider distribution among more people and places, for example, from universities to businesses, from large businesses to smaller ones, from governments to businesses, etc.

TECTONIC BOUNDARY: A boundary between two or more plates, where the plates can be moving toward each other, called **convergent** or **destructive plate boundary**; away from each other, called **divergent** or **constructive plate boundary**; or past each other, called **conservative plate margin**. Most earthquakes and volcanic eruptions occur along or near tectonic boundaries.

TECTONICS: The scientific study of the deformation of the rocks that make up the Earth's crust and the forces that produce such deformation. It deals with the folding and faulting associated with mountain formation; the large-scale, gradual, upward and downward movements of the crust; and sudden horizontal displacements along faults.

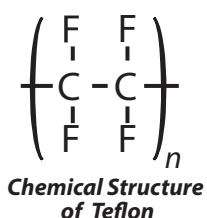
TECTUM: 1. Also known as **tectum mesencephali**, it is the roof of the midbrain, behind and above the cerebral aqueduct. From the nerve tissue protrude two pairs of rounded swellings called the superior and inferior colliculi, which contain cells concerned with reflexes involving vision and hearing, respectively. 2. A bodily structure resembling or serving as a roof. 3. The outermost layer of a pollen grain.

TEDDING: The process of spreading by lifting the swaths of newly-mown grass in haymaking, in order to expose more grass to the Sun and air and make it dry more quickly.

TEESWATER: A breed of long wool sheep with a dark muzzle. It is used to provide rams for cross-breeding with Swaledale ewes to produce the hybrid *Masham*. They are bred primarily for meat.

TEETHING: The period when a baby cuts its primary teeth or milk teeth, usually occurring at about six months of age. It may be characterised by restlessness and irritability. Inflammation of the gingival tissues before complete emergence of the crown may cause a temporary painful condition.

TEFLON: Trademark for polymer of tetrafluoroethylene fluorocarbon (polytetrafluoroethylene [PTFE]). Teflon is a tough, waxy, nonflammable organic compound with a slippery surface. It is stable over a wide temperature range and attacked by very few chemicals. Its qualities make it useful in gaskets, bearings, as container and pipe linings, as electrical insulation, as parts for valves and pumps used for corrosive fluids and as protective non-stick coatings on cooking utensils, saw blades and other articles.



TEGMENTUM: The region of the midbrain below and in front of the cerebral aqueduct. It contains the nuclei of several cranial nerves, the reticular formation and other ascending and descending nerve pathways linking the forebrain and spinal cord.

TEGULE: One of the bracts of the involucre in the Compositae (Asteraceae).

TEICHOIC ACIDS: Any of a class of strongly acidic polymers found in the cell walls, capsules, and membranes of all gram-positive bacteria that contains residues of the phosphates of glycerol and adonitol.

TEKTITE: A small black, yellowish or greenish glassy object found in groups on the Earth's surface. It consists of a siliceous material which is not related to the geological formations in which it is found. Their formation is attributed to the impact of meteorites.

TELE-SUBSTITUTION: A type of substitution reaction in which the entering group takes a position on an atom that is more than one atom away from the atom to which the leaving group is attached.

TELECOMMUNICATION: The science and technology of transmitting information electronically by wires or radio signals or the study and application of means of transmitting information, either by wires or by electromagnetic radiation with integrated encoding and decoding equipment.

TELECOMMUNICATIONS RELAY SERVICE (TRS; RELAY SERVICE; IP-RELAY): A service that allows hearing or speech impaired users to use a telephone, computer or other device to place calls to standard telephones.

TELEGRAPH: An outdated communication system that transmitted and received simple unmodulated electric impulses, especially one in which the transmission and reception stations were directly connected by wires.

TELEMETRY: A system of sending data, usually measurements, over a distance. Telemetered data may be physical, environmental or biological. Telemetry is typically used to gather data from distant, inaccessible locations or when data collection would be difficult or dangerous for a variety of reasons. In telemetry, specialised instruments carry out measurements of physical quantities and store or transmit the resulting signal, often after some initial signal processing or conversion.

TELEOSTEI: The major super order of the bony fishes with rayed fins containing about 20,000 species. They are distinguished by paired bracing bones in the supporting skeleton of the caudal fin, a homocercal caudal fin, thin cycloid scales and a swim bladder with a hydrostatic function. Teleosts have colonised an extensive variety of habitats and show great diversity of form. The group includes eel, seahorse, plaice and salmon. They have been the dominant fish since the cretaceous period.

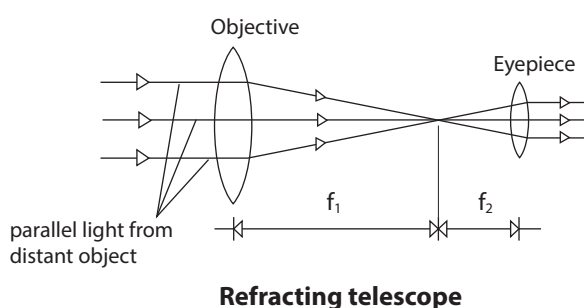
TELEPHONE: A device that allows people in different locations to communicate. It carries speech and data as electric signals in a wire, as light signals in a fibre or as radio signals or as an electronic apparatus containing a receiver and transmitter that is connected to

a telecommunications system, which enables the user to speak to and hear others with similar equipment or an instrument designed for simultaneous transmission and reception of the human voice. It works by converting the sound waves of the human voice to pulses of electrical current, transmitting the current and then retranslating the current back to sound.

TELEPHONY APPLICATION PROGRAMMING INTERFACE (TAPI): A standard developed by Microsoft and Intel, that enables a PC to support a broad range of communications devices, for example, support for a modem.

TELEPHOTO LENS: A lens in which the focal length is greater than the dimensions of the lens itself. This is achieved by designing the lens to have a negative rear element, which increases the focal length by moving the rear nodal plane (from which the focal length is measured) forward to a position in front of the lens itself. It is a camera lens used to produce an image of a distant object without the need for a lens of very long focal length, which makes it different from a telescope.

TELESCOPE: Any device for collecting and focusing light and other forms of electromagnetic radiation from a distant object in order to produce an image of it or to enable the radiation to be analysed or any of various optical devices, such as a radio telescope, used to detect and observe distant objects by their emission, transmission, reflection or other interaction with invisible radiation. It is a major research tool in astronomy and other fields. The part of the telescope that gathers the light, called the **objective**, determines the type of telescope. There are two basic types of telescopes: (i) refracting telescopes, which use lenses as the objective: light from an object is gathered and focused by lenses, with the resulting image magnified by the eyepiece. Examples of refracting telescopes are the Galilean telescope and terrestrial telescope. It is a type of telescope, which produces an erect image and is suitable for terrestrial use. It has one convex lens and one concave lens. The concave lens serves as the ocular lens or the eyepiece, while the convex lens serves as the objective. It gives erect images and is shorter than the astronomical telescope with the same power. (ii) Reflecting telescopes use mirrors as the objective. The mirror is close to the rear of the telescope and light is bounced off (reflected) as it strikes the mirror. An astronomical telescope is a telescope that is designed to observe celestial bodies and that requires no image-erecting system. It consists of a long focal length (low power) convex lens, which is the object lens or objective and a short focal length (high power) convex lens, which is the eyepiece or eye lens. Its length is equal to the sum of the focal lengths of the objective and eyepiece, giving an inverted image. The angular magnification (also sometimes called magnifying power), M of a telescope is the ratio of the angle subtended at the eye by the image formed to the angle subtended at the eye by the object being viewed. It is given by $M = f_o/f_e$, where f_o is the focal length of the objective lens and f_e is the focal length of the eyepiece.



TELESCOPING SERIES: A series whose terms may be expressed in the form $a_n = b_n - b_{n-1}$ and whose partial sums eventually only have a fixed number of terms after cancellation.

TELETYPEWRITER (TELETYPE; TTY): In computer science, it is an input device that allows alphanumeric character to be typed in

and sent, usually one at a time as they are typed, to a computer or a printer. Many communications programs still include a TTY mode to provide the simplest level of communication with a remote computer.

TELEUTOSPORE (TELIOPORE): A dark two-celled thick-walled binucleate spore formed towards the end of the growing season by rust and smut fungi. Teleutospores develop in a black sorus, the teleutosorus or telium and are the form in which the fungus overwinters.

TELEVISION: 1. The transmission and reception of still or moving monochrome (black-and-white) or colour images or images with or without accompanying sound by means of radio waves or cable. 2. An electronic device for receiving and reproducing the images and sounds of a combined audio and video signal.

TELLURIC LINES: Lines or bands in the spectrum of a celestial object that are due to absorption by gases such as oxygen, water vapour or carbon dioxide in Earth's atmosphere. The most prominent are the A and B bands.

TELLURIDE: Binary compound that contains tellurium. Compounds of tellurium with non-metals are covalent, for example, H_2Te and Na_2Te . Metal tellurides can be made by direct combination of the elements and are ionic or non stoichiometric interstitial compounds.

TELLURIUM: A brittle silvery-white semi-metallic element in Group 16 (VIA) of the Periodic Table with the symbol Te, atomic number 52, atomic weight 127.60, melting point $449.5^\circ C$ ($722.5 K$), boiling point $989.8^\circ C$ ($1263 K$), relative density 6.24 and valences of 2, 4 and 6. It has eight natural isotopes and nine radioactive isotopes. It is obtained as a by-product of the extraction of gold, silver and copper. Its chemical properties are similar to that of sulfur. It is used as a catalyst and to add hardness to alloys of lead or steel and also used in semi conductors.

TELNET: A program which allows a user to log in to a remote host computer and carry out the same commands as if the user is using a terminal at the host site. It enables a user to manage an account or device remotely. For example, a user may telnet into a computer that hosts their website to manage his or her files remotely. To use a Telnet, a user needs to use CMD interface (command line interface).

TELOCENTRIC: A term used to describe a chromosome that has the centromere at or very close to, one end so that only one chromosome arm is visible.

TELOLECITHAL: Of an egg, having a yolk concentrated in one part of the cytoplasm. The area with less yolk and prominent haploid nucleus is the **animal pole** and the area with more yolk is the **vegetal pole**. Mesolecithal and macrolecithal eggs are telolecithal.

TELOME THEORY: The theory that the leaves of ferns and seed plants evolve by the modification of terminal branches of stems or fusions and rearrangements of branching stem systems. It envisages that instead of the primitive equal branching of the stem there developed a main axis with lateral side branches.

TELOMERASE: The enzyme that maintains the ends of chromosomes by catalysing for the synthesis of telomeres. Unlike many other enzymes, which are made of only proteins, telomerase also contains RNA.

TELOMERE: Structures found at the ends of chromosomes in the cells of eukaryotes. Telomeres function by protecting chromosome ends from recombination, fusion to other chromosomes or degradation by nucleases. They permit cells to distinguish between random DNA breaks and chromosome ends. They also play a significant role in determining the number of times that a normal cell can divide.

TELOPHASE: The final stage of mitosis or meiosis during which the chromosomes of daughter cells are grouped in new nuclei.

TELSON: The hindmost segment of the arthropod abdomen. It may be flat and paddle-like, button-like or long and spiny, as in the horseshoe crabs. In insects the telson is present only in the embryo. It is meant for backward swimming.

TEMPERAMENT: 1. The characteristic aspects of a person or an animal, especially those that permanently affect behaviour and which are often displayed. It is inherent and is a combination of mental, physical and emotional traits. An example is aggression. 2. The way in which the intervals between notes on keyboard instruments are distributed throughout the scale to ensure that music in all keys sound in tune.

TEMPERATE DECIDUOUS FOREST: A biome located throughout mid-latitude regions where there is sufficient moisture to support the growth of large, broad-leaf deciduous trees.

TEMPERATE PHAGE: A DNA phage that does not necessarily induce lysogeny of an infected bacterial cell. Instead, the phage DNA may be incorporated into the bacterial chromosome, where it replicates along with the bacterium. An infected bacterium is described as **lysogenic**.

TEMPERATE REGION: A geographic area characterised by its climate, plant life and ecological community, extending across the Tropic of Cancer (latitude 23.5° North) and the Arctic Circle (latitude 66.67° North) or between the Tropic of Capricorn (latitude 23.5° South) and the Antarctic Circle (latitude 66.67° South).

TEMPERATE VIRUS: A virus that does not cause immediate lysis following entry to its host but remains in a latent state, replicating its genome along with the host's genome.

TEMPERATURE: The measure of the average kinetic energy of molecules of a substance. In other words, it is the measure of a number that expresses the degree of hotness or coldness on a chosen temperature scale. It is a measurement that describes the state of hotness or coldness or energy level of a body or substance and the condition that determines whether or not it will transfer heat to or receive heat from another body according to the laws of thermodynamics. Temperature plays an important part in determining the conditions in which living matter can exist. The properties of all materials are also markedly affected by temperature changes. At Arctic temperatures, for example, steel becomes very brittle and breaks easily and liquids either solidify or become very viscous, offering high frictional resistance to flow. At temperatures near absolute zero, many materials exhibit strikingly different characteristics. At high temperatures, solid materials liquefy or become gaseous; chemical compounds may break up into their constituents. Thermometer is an instrument used to measure temperature. It is often calibrated in degrees Celsius, Kelvin or Fahrenheit.

TEMPERATURE COEFFICIENT: A coefficient that determines the rate of change of some physical properties with change in temperature or the fractional change of a physical quantity per degree of temperature change. For example, the temperature coefficient of resistance of a metal is the increase in the resistance of the metal per degree rise in temperature. Its unit is K^{-1} . Temperature coefficient of resistance (α) = Increase in resistance divided by resistance at 0 °C: $\alpha = \frac{R_2 - R_1}{R_0(\theta_2 - \theta_1)}$, where R_2 and R_1 are the final and initial resistances, respectively, R_0 the resistance of the conductor at 0 °C and θ_2 and θ_1 are the final and initial temperatures, respectively.

TEMPERATURE EFFECT: A term used for changes in the luminescence parameters caused by changes in temperature, since the type of solvent and its temperature can effect the luminescence yield from an analyte as a result of quenching, exciplex formation, aggregation, etc.

TEMPERATURE GRADIENT: The ratio of temperature rise between two points to distance between the points. In other words, it is the

change in temperature divided by the change in distance, expressed as: $\frac{d\theta}{dx}$. Its unit is kelvin per metre (K/m or Km^{-1}). It gives rise to heat transfer, when two bodies are in thermal contact.

TEMPERATURE INVERSION: An abnormal increase in air temperature that occurs in the troposphere (the lowest level of the earth's atmosphere) with altitude. Normally temperature decreases with distance in the troposphere but sometimes the temperature starts to increase at a point. When this occurs there is a temperature inversion. This can lead to pollutants becoming trapped in the troposphere. Strong wind shears often occur across inversion layers and abrupt changes in concentrations of atmospheric particulates and atmospheric water vapour may be encountered on ascending through the inversion layer.

TEMPERATURE JUMP: A relaxation technique in which the temperature of a chemical system is suddenly raised. The system then relaxes to a new state of equilibrium and analysis of the relaxation processes provides rate constants.

TEMPERATURE LAPSE RATE: The rate of temperature decrease with increase in altitude which is expected to occur in an unperturbed dry air mass is $9.8 \times 10^3 \text{ } ^\circ\text{C min}^{-1}$. This is called the dry adiabatic lapse rate. The lapse rate is taken as positive, when temperature decreases with increasing height. For air saturated with H_2O , the lapse rate is less because of the release of the latent heat of water as it condenses. The average tropospheric lapse rate is about $6.5 \times 10^3 \text{ } ^\circ\text{C min}^{-1}$. The lapse rate has a negative value within an inversion layer.

TEMPERATURE SCALES: Empirical scales in temperature measuring instruments. These scales both rely on the use of fixed points, such as the freezing point and the boiling point of water and the division of the fundamental interval between these two points into units of temperature (100 degrees in the case of the Celsius scale). Examples are the Celsius scale, the Kelvin scale and the Fahrenheit scale. The relationship between °C and the Kelvin scale is $K = ^\circ\text{C} + 273.15$. That between degrees Fahrenheit and degree Celsius are $^\circ\text{C} = 5/9(^\circ\text{F} - 32)$ or $^\circ\text{F} = 9/5(^\circ\text{C}) + 32$. For scientific purposes, the International Practical Temperature Scale (IPTS) is in use and designed to conform as closely as possible to thermodynamic temperature and is expressed in the unit of the Kelvin.

TEMPERATURE-INDEPENDENT PARAMAGNETISM (TIP): Orbital paramagnetism that is independent of temperature. In some substances, TIP can make the molecules paramagnetic, although the electrons in the ground state are paired, if there are low-lying excited states to which electrons can readily move.

TEMPERING: The process of increasing the toughness of an alloy such as steel, by heating it to a predetermined temperature, maintaining it at this temperature for a predetermined time and cooling it to room temperature at a predetermined rate.

TEMPLATE: 1. A molecule that acts as a pattern for the synthesis of a new molecule. For example, the two nucleotide chains of a DNA molecule can separate, each acting as a template for the synthesis of the missing chain. 2. A computer file with a preset format that is used as a master for creating others similar to it. 3. In computing, it is an overlay that fits over all or part of a keyboard and has labels describing the functions of each key within a particular application. 4. A mechanical pattern or mould with one or more shapes, used to guide the manufacture or drawing of objects with similar shapes. 5. A short beam of metal, wood or stone, used to distribute weight or pressure in a structure.

TEMPLE: The area between the temporal line and the zygomatic arch. In this region, the skull is more thickly covered than in the scalp, since, in place of the areolar tissue of the scalp, there is the temporal muscle covered by the temporal fascia. The other layers are the same as in the scalp, including an extension of the epicranial aponeurosis in which the extrinsic muscles of the auricle (or pinna), the external flap of the ear, are embedded.

TEMPORAL BONE: One of the two irregular bones on either side of the skull which form part of the lateral surfaces and base of the skull (the inferior sides of the cranium and part of the cranial floor). The lower region of the temporal bone is called the **mastoid bone**, a projecting portion of which is the mastoid process.

TEMPORAL LOBE: The lower lateral lobe of either cerebral hemisphere, located in front of the occipital lobe and containing the sensory centre responsible for processing auditory signals.

TEMPORARY HARDNESS: Hardness of water that is removed by boiling or portion of the total hardness of water that can be removed by boiling, so that the soluble calcium and magnesium bicarbonate are precipitated as insoluble carbonates.

TEMPORARY MAGNETISM: Magnetism in a body that is present when the body is in a magnetic field but disappears when it is removed from the field.

TEMPORARY POISONING: Weak and reversible adsorption of a poison, such that removal of the poison from the fluid phase results in restoration of the original catalytic activity.

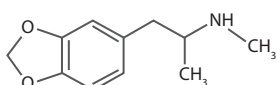
TEMPORARY STAIN: A type of stain used for immediate observation through the light microscope. Such stains often damage the section or the colour of the stain fades after a short time. For example, when plant sections are stained with Schultze's solution and mounted in 50% glycerol, the stain eventually causes swelling and dissolves the walls. Some specimens can be mounted directly in the stain, for example, when using ruthenium red.

TEMPORARY STORAGE (INTERIM STORAGE): A part of computer memory usually a RAM, in which data is stored temporarily stored. The data is erased or lost when the device is powered off.

TEMPOROMANDIBULAR JOINT: The joint between the head of the mandible (lower jaw bone) and the temporal bone of the skull. It is a synovial joint and its cavity is separated into an upper and a lower part by an articular disk.

TEN: The base of the familiar number system, which stems directly from the fact that we have ten fingers on which to count. Ten is the only triangular number that is a sum of consecutive odd squares ($10 = 1^2 + 3^2$) and the only composite integer such that all of its positive integer divisors other than 1 are of the form $x^2 + 1$ ($2 = 1^2 + 1$, $5 = 2^2 + 1$, $10 = 3^2 + 1$).

TENAMFETAMINE (3,4-METHYLENEDIOXY-METHAMPHETAMINE, MDMA; ECSTASY): A synthetic drug, with the chemical formula $C_{11}H_{15}NO_2$, initially intended as an appetite suppressant, which produces a feeling of euphoria and hallucinogen. It is derived from the amphetamine family and a relative of the stimulant methamphetamine. Apart from increasing the amount of the neurotransmitter dopamine, it also increases the production of the neurotransmitter serotonin and blocks its re-absorption in the brain. It stimulates the central nervous



Chemical Structure of Ecstasy

system and gives users a feeling of increased energy and lower social inhibition. It is now an illegal drug used as a stimulant and relaxer of inhibitions.

TEND TO: 1. In mathematics, to have as a limit, especially (of a dependent variable) implying that the independent variable is arbitrary getting closer and closer to the limit value. 2. **Tend to infinity** is to increase without limit. The value of an expression tends to infinity if it is such that for any number N , however large, a value of the expression can be found that is greater still. A limit of the function $f(x)$ as x tends to infinity is a value to which the function becomes arbitrarily close as the independent variable increases without bound. It is written $\lim_{x \rightarrow \infty} f(x) = a$.

TENDER: 1. Soft, susceptible to damage or not hardy. 2. Sensitive to frost or severe cold, not hardy, for example, tender green shoots.

TENDON (SINEW CORD): Connective tissue attaching a muscle to other body parts, usually bones, to transmit the mechanical force of muscle contraction to the other part. Much like ligaments, tendons are composed of dense, fibrous connective tissue with a high collagen content, which makes them remarkably tough and strong, with great tensile strength to withstand the stresses generated by muscle contraction.

TENDRIL: A slender coiled wiry structure found in many climbing plants. It may be a modified leaf, stem leaflet or petiole and usually in the form of a thread that coils around and attaches the plant to supporting objects. The cells that touch the object lose water and decrease in volume in comparison to the outer cells, thus causing the tendrils to curve. **Tendrill-pinnate** means pinnately compound but ending in a tendril, as occurs in the sweet pea.



Tendrill pinnate

TENSE LOGIC: The study of the logical properties of tense operators such as past, present and future and of the logical relations between tensed sentences, by means of consideration of appropriate formal systems.

TENSILE NORMAL STRESS: A normal stress that is in the same direction as the outward normal at a point on a surface.

TENSILE STRENGTH: The resistance that a material offers to tensile stress or the maximum stretching force that the material can withstand before breaking. It is measured as the force per unit cross sectional area or stress required to break it. It is measured with tensiometer.

TENSIMETER: An apparatus for measuring differences in vapour pressure. It is a form of differential manometer with two sealed bulbs attached to the limbs. If the vapour pressure of one of the liquids involved (normally water) is known, that of the other can be determined.

TENSIOMETER: 1. An apparatus for measuring surface tension of liquids. 2. An instrument used to measure the moisture content of soils. 3. An instrument used to measure tensile strength or tension in a wire, fibre or beam.

TENSION: 1. The magnitude of a stretching or pulling force or a reaction force set up in a body that is subjected to stress; or the magnitude of the pulling force exerted by a string, cable, chain or similar object on another object. It is measured in newton. 2. Voltage or electromotive force, such as in high-tension wires.

TENSIO METER: 1. An instrument for measuring the negative pressure of water in soil at its location. It is a porous, permeable ceramic cup connected through a tube to a manometer or vacuum gauge. 2. An instrument for measuring longitudinal stress in wires, structural beams, etc. 3. An instrument for measuring the surface tension of a liquid.

TENSOR: Geometric objects that describe linear relations between geometric vectors, scalars and other tensors; they are simply arrays of numbers or functions, that transform according to certain rules under a change of coordinates. A tensor may be defined at a single point or collection of isolated points of space (or space-time) or it may be defined over a continuum of points. In other words a tensor is defined at every point within a region of space (or space-time), rather than just at a point or collection of isolated points. The rank or order of a tensor is defined by the number of directions (and hence the dimensionality of the array) required to describe it and may consist of a single number, referred to as a tensor of order zero or a scalar. A first rank tensor is a vector that can be fully described by a 3×1 column vector and properties that require two directions, called second rank tensors, can be described by 9 numbers, as a 3×3 matrix. As such, in general an n^{th} rank tensor can be described by $3n$ coefficients.

TENT POLE: In certain gymnosperms, an extension of the female gametophyte towards the pollen chamber. It is seen in Ginkgo biloba and is also characteristic of many fossil genera.

TENTACLE: 1. Unsegmented extension, as one of those surrounding the mouth or oral cavity of the squid, used for feeling, grasping or locomotion. 2. In botany, it is the sensitive filament such as the glandular hairs of Drosera.

TENURE: Holding of land or other property in one's possession for a period.

TEPAL: An individual perianth part in flowers that have no distinct calyx and corolla, as occurs in most monocotyledons. In the tulip the tepals are all highly coloured.

TEPHRA: A general term for fragments of volcanic rock and lava regardless of size that are blasted into the air by explosions or carried upward by hot gases in eruption columns or lava fountains. Tephra includes large dense blocks and bombs and small light rock debris such as scoria, pumice, reticulite and ash.

TERA-: A prefix used in the metric system to denote one trillion times. For example, 10^{12} volts = 1 teravolt.

TERABIT (TB): 1. For processor storage, it is real and virtual storage and for channel volume, it is 240 or 1 099 511 627 776 bits. 2. For disk storage capacity and communications volume, it is 1 000 000 000 000 (10^{12}) bits.

TERABYTE (TB): 1. For processor storage, real and virtual storage and channel volume, it is 240 or 1 099 511 627 776 bytes. 2. For disk storage capacity and communications volume, it is 1 000 000 000 000 bytes.

TERAFLOPS: In computing, a term for a unit of computing speed that is equal to one trillion (10^{12}) floating point operations (computations) per second.

TERATOGEN: Any environmental factor that acts on foetus to cause congenital abnormality. Examples include ionising radiation, nutritional deficiencies, virus infection and toxic chemicals.

TERATOLOGY: The study of malformations and abnormal growth.

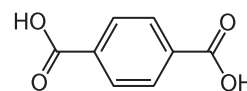
TERATOMA: A tumour that mostly affects the ovary or testis composed of tissues such as hair, teeth, muscle and bone that are not normally found in the affected part.

TERBIUM: A very rare silvery-grey metallic element, one of the rare earth elements in the lanthanide series of the Periodic Table with the symbol Tb, atomic number of 65, atomic weight 158.924, melting

point about 1356°C (1629 K), boils at about 3230°C (3503 K) and has a relative density of 8.23. It occurs in apatite and xenotime, from which it is obtained by an ion-exchange process. It has only one natural isotope (terbium-159), which is stable. However, seventeen artificial isotopes have been identified. It is used in lasers, x-rays, television tubes, etc.

TEREPHTHALIC ACID (PARA-PHTHALIC ACID; TPA; BENZENE-1,4-DICARBOXYLIC ACID; PARA-PHTHALIC ACID): A colourless

or white crystalline aromatic compound with the chemical formula $\text{C}_6\text{H}_4(\text{COOH})_2$, density 1.522 g/cm^3 and sublimates above 300°C (573 K). It is insoluble in water and soluble in alkalis. Terephthalic acid is produced by oxidation of *p*-xylene by oxygen in the air. It is used in the manufacture of plasticisers, polyester fibres and films such as dacron and terylene.



Chemical Structure of Terephthalic Acid

TERETE: In botany, it is any structure that is smooth, cylindrical and tapering as, for example, a grass stem.

TERGEMINATE: Thrice divided into equal pairs, such as paired leaflets ternately compound.

TERGUM: The protective cuticle found on the dorsal surface of the body segments in arthropods or the movable body segment of a barnacle's shell.

TERM: 1. The time at the end of a woman's pregnancy when the baby is due. 2. A mathematical expression that forms part of a proportion or fraction, which is part of a series or is associated with another by a plus or minus sign. 3. In logic, in predicate calculus, a name or variable, as opposed to a predicate; it is qualified by a predicate. 4. A set of levels having the same electron configuration and the same value of the quantum numbers for total spin and total orbital angular momentum L . 5. Energy divided by the product of the Planck constant and the speed of light, when it is of wavenumber dimension. It is energy divided by the Planck constant, when it is of frequency dimension. Its symbol is T .

TERM SYMBOLS: Symbols characterising the states of atoms and molecules in terms of multiplicity, symmetry of the total electronic wavefunction and sometimes total (orbital + spin) angular momentum. For example, $^2\text{P}_{1/2}$ denotes an atomic state of multiplicity 2 (doublet), electron orbital angular momentum quantum number $L = 1$ and orbital plus spin angular momentum quantum number $J = \frac{1}{2}$.

TERMINAL: 1. A position in a circuit or device at which a connection is normally established or broken. 2. In computing, it is a data input-output device, usually consisting of a monitor and keyboard through which data or commands are sent to or received from another system such as another computer or a website. A terminal may be classified as **intelligent**, having its own processor to perform functions such as display operations; **dumb**, with no processing capabilities and relying entirely on the main computer; or **smart**, with some processing capabilities, but not as much as an intelligent terminal. 3. Any entrance or exit node for input or output, for example, a scanner or camera for input or a monitor or printer for output. 4. A connector used to attach a wire. The wire is fastened by pressure, by a clip of some type or by tightening a screw.

TERMINAL ADAPTER (TA): A device similar to a modem that enables computers or other devices to communicate with one another over an Integrated Services Digital Network phone line.

TERMINAL DECIMAL: Decimal quantity that has a finite number of digits after the decimal point. Any decimal that has only a finite number of nonzero digits is a terminating decimal. For example, 36, 0.72 and 3.005 are terminating decimals.

TERMINAL ELECTRON ACCEPTOR: A compound that receives or accepts an electron during oxidation of carbon source, for example, during cellular respiration or photosynthesis. In cellular respiration under aerobic conditions, oxygen is the terminal electron acceptor.

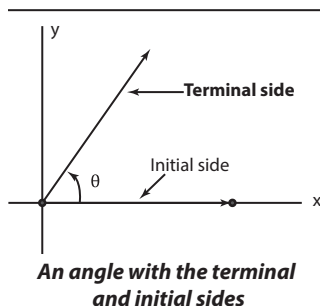
TERMINAL EMULATOR: A program that allows a device such as a microcomputer or personal computer to enter and receive data from a computer system as if it were a particular type of attached terminal.

TERMINAL MORaine: A deposition of rocks, soil and sediment at the furthest point reached by a glacier.

TERMINAL NETWORK SYSTEM: A network system in which there is a central computer with high processing power and the connected nodes are either dumb terminals or with little processing capability.

TERMINAL SERVER: A hardware device or server that provides terminals such as computers and printers, with a common connection point to a local or wide area network without the need for each terminal to have a network interface card (NIC) or modem.

TERMINAL SIDE OF AN ANGLE: The ray where measurement of an angle stops. Where it begins on the x-axis is called **initial side of an angle**.



TERMINAL VELOCITY: The maximum velocity attainable, by a body falling freely in fluids such as air. A free-falling body achieves this terminal velocity, when the downward force of gravity (F_g) equals the sum of the upward forces of drag (F_d) and the upthrust (F_u).

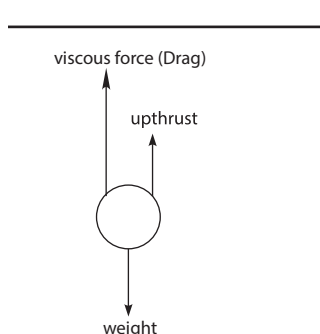


Fig. 1: Forces acting on a body falling in a fluid

TERMINAL VOLTAGE: Potential difference (pd) or voltage, across the terminals of a power supply, such as a battery of cells. When the supply is not connected to a load, its terminal voltage, V_o is the same as its electromotive force (emf). When it is connected to a load, the terminal voltage, V falls because of its internal resistance (r). Its equation is $v = V_o - I_r$, where V is the terminal voltage when it is connected to a load, V_o the open circuit voltage, r the internal resistance and I the current flowing at the time measurement.

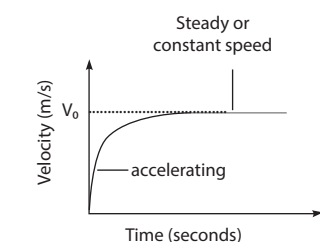


Fig. 2: Velocity-Time Graph for the motion of a body falling freely in a fluid

TERMINATE: 1. In mathematics, of a decimal expansion, having only a finite number of digits. For example 0.75 is a terminating decimal because it has a finite number of digits after the decimal point. Terminating decimals can be written in fractions for example 0.75 in fraction is $3/4$. 2. In computing, also known as **kill** or **abort**, it is used to describe the process of stopping a program from running. Alternatively, it is a device added to the end of a bus to let the system know when to stop looking for other devices.

TERMINATE AND STAY RESIDENT (TSR): A term given to an MS-DOS program that remains in the memory of a device such as, a clock, calculator or thesaurus. The program is recalled by the use of a hot key.

TERMINATION: 1. The end of translation, which occurs when the ribosome reaches the stop codon on the messenger RNA molecule, when the polypeptide, the messenger RNA and the transfer RNA molecule are released from the ribosome. 2. The steps in a chain reaction in which reactive intermediates are destroyed or rendered inactive, thus ending the chain.

TERMINATION CODON: One of the three-base sequences that initiate termination of the protein synthesis process by signalling the end of coding.

TERMINATION CRITERION (STOPPING RULE): A criterion that determines when to stop an approximative procedure, either because enough accuracy has been obtained (measured variously) or because a solution can be ruled out or too much effort has been expended.

TERMINATION FACTORS: Proteins that are exclusively involved in the termination of reactions of protein synthesis on the ribosome. In eukaryotes, a termination factor is required to release the newly made (nascent) RNA from the transcription complex. Prokaryote mRNAs often do not require a termination factor: an inverted repeat followed by a string of uracils (Us) in the mRNA template strand forms a stem-loop structure which destabilises binding by the RNA polymerase and causes Rho-independent transcription termination.

TERMINATOR: 1. A sequence of DNA lying beyond the 3' end of the coding segment of a gene which is recognised by RNA polymerase as a signal to stop synthesising mRNA. 2. The boundary on the surface of the Moon or a planet between the sunlit area and the dark area or the boundary between the part of a Moon or planet that is illuminated and the part that is dark.

TERMITARIUM (TERMITARY): A nest built by a colony of termites or white ants.

TERMITE (WHITE ANT): A light-coloured social insect, belonging to the order Isoptera that forms large colonies. Many species live in warm or tropical regions, feed on wood and are highly destructive to trees and wooden structures.

TERMOMOLECULAR REACTION: An elementary reaction involving the simultaneous collision of any combination of three molecules, ions or atoms. It involves three steps to give a product:

$A + A + A \rightarrow \text{Products}$. Its rate law is: $\text{rate} = k[A]^3$.

TERNARY: 1. In mathematics, the number system that uses 3 as a basis for counting or ordering, the numbers being expressed as sequences of the digits 0, 1 and 2; a ternary logarithm. 2. Referring to, involving or having three variables. 3. In metallurgy, it describes an alloy that consists of three components. 4. In chemistry, it describes chemical compounds consisting of three active elements, for example, three atoms, molecules or radicals.

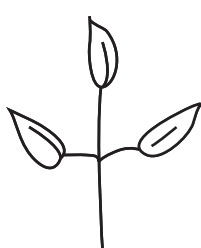
TERNARY ACID: A ternary compound containing H, O and another element, often a nonmetal. The acid does not have hydro-, this is only for binary acid.

TERNARY COMPOUND: A compound of atoms consisting of three different elements, which may be ionic or covalent. An example of this is sodium phosphate (Na_3PO_4). The sodium ion has a charge of 1^+ and the phosphate ion has a charge of 3^- . Therefore, three sodium ions are needed to balance the charge of one phosphate ion. Another example of a ternary compound is calcium carbonate (CaCO_3).

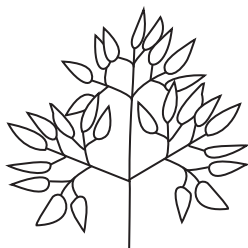
TERNARY FORM (SONG FORM): In music, a three-part composition in which the outer parts form a near-symmetrical pair separated by a contrasting middle section. Examples include the minuet and trio. Ternary form can be recognised in popular music by the middle-8 or eight-bar middle section, preceding the verse and chorus reprise.

TERNATE: Arranged in three sections or describing parts such as branches or flowers divided into three, for example, a compound leaf

which is divided into three leaflets. If the three divisions of a ternate leaf are subdivided into three leaflets each, the leaf is **bitermate** and a further subdivision produces a **tritermate** leaf.



Ternate leaves

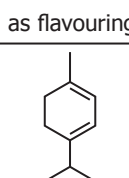
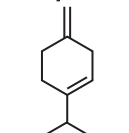
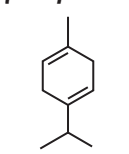
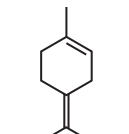


Tritermate

TERPENE: A member of a class of unsaturated hydrocarbons ($C_{10}H_{16}$) found in essential oils and oleoresins of plants with a structure based on the isoprene unit used in organic synthesis.

TERPENOIDS: Natural products and related compounds formally derived from isoprene units. They contain oxygen in various functional groups. This class is subdivided according to the number of carbon atoms in the same way as are terpenes. Terpenoids based on one, two, three or four isoprene subunits are called hemiterpenoids such as apiose, monoterpenoids such as camphor, sesquiterpenoids such as zingiberine and diterpenoids such as phytol, respectively. The terpenoids are a very diverse class of compounds.

TERPINENE: A cyclic terpene used in perfumery and as flavouring with the chemical formula $C_{10}H_{16}$. It exists in three forms: (i) **α -terpinene** is an oily liquid that smells of lemons with a boiling point of 182°C (455 K) and occurring in the herbs cardamom, coriander and marjoram, although it is usually extracted from the essential oils of other plants; (ii) **β -Terpinene** with a boiling point of 174°C (447 K) is generally synthesised from oil of savin; (iii) **γ -Terpinene** with a boiling point of 182°C (455 K) occurs in various plant oils, including those of coriander, cumin, lemon and samphire. Delta (δ -terpinene) also known as terpinolene is a natural product and has been isolated from a variety of plant sources.

 α -terpinene β -Terpinene γ -Terpinene δ -terpinene

TERQUEM'S CIRCLE (EULER'S CIRCLE; NINE POINT CIRCLE; FEUERBACH'S CIRCLE; NINE-POINT CIRCLE): The circle that passes through nine significant midpoints in a triangle. The points are: the midpoint of each side of the triangle, the foot of each altitude and the midpoint of the line segment from each vertex of the triangle to the orthocentre, where the three altitudes meet.

TERRA ROSSA: A type of intrazonal soil, rich in lime (calcmorphic) from the underlying limestone. It is a clayey soil with a low humus content and its bright red colour results from large amounts of iron oxides. It may be found in regions of Spain, southern France and southern Italy.

TERRACE: 1. A flat strip of land across a sloping hillside, lying level along the contours. 2. A raised level area of ground or a series of these into which a hillside is shaped so that it can be cultivated. 3. Soil or wall structure built across a slope to capture water and move it safely to areas where it will not cause erosion. 4. A flat, narrow stretch of ground, often having a steep slope facing a river, lake or sea

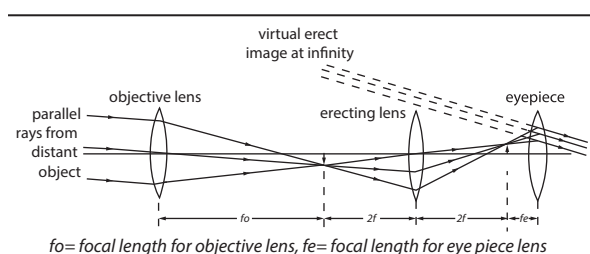
TERRACE CULTIVATION (TERRACING): Hilly slopes cut to form a system of steps or benches around the sides of a hill or on sloping ground. They rise in steps one above the other and are cultivated, often with the aid of irrigation in order to control soil erosion.

TERRESTRIAL: Living or growing on land or of the planet Earth such as a terrestrial plant or animal; not aquatic.

TERRESTRIAL MAGNETISM (EARTH'S MAGNETIC FIELD; GEOMAGNETISM SURFACE MAGNETIC FIELD): Terrestrial magnetism was the former name for Earth's magnetic field or geomagnetism surface magnetic field. It is a magnetic field that surrounds Earth and shields the planet from harmful cosmic rays and the Sun's solar wind. It is created by Earth's liquid iron core, with the magnetic field South Pole near the Earth's geographic North Pole and the North Pole near the Earth's geographic South Pole. The magnetic poles are not stationary and are not always in the same place. The magnetic poles are used in navigation.

TERRESTRIAL TELESCOPE: A refracting telescope used for viewing objects on Earth rather than for making astronomical observations. The difference between an astronomical telescope and a terrestrial one is that an astronomical telescope inverts the image, whilst a terrestrial telescope does not invert the image because it has an extra lens to re-invert the image. The terrestrial telescope consists of three or four converging lenses, namely the objective lens, erecting lens and the eyepiece lens. The objective lens forms an image of a distant object at its principal focus. The erecting lens is placed so that the image is at $2f$ of it and thus forms another image, which is real and the same size, but inverted at $2f$ on the other side. This image is taken as an object for the eyepiece. The final image produced by the telescope is erect, virtual and the same size as the object.

TERRESTRIAL PLANET (INNER PLANET): Any of the four highly dense planets whose orbit lies nearest the Sun, or with orbit lying within the asteroid belt. The planets are Mercury, Venus, Earth, and Mars. The inner planets have similar size and composed primarily of silicate rocks or metals.



Terrestrial telescope

TERRITORIAL BEHAVIOUR (TERRITORIALITY): The methods by which an animal or group of animals, protects its territory from intruders of its species by various types of behaviours such as scent markings, songs, etc. Territorial behaviour is adaptive in many ways; it may permit an animal to mate without interruption or to raise its young in an area where there will be little competition for food. It can also prevent overcrowding by maintaining an optimum distance among members of a population.

TERRITORY: A fixed area that an animal or group of animals defends against intrusion from intruders of its species by various types of behaviours such as scent markings, songs, etc. Territories may be seasonal; in many songbirds the mated pair defends the nest and feeding area until after the young are fledged. In communally nesting birds such as gulls, the territory may simply consist of the nest itself.

TERTIARY: 1. Third in rank order, occurrence or importance. 2. Tertiary feathers are the few flight feathers nearest to a bird's body on the rear edge of a wing, making up the third row of feathers. 3. Relating to or derived by the replacement of three atoms or groups, especially the replacement of the three hydrogen atoms in ammonia with alkyl groups to form amines or containing or relating to a carbon atom that has direct bonds to three other carbon atoms. 4. A period of geological time, 65 to 1.64 million years ago divided into five epochs or the third period in the formation of rocks.

TERTIARY AMINE: A derivative of ammonia formed by replacing three hydrogen atoms in ammonia by alkyl radicals. An example is trimethylamine ($\text{CH}_3)_3\text{N}$.

TERTIARY COLOUR: A colour obtained by mixing two secondary colours. It may also be obtained by mixing a primary colour with either of the secondary colours adjacent to it on a colour wheel.

TERTIARY CONSUMER: A member of a trophic level of an ecosystem consisting of carnivores that eat mainly other carnivores.

TERTIAL FLIGHT FEATHERS: Three large stiff feathers adapted for flight that are closest to the bird's body along the wing, next to the secondary flight feathers. Primary, secondary and tertiary flight feathers are collectively known as **remiges**.

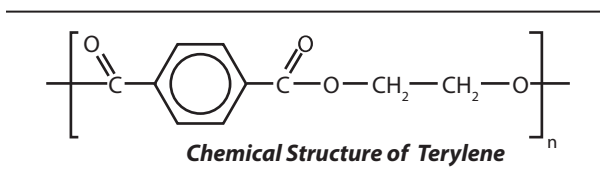
TERTIARY HYDROGEN ATOM: A hydrogen atom that is bonded to a tertiary carbon atom (a carbon atom bonded to 3 others with a single bond).

TERTIARY INDUSTRY: The field of industry providing services such as finance, banking, transportation, legal services, etc, rather than manufacturing or extraction of raw materials; or economic activity concerned with the sale and use of economic goods and services (service industry).

TERTIARY PERIOD: A period of geological time 65 to 1.64 million years ago, divided into five epochs: Palaeocene, Eocene, Oligocene, Miocene and Pliocene. During **Palaeocene Period**, placental mammals, took over all the ecological niches left vacant by the extinction of the dinosaurs and became the prevalent land animals; it also marked continental collisions leading to the formation of the Rocky Mountains and the Himalayas. The continents took on their present positions and climatic and vegetation zones as we know them became established. The **Eocene period** was the time of rising temperatures and an abundance of mammals, including the first horses, bats and whales. The **Oligocene Period** was characterised by the continued rise of mammals. The first to appear were pigs, rhinoceroses, etc. In the **Miocene Period**, the modern ocean current swere established and Antarctica was frozen. In the **Pliocene Period** the earliest hominid, the humanlike ape 'australopithecies' evolved in Africa and mammals grew larger than those of earlier epochs. Within the geological time column the Tertiary follows the Cretaceous period and is succeeded by the Quaternary period.

TERTIARY STRUCTURE: The final folded form of the polymer chain in a protein or nucleic acid or the three-dimensional structure of a protein or nucleic acid.

TERYLENE: Polyester which is widely used in the clothing industry and household furnishing industries. It is also a trade name for this product. It is made by acid-catalysed esterification of ethylene glycol and terephthalic acid.



TESLA: The SI unit of magnetic flux, which is equal to the magnitude of the magnetic field vector necessary to produce a force of one newton on a charge of one coulomb moving perpendicular to the direction of the magnetic field vector with a velocity of one metre per second. It is equivalent to one weber per square metre. It is denoted by T. $1\text{ T} = 1\text{ Wb m}^{-2}$ or $\text{kg s}^{-2}\text{ A}^{-1}$.

TESLA COIL: A step-up transformer with an air core for producing low current, high frequency alternating current electricity. It was invented by Nikola Tesla around 1891. It is mainly used to excite luminous discharges in glass vacuum apparatus in order to detect leaks. Tesla coils produce higher current than the other source of high voltage discharge electrostatic machines.

TESSELATE: Having a checkered pattern.

TESSELATION (TILING): A collection of smaller shapes that precisely covers a larger shape, without any gaps or overlaps. Usually, the shape to be tiled is a flat plane but other shapes and

three-dimensional objects can be tiled, too. **Regular tessellation** is a tiling method in which the tiles are all congruent regular polygons. There are only three regular tessellation types: regular triangles, regular squares and regular hexagons. A **semi-regular tessellation** is made of two or more regular polygons; it is a tessellation made with congruent copies of two or more different regular polygons that must meet in the same order at each vertex point, and the angles at each vertex point must add up to 360° .

TESSERACT: A four-dimensional equivalent of a cube, a regular four-dimensional hypercube consisting of 8 cubical cells. It is one of the six convex regular 4-polytopes.

TEST: 1. Examination or trial of the qualities of a person or thing or such an examination conducted for medical purposes. 2. To examine something in order to ascertain the presence of or the properties of a substance. 3. A trial run-through of a process or an equipment to find out if it works. 4. An examination of the body or of a body fluid or specimen to determine whether it is functioning properly or is infected. An example is a pregnancy or eye test. 5. A procedure to ascertain the presence or properties of a substance or reagent that reacts in a specific way to show the presence of a substance.

TEST CROSS: A cross between an individual exhibiting a dominant character with a known recessive parent with the same character. It is done to identify the genotype of the individual with the dominant character, whether homozygous dominant or heterozygous dominant.

TEST OF SIGNIFICANCE: A way of finding whether the results obtained from an experiment differ from the hypothesis the experiment was constructed to test because of sampling error or because the original hypothesis is invalid.

TEST PORTION (ANALYTICAL PORTION): The amount or volume of the test sample taken for analysis, usually of known weight or volume.

TEST SAMPLE: The sample, prepared from the laboratory sample, from which test portions are removed for testing or for analysis.

TEST SOLUTION (ANALYTICAL SOLUTION): The solution prepared from the test portion for the analytical procedure. The proportions of test portion and solvent are normally known.

TEST STATISTIC: A statistic that has a known distribution under the null hypothesis of a test and a different distribution under an alternative hypothesis and determines the likelihood of obtaining sample outcomes if the null hypothesis were true. For example, a test statistic may have a small numerical value under the null hypothesis, but a large value under the alternative.

TEST RULE: A test statistic, T , together with a function, δ_A , on the set, such that the hypothesis $\delta_{A(T)} = 0$ is accepted and rejected otherwise.

TEST TUBE: A cylindrical clear tube made of glass, usually open at one end and rounded at the other that is used to mix, heat, or store substances in laboratory tests and experiments. A **test tube holder** is an apparatus used in the laboratory to hold a test tube in place when the tube is hot or should not be touched. For example, a test tube holder can be used to hold a test tube while it is being heated. A **test tube rack** is an apparatus used in a laboratory to hold and transport test tubes during experiments.

TEST TUBE BABY: A baby that is conceived by artificial insemination or that develops elsewhere other than in a mother's body. This is achieved by in vitro fertilisation (IVF), which is a procedure in which eggs (ova) from a woman's ovary are removed, fertilised with sperm in a laboratory procedure and then the fertilised egg (embryo) is returned to the woman's uterus.

TESTA (SEED COAT): The protective covering of a seed that develops from the integuments of the ovule after fertilisation.

TESTICLE (TESTIS; TESTES): One of a pair of glands that produce the male reproductive cells or sperm. In foetal life the testes develop in the abdomen and in some species descend into an external sac, the scrotum. A testis is composed of about 800 coiled seminiferous tubules whose linings contain cells that develop into sperm. These tubules merge into a larger tube called the epididymis that leads out of the

testis into the vas deferens. Between the seminiferous tubules there is interstitial tissue that secretes the male sex hormone testosterone, which stimulates the development of the male reproductive system and secondary sex characteristics.

TESTIS-DETERMINING FACTOR (TDF; SRY PROTEIN): A protein that plays a crucial role in sex determination in mammals. It is encoded by the SRY (sex reversal on Y) gene on the Y chromosome and switches embryonic development from the default female pathway to the male pathway, by driving testis formation. The male developmental cascade is then consolidated by secretion of the male sex hormone testosterone by the testes.

TESTOSTERONE: The main male sex hormone, made in the testes of male mammals. It is connected with the development of the male sex organs' secondary sexual characteristics such as facial hair, masculine musculature, deep voice and male-pattern baldness.

TETANUS: An infectious disease caused by the bacterium *Clostridium tetani* which generally enters the body through broken skin or open wounds. It causes severe muscular spasms and contractions, especially around the neck and jaws and inability to open the jaws, called lockjaw or trismus. The spasm is caused by a toxin released by the bacterium.

TETHER: 1. A rope, chain or similar restraint for holding an animal in place, allowing a short radius in which it can move about. 2. A similar rope-like restraint used as a safety measure, especially for young children and astronauts.

TETHERING: Sharing of Internet connection on a smartphone with a computer, with the smartphone being physically connected to the computer using a USB cable or similar cable or transmit a Wi-fi or Bluetooth signal for the computer to connect to and receive the Internet connection.

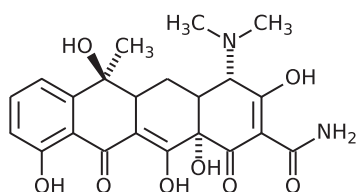
TETRA: 1. A small tropical freshwater fish of the family Characidae that is typically brightly coloured. It is native to Africa and South America and popular in aquaria. 2. A prefix used to denote four. For example, a **tetragon** is a polygon having four angles or side (4). **Tetrad** means a group of four (4). **Tetrاندrous** means four stamens. **Tetragonal** means four angles; **tetrasepalous** means four sepals; and **trapetalous** means four petals.

TETRACHLOROETHENE (TETRACHLOROETHYLENE; PERCHLOROETHENE; PERCHLOROETHYLENE; PERC; PCE): A colourless, nonflammable volatile liquid halogenoalkene with the chemical formula C_2Cl_4 , density 1.622 g/cm³, melting point -19 °C (254 K) and boiling point 121 °C (394 K). Tetrachloroethene is produced mostly by high temperature chlorinolysis of light hydrocarbons. It is used as a solvent, especially as a de-greasing agent.

TETRACHLOROMETHANE (CARBON TETRACHLORIDE): A dense, colourless, nonflammable, volatile toxic liquid with the chemical formula CCl_4 that boils at 76.8°C (349.8 K). It is made by the chlorination of methane. It reacts at high temperatures to form the poisonous gas phosgene. It is immiscible with water but miscible with many organic liquids such as benzene and ethanol. It is used as a solvent, refrigerant, dry cleaning agent, in fire extinguishers, etc.

TETRACYCLIC: 1. In botany, it is cyclic flowers having four whorls. 2. In chemistry, it is a compound having four rings in its molecular structure.

TETRACYCLINE: Any of a class of broad-spectrum antibiotics with a common basic structure, including doxycycline. They may be isolated directly from several species of actinomycetes of the genus *Streptomyces* or modified from the compounds isolated. They are used in the treatment of cholera, rickettsial infections, psittacosis, brucellosis, tularemia and acne.



Chemical Structure of Tetracycline

TETRAD: 1. A group or set of four. 2. A tetravalent atom, radical or element. 3. A four-part structure that forms during the prophase of meiosis and consists of two homologous chromosomes each composed of two sister chromatids. 4. A group of four cells, as of spores or pollen grains, formed by division of one mother cell.

TETRAD ANALYSIS: The genetic analysis of the four cells resulting from a meiotic division in order to gain information on the nature and extent of recombination. Tetrad analysis is only possible when the meiotic products remain together in groups (tetrads). This occurs in certain lower plants notably ascomycete fungi, such as *Neurospora* and some bryophytes, such as the liverwort *Sphaerocarpus*.

TETRADECANOIC ACID (MYRISTIC ACID): A saturated carboxylic acid with the chemical formula $C_{14}H_{28}O_2$, density 0.8622 g/cm³, melting point 54.4 °C (327.55 K) and boiling point of 250.5 °C (523.65 K) at 100 mmHg. Its glyceride is found in nutmeg, palm kernel oil, coconut oil, butter fat and a minor component of many other animal fats. It is also found in spermacetin, the crystallised fraction of oil from the sperm whale. It is used in cosmetics and skin care preparations.

TETRADYNAMOUS: Having four long stamens and two short stamens, as in the flowers of plants in the family Cruciferae (Brassicaceae), which includes mustards, cabbages, broccoli and turnips.

TETRAETHYL LEAD: A viscous colourless, poisonous, oily liquid with the chemical formula $Pb(C_2H_5)_4$. It is added to petrol to make it burn more efficiently in engines. The use of this compound as anti-knocking agent in automobile fuel has been banned worldwide, due to the toxicity of lead.

TETRAFLUOROETHENE, TFE (PERFLUOROETHYLENE): A flammable, colourless, heavy gas with the chemical formula C_2F_4 , density 1.519 g/cm³ at -76 °C (197 K), melting point -142.5 °C (131 K) and boils at 78°C (351 K). TFE is manufactured from chloroform. It is insoluble in water and used as a monomer to make polytetrafluoroethylene polymers, for example, teflon.

TETRAGONAL: A method of classifying crystalline substances on the basis of their unit cell. The tetragonal crystal system is one of the 7 lattice point groups. Tetragonal crystal lattices result from stretching a cubic lattice along one of its lattice vectors, so that the cube becomes a rectangular prism with a square base and height. The two types of tetragonal Bravais lattices are the **simple tetragonal**, which results from stretching the simple-cubic lattice; and the **centred tetragonal**, which results from stretching either the face-centred or the body-centred cubic lattice.

TETRAHEDRAL ANGLE: The angle between the bonds in a tetrahedral compound, which is approximately 109° for a regular tetrahedron.

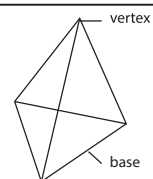
TETRAHEDRAL COMPOUND: A compound in which four atoms or groups are arranged at the corners of a tetrahedron and linked to atoms at the centre of it by covalent or coordinate bonds.

TETRAHEDRO-: An affix used in names to denote four atoms bound into a tetrahedron.

TETRAHEDRON: A three dimensional geometric figure or a polyhedron with four triangular faces, all of which are equilateral triangles. In a regular tetrahedron all four triangles are congruent equilateral triangles. It constitutes a regular triangular pyramid.

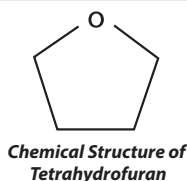
TETRAHYDRATE: A chemical or a hydrate containing four molecules of water of crystallisation per molecule of compound. An example is sodium tartrate, $KNaC_4H_4O_6 \cdot 4H_2O$.

TETRAHYDROCANNABINOL (THC): The main compound (cannabinoid) found in *Cannabis sativa* and is responsible for the psychoactive effect observed in people who use marijuana.



Tetrahedron

TETRAHYDROFURAN, THF (OXOLANE; 1,4-EPOXYBUTANE; BUTYLENE OXIDE; CYCLOTETRAMETHYLENE OXIDE; OXACYCLOPENTANE; DIETHYLENE OXIDE; FURANIDINE; HYDROFURAN; TETRA-METHYLENE OXIDE): A colourless volatile water miscible organic liquid with the chemical formula C_4H_8O , density 0.8892 g/cm³ (20 °C, liquid), melting point -108.4 °C (165 K) and boiling point 66 °C (339 K). It is made by the acid hydrolysis of polysaccharides in oat husks. It is widely used as a solvent.



TETRAMER: A structure resulting from the association of four subunits or identical monomers.

TETRAMEROUS (4-MEROUS): Flowers in which the parts of each whorl are inserted in fours or multiples of four, as in the willow herbs (Epilobium) and members of the Cruciferae.

TETRAOXOCHLORATE(VII) ION: The International Union of Pure and Applied Chemistry (IUPAC) name for **perchlorate ion** with the chemical formula ClO_4^- . It is one of the competing ions which block the uptake and transport of iodine ions. Most perchlorate salts are soluble in water, except for potassium perchlorate which has the lowest solubility of any alkali metal perchlorate (1.5 g in 100 mL of water at 25 °C). **Perchlorates** are the salts derived from perchloric acid ($HClO_4$). They occur both naturally and through manufacturing. They have been used as medicine to treat thyroid gland disorders. They are used extensively within the pyrotechnics industry and ammonium perchlorate is also a component of solid rocket fuel. Lithium perchlorate, which decomposes exothermically to give oxygen, is used in oxygen candles on spacecraft, submarines and in other esoteric situations where a reliable backup or supplementary oxygen supply is needed.

TETRAPLOID: The nucleus cell of organism that has four times the haploid number of chromosomes. The fusion of diploid gametes will result in the formation of a tetraploid. Such gametes initially arise because of the failure of homologous chromosomes to separate at meiosis. Alternatively a tetraploid segment of a plant may arise because of the failure of chromatids to separate at mitosis. If this segment develops reproductive structures then diploid gametes will be formed.

TETRAPOD: Terrestrial vertebrate animals (amphibians, reptiles, birds or mammals) with four limbs or legs used for swimming, walking, running and flying.

TETRAPODA: Superclass of jawed chordates (phylum Chordata) comprising of all vertebrate animals that possess four limbs. This includes the amphibians, birds, reptiles, birds and mammals.

TETRAPTEROUS: Having four wings or wing-like appendages.

TETRAPYRROLES: Natural pigments containing four pyrrole rings joined by one-carbon unit linking position 2 of one pyrrole ring to position 5 of the next. Porphyrins are macrocyclic tetrapyrroles, for example, bilin, a linear tetrapyrrole.

TETRASPORANGIUM: A sporangium in which four spores (the tetraspores) are formed. It is seen in some of the red algae, for example, Polysiphonia.

TETRASPORE: Any of four spores produced by meiosis in an independent diploid sporophyte in the life cycle of certain algae. They are produced within a tetrasporangium and germinate to produce the sexual haploid generation.

TETRASTICHOUS: Flowers, leaves or other structures, which occur in four vertical ranks or rows on an axis.

TETRAVALENT: 1. Having a valency of four. 2. Having four sites of attachment, as of an antibody or antigen; or containing antigens from four strains of a microorganism or virus, as of a vaccine or serum.

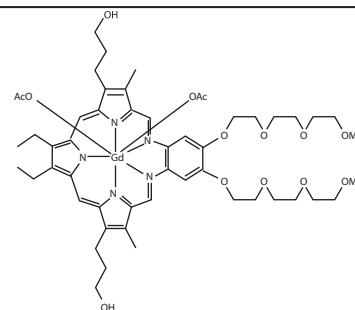
TETRODE: A four-element electron tube containing an anode, cathode, control grid and additional electrode. It is a thermionic valve with a screen grid placed between the anode and the control grid of

a triode to reduce the capacitance between these two electrodes and so improve the valve's performance as an amplifier or oscillator at high frequencies. The screen grid is maintained at a fixed potential.

TETROMINO: A four-square polyomino; one of the five plane figures that can be formed by joining four unit squares along their sides.

T_EX: A public domain text formatting and typesetting system, developed by Donald Knuth and widely used for producing high-quality mathematical and technical documents such as mathematics, physics, computer science, statistics and economics. Unlike desktop publishing applications, TeX is not WYSIWYG (What You See Is What You Get), although in some implementations a screen preview of pages is possible. A WYSIWYG application is one that enables you to see on the display screen exactly what will appear when the document is printed.

TEXAPHYRIN: A synthetic molecule similar to a porphyrin but containing five central nitrogen atoms rather than four, thus increasing the size of the central 'hole' and enabling larger cations, such as cadmium to be bound stably.



Chemical Structure of Texaphyrin

TEXT EDITOR: A word processing program that is used to type and edit text.

Examples are Word Pad and NotePad for Windows and SimpleText and TextEdit for the Mac. Microsoft Word and Word Perfect are also text editors, but they have many more features.

TEXT FILE (TEXTFILE): A data file consisting entirely of text and has no special formatting such as bold text, italic text, images, etc. Text files often have a .txt Extension after the filename such as view.txt and their contents can be viewed using programs such as Windows Notepad.

TEXTURE: 1. The detailed geometry of the void space in the particles of a catalyst. 2. The way a surface, substance or fabric looks or feels to touch such as its thickness, fairness and roughness.

TFT LCD, THIN FILM TRANSISTOR LCD (ACTIVE MATRIX LCD): A type of colour liquid crystal display commonly used in laptop computers. TFT screens have one or more transistors to control each pixel on the screen, as opposed to the less expensive passive matrix displays where each pixel is not matched with a transistor. TFT LCDs have a faster refresh rate than other types of LCD, reducing the lag between computer and display.

THALAMUS: 1. A part of the vertebrate forebrain that lies above the hypothalamus or either of a pair of egg-shaped masses of grey matter lying beneath each cerebral hemisphere in the brain that relays sensory information to the cerebral cortex. It is concerned with awareness of all the main senses except for smell. 2. The receptacle, which is the flower bearing or reproductive part of a plant.

THALASSEMIA (THALASSAEMIA; THALASSANAEMIA): Any of a group of diseases or disorders that are inherited genetically, caused by alpha and beta haemoglobin substituted mutation synthesis in the blood leading to severe anaemia and growth retardation.

THALES THEOREM: A theorem by Thales of Miletus (c. 624 – c. 546 BC) that if an angle is drawn from the diameter of a circle to the point on its circumference, then, the angle formed is a right angle. In other words, the diameter of a circle always subtends a right angle to any point on the circle.

THALIDOMIDE: A hypnotic drug, $C_{13}H_{10}N_2O_4$, developed in the 1950s for use as a sedative. When taken in early pregnancy, it caused malformation such as abnormalities in the limbs or absence of limbs of the foetus and the drug was withdrawn. It is now mainly as a treatment of certain cancers is used mainly as a treatment of certain cancers such as multiple myeloma and leprosy.

THALLIUM: A silver-grey metallic element in Group 13 (IIIA) of the Periodic Table with the chemical symbol Tl, atomic number 81, atomic weight 204.38, melting point 303.5 °C (576.5 K), boiling point 1,457 °C (1730 K), relative density 11.85 and valences of 1, 3. It occurs in zinc blende and some iron ores and is recovered in small quantities from lead and zinc concentrates. Eleven radioisotopes have been identified and naturally occurring isotopes are thallium 203 and thallium 205. It is used in electronic equipment and in pesticides.

THALLOPHYTA: A former division of the plant kingdom containing relatively simple plants which consist of thallus or those with no leaves, stems or roots. Examples are algae, bacteria, fungi and lichens.

THALLUS: A relatively undifferentiated plant body with no true roots, stems, leaves or vascular system. It is found in algae, lichens, mosses and liverworts and also in gametophyte generation of club mosses, ferns and horse tails.

THATCH: 1. A loose, intermingled organic layer of dead and living shoots, stems and roots that develops between the zone of green vegetation and the soil surface. Thatch build up begins when turf produces organic debris faster than it can be broken down. 2. A plant material used for roofing a house.

THEAVES: An ewe or female sheep between the first and second shearing or an ewe of the first year; it is also a three year old sheep.

THECA: 1. An enveloping sheath or case of an animal or animal part. 2. Half of anther containing two pollen sacs or male sporangia.

THECODONT: 1. Any of various extinct primitive reptiles of the order Thecodontia of the late Permian and Triassic periods, having teeth in sockets. It is believed dinosaurs, crocodiles, pterodactyls and birds descended from them. 2. Having teeth that arise from sockets in the jaw bones, as found in mammals and some reptiles.

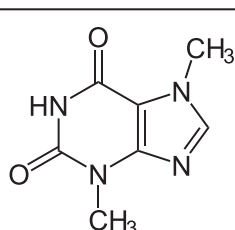
THELEPHOROUS: Having nipple-like protuberances.

THELYTOKY: A reproduction in which fertilised eggs hatch into males and unfertilised eggs hatch into females. Males are therefore diploid, whereas females are haploid. It occurs in aphids and mites.

THENARDITE: A whitish vitreous mineral that consists of anhydrous sodium sulfate. Its chemical formula is Na_2SO_4 . It occurs in arid evaporite environments and also in dry caves and old mine workings as an efflorescence and as a crusty deposit around fumaroles.

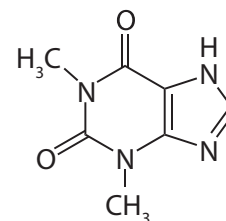
THEOBROMA CACAO (CACAO TREE; COCOA TREE): A small evergreen tree in the family Sterculiaceae (alternatively Malvaceae), native to the deep tropical region of the Americas, but grown in most tropical forests, especially in South America and West Africa. The flowers are produced in clusters directly on the trunk and older branches. The fruit, called a cocoa or cacao pod is ovoid, 15-30 cm long and 8-10 cm wide, ripening yellow to orange and weighs about 500 g when ripe. The pod contains 20 to 60 seeds, usually called beans, embedded in a white pulp. The seeds are the main ingredient of chocolate, while the pulp is used in some countries to prepare a refreshing juice. Each seed contains a significant amount of fat (40-50%) as cocoa butter. The seeds are used to make cocoa powder and chocolate.

THEOBROMINE (3,7-DIMETHYLXANTHINE; 3,7-DIHYDRO-3,7-DIMETHYL-1H-PURINE-2,6-DIONE; 3,7-DIMETHYLXANTHINE; THEOBROMIN; XANTHOSE): A bitter, colourless alkaloid with the chemical formula $\text{C}_7\text{H}_8\text{N}_4\text{O}_2$, melting point 357 °C (630 K). It is derived from the cacao bean and found in chocolate products. It is used in medicine as a diuretic, vasodilator and myocardial stimulant.



Chemical Structure of Theobromine

THEODOLITE: An optical surveying instrument for measuring horizontal and vertical angles. It is made up of a sighting telescope with crosshairs in the eyepiece for focusing on the target and which can be rotated in both the horizontal and vertical planes. It is mounted on a tripod and a spirit level is used to indicate that the instrument is horizontal. The angles are then read off graduated circles which are seen through a second eyepiece in the instrument.



Chemical Structure of Theophylline

THEOPHYLLINE

(1,3-DIMETHYLXANTHINE): An alkaloid, structurally related to caffeine with the chemical formula $\text{C}_7\text{H}_8\text{N}_4\text{O}_2$. It is found in small quantities in tea and also in cocoa beans. It is used medicinally to prevent and treat wheezing, shortness of breath and difficulty in breathing caused by asthma, chronic bronchitis, emphysema and other lung diseases. It relaxes and opens air passages in the lungs, making it easier to breathe.

THEOREM: 1. A proposition, statement or formula that is proved by a chain of reasoning or a truth established by means of accepted truths. 2. In logic, a proposition that can be deduced from the premises or assumptions of a system by recursive application of its rules of inference.

THEORETICAL FIELD CAPACITY: The rate of work that would be achieved if a machine were performing its function at its full-rated forward speed for 100% of the time.

THEORETICAL PHYSICS: A branch of physics that formulates and analyses theories to describe natural processes. It is complementary to the study of physics by experiment, which is known as **Experimental Physics**. That branch of theoretical physics is concerned with the mathematical aspects of theories in physics is known as **Mathematical Physics**.

THEORETICAL PROBABILITY: A predicted probability calculated by using the formula: number of particular outcomes that can happen divided by number of outcomes that are possible from the task: $P(\text{event}) = \text{no. of favourable outcomes} / \text{total no. of possible outcomes}$.

THEORETICAL YIELD: The amount of product obtained from the complete conversion of the limiting reactant in a chemical reaction. It is usually expressed in terms of grams or moles.

THEORY: A set of ideas, facts, propositions, concepts, principles or methods in their relation to one another, used to explain a wide set of observed facts or the causes of connected observations especially one that has been repeatedly tested or is widely accepted and can be used to make predictions about natural phenomena. An example is the kinetic theory of matter.

THEORY OF EVERYTHING, ToE (GRAND UNIFIED THEORY, GUT): Attempts to combine the three fundamental interactions: strong, weak and electromagnetic interactions into a single gauge theory with a single symmetry group. There are a number of different theories, most of which postulate that the three forces, with the exception of gravity, can be unified into one at some very high energy. The energy above which the interactions are the same is around 10^{15} GeV, which is much higher than those obtainable with existing accelerators.

THEORY OF EVOLUTION: The process by which organisms change over time as a result of changes in heritable physical or behavioural traits. Changes that allow an organism to better adapt to its environment will help it survive and have more offspring. It is a theory of natural selection, first formulated in Darwin's book "On the Origin of Species" in 1859.

THEORY OF RELATIVITY (RELATIVITY; RELATIVITY THEORY): Albert Einstein's two theories, comprising the **special theory of relativity**, which describes the causal structure of spacetime without

gravity and the **general theory of relativity**, in which the causal structure becomes a dynamical entity that is partly determined by the distribution of matter and energy.

THERAPEUTIC HALF-LIFE: The time taken for half the dose of a drug to be excreted. It is used to calculate the most effective and non-toxic dosing interval. It can be determined by administering a therapeutic dose of the drug labelled with the radioisotope and measuring the time for half of it to be excreted in urine.

THERAPEUTIC INDEX: The therapeutic usefulness of a drug, expressed as a ratio of the maximum effective dose of the drug. The therapeutic index is commonly used as a measure of the range of the dose considered to be safe. However, due to variation in individual response to a given drug it cannot be computed accurately.

THERM: A practical unit of energy defined as 10^5 British thermal units. 1 therm is equal to 1.055×10^8 joules.

THERMAL: 1. Of heat, warmth or hotness or relating to, using, producing or caused by heat. 2. Intended or designed in such a way as to help retain body heat.

THERMAL ANALYSIS: 1. Analytical techniques developed to continuously monitor physical or chemical changes of a sample which occur as the temperature of a sample is increased or decreased. Examples are differential scanning calorimetry, thermogravimetry and differential thermal analysis. 2. A method of chemical analysis that depends on the change of weight of a compound as a function of temperature (thermogravimetry). 3. A method of analysing metals and alloys by studying their heating and cooling curves. 4. A method of chemical analysis and the investigation of the products formed by heating a substance. In differential thermal analysis, a sample is heated in an inert atmosphere and the weight is plotted against the temperature. In differential scanning calorimetry, heat is added to or removed from a sample electrically as the temperature is increased. This allows the enthalpy changes due to thermal decomposition to be studied.

THERMAL BARRIER: The speed at which frictional heat, generated by rapid passage of an object through the atmosphere, exceeds endurance compatible with the function of the object.

THERMAL BLACK: A special type of carbon black produced by pyrolysis of gaseous hydrocarbons in a preheated chamber in the absence of air. It consists of relatively large individual spheres (100 - 500 nm diameter) and aggregates of a small number of pseudospherical particles. The preferred alignment of the layer planes is parallel to the surface of the spheres.

THERMAL CAPACITY (HEAT CAPACITY): The ability of an object to absorb thermal energy or the energy involved in changing its temperature by one degree. The unit is the joule per kelvin, J/K or J/°C.

THERMAL COMPOUND (HEAT DOPE; SILICON COMPOUND; THERMAL COMPOUND): A sticky paste that is placed directly onto the central processing unit (CPU) to allow for a more direct heat transfer between the CPU and heat sink and to prevent air gaps from forming between the CPU and heat sink.

THERMAL CONDUCTANCE: Heat flow rate divided by the temperature difference. Its symbol is G .

THERMAL CONDUCTION: The transmission of heat from places of higher temperature to places of lower temperature in a substance, by the interaction of atoms, molecules or ions or electrons possessing greater kinetic energy with those possessing less. In gases, the heat energy is transmitted by collision of the gaseous molecules, those possessing the greater kinetic energy imparting, on collision, some of their energy to molecules having less. Conduction in liquids is mainly due to the same process. In solid electrical conductors, the chief contribution to thermal conduction arises from a similar process taking place between the free electrons present. The interaction of

the molecules responsible for thermal conduction in solid electrical insulators arises from the elastic binding forces between the molecules, which are effectively fixed in space.

THERMAL CONDUCTIVITY (COEFFICIENT OF CONDUCTIVITY; HEAT CONDUCTIVITY): The rate at which heat flows through a material between points at different temperatures, measured in watts per metre per degree. It is the heat flow across a surface per unit area per unit time, divided by the negative of the rate of change of temperature with distance in a direction perpendicular to the surface.

THERMAL CONDUCTIVITY DETECTOR: In general, two cells arranged in a bridge configuration detect the change in thermal conductivity of the gas at the output of the column. This detector is sensitive to any substance with thermal conductivity different from that of the carrier gas. The lowest detectable limit is between 0.5 and 100 ppmv. The linear dynamic range is of the order of 10^3 . This type of detector is often used for measuring components at relatively high concentrations.

THERMAL CONDUCTOR: A material that is a good conductor of heat or has a relatively high thermal conductivity. In other words, it is a material that freely and quickly transfers energy in the form of heat from a warmer area through the material to its colder area. Most metals are good thermal conductors because they have freely moving electrons that can transfer thermal energy quickly and easily. However, the best thermal conductors are diamond and carbon nanotubes because both diamond and carbon nanotubes contain strong molecular bonds in very regular order making, it easy for molecular vibrations to travel quickly and efficiently through the materials.

THERMAL CRACKING: A petroleum refining process that decomposes, rearranges or combines hydrocarbon molecules by the application of heat, without the aid of catalysts.

THERMAL CROSS-SECTION: The cross section of a thermal neutron. A neutron in thermal equilibrium with the surrounding medium, especially one produced by fission and slowed by a moderator.

THERMAL DECOMPOSITION (THERMOLYSIS): A type of chemical reaction in which a chemical substance breaks up into simpler substances or its constituent atoms when heated. For example, water decomposes to hydrogen and oxygen when heated over $2,000^\circ\text{C}$ (2273 K).

THERMAL DIFFUSION (THERMODIFFUSION): A phenomenon in which a temperature gradient in a mixture of fluids gives rise to a flow of one constituent relative to the mixture as a whole.

THERMAL DISSOCIATION: Temporary reversible decomposition of a chemical compound into component molecules or ions resulting from the application of heat. Thermal dissociation occurs at high temperatures. For example, hydrogen molecules (H_2) dissociate into atoms (H) at very high temperatures. At $5,000^\circ\text{K}$ about 95% of the molecules in a sample of hydrogen are dissociated into atoms.

THERMAL EFFICIENCY: The ratio of work done by a heat engine to the thermal energy required to operate it.

THERMAL ENERGY (HEAT): Energy of a system in thermodynamic equilibrium by virtue of its temperature. It is the portion of internal energy of a system associated with molecular translation, rotation, vibration, electronic transition and spin of molecules as well as the phase of the system. A hot body has more thermal energy than a similar volume of cold body; but a large tub of cold water may have more thermal energy than a cup of boiling water. Thermal energy can be transferred from one body, usually hotter, to a second body, usually colder.

THERMAL EQUILIBRIUM: Property of a system in which all parts have attained a uniform temperature which is the same as that of the system's surroundings.

THERMAL EROSION: The erosion of frozen rocks and ice by the combined thermal and mechanical action of moving water.

THERMAL EXPANSION: The increase in size of an object as its temperature is increased. It is the dimensional changes exhibited by solids, liquids and gases for changes in temperature while pressure is held constant. It is usually expressed as a fractional change in length or volume per unit temperature change. A linear expansion coefficient is usually employed in describing the expansion of a solid, while a volume expansion coefficient is more useful for a liquid or a gas. Liquids have higher expansivities than solids.

THERMAL IMAGING: The production of images by detecting, measuring and recording the thermal (infrared) radiation emitted by objects. Some application of thermal imaging are mapping of the Earth's surface, thermography in medicine and mapping the weather.

THERMAL INSULATOR: A material such as plastics, wood, cork and some fabrics that is not a good conductor of heat. It has a relatively low thermal conductivity. The reason is that they contain weak molecular bonds in disorderly arrangements. Heat is transferred in a material by the vibration of the atoms and molecules. A disorderly arrangement of particles and bonds slows down the passage of heat through the material.

THERMAL IONISATION: The ionisation that takes place when an atom or molecule interacts with a heated surface or is in a gaseous environment at high temperature. Examples of the latter are capillary arc plasma, a microwave plasma or an inductively coupled plasma.

THERMAL LENSING: A technique that determines the alteration in the refractive index of a medium as a result of the temperature rise in the path of a laser beam absorbed by the medium. The lens produced (usually divergent) causes a change (usually a decrease) in the irradiance measured along the laser beam axis.

THERMAL NEUTRON (SLOW NEUTRON): A neutron whose energy distribution is governed primarily by the kinetic energy distribution of molecules of the material in which the neutrons are found and has an energy of about 0.025 electron volts (eV).

THERMAL OXIDE REPROCESSING PLANT (THORP): A nuclear plant built at Sellafield, UK, for reprocessing spent fuel from countries around the world to separate uranium, plutonium and waste products. 96% of uranium and 1% of plutonium are recovered to process into new fuel. The plant began operating 1994, despite a court action by Greenpeace to stop operations.

THERMAL POLLUTION: A temperature increase in a body of water that is caused by human activity such as deforestation, water for cooling in factories, burning of natural fuels and runoff from paved areas. This has a harmful effect on water quality and on the ability of that body of water to support life.

THERMAL PROBE (CRYOBOT): A long, pencil-shaped probe with a heated tip that can melt its way down through a thick layer of ice and deploy equipment for returning data from any watery environment that may lie below.

THERMAL RADIATION: Electromagnetic radiation resulting from interactions between electrons and atoms or molecules in a hot dense medium.

THERMAL REACTOR: 1. A nuclear reactor in which fission is induced primarily by neutrons of such low energy that they are in substantial thermal equilibrium with the material of the core. 2. A device, system or vessel in which chemical reactions take place because of heat (no catalysis). Examples are thermal cracking, thermal reforming or thermal polymerisation.

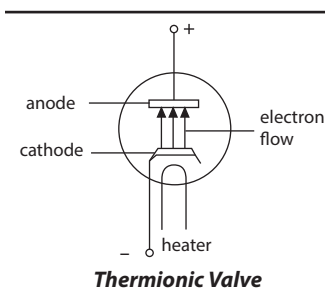
THERMALISATION (THERMALIZATION): The reduction of the kinetic energy of neutrons in a thermal nuclear reactor by means of a moderator or the process of producing thermal neutrons. When a

molecule absorbs thermal energy, as in the technique of molecular fluorescence, the lifetime of the excited state is $\sim 10^{-12}$ sec. Then it rapidly loses energy to the lowest level of the lowest excited state.

THERMALLY-INDUCED TRANSITION: A transition that is induced by a change in temperature.

THERMIONIC EMISSION: The release of electrons by some substances when heated or the emission of electrons or ions from a solid or liquid as a result of its thermal energy or the emission of electrons into a vacuum from a heated conductor. Thermionic emission is used in cathode ray tubes. The emitted current density, is given by Richardson equation: $j = AT^2 \exp(-W/kT)$, where T is the thermodynamic temperature of the emitter, W is its work function, k is the Boltzmann constant and A is the constant.

THERMIONIC VALVE (THERMIONIC TUBE): An electronic device that consists of an evacuated glass tube containing a heated cathode that emits electrons and an anode that collects the electrons. The cathode can also be heated by a separate heater such as a heating filament inside the tube. It converts a high-frequency oscillating signal into a unidirectional current capable of registering as a sound. It can be used as a switch or an amplifier in radio receivers.



THERMIONICS: The branch of electronics mainly concerned with the design of thermionic valves and electron guns of cathode-ray tubes as well as other devices.

THERMISTOR: A type of resistor or semiconductor electronic device, that is very sensitive to temperature and whose resistance decreases as its temperature increases. It is made up of a bead, rod or disc of various oxides of manganese, nickel, cobalt, copper, iron or other metals. Thermistors are used as thermometers, which often forms one element in a resistance bridge. They are also used in such applications as bearings, cylinder heads and transformer cores. They are also used to compensate for the increased resistance of ordinary resistors when hot and in vacuum gauges, time-delay switches and voltage regulators.

THERMITE (THERMITE REACTION): A pyrotechnic composition of a metal powder and a metal oxide which produces an aluminothermic reaction known as thermite reaction. The most common thermite is aluminium-iron(III) oxide. It is not explosive but can create short bursts of extremely high temperatures focused on a very small area for a short period of time. The chemical reaction involved is the reduction of Fe_2O_3 to Fe and oxidation of Al to Al_2O_3 accompanied by a large amount of energy. Thermites are therefore used as pyrotechnic initiators such as fireworks, welding and for incendiary bombs. The fuels are often aluminium, magnesium, calcium, titanium, zinc, silicon and boron. The oxidisers can be boron(III) oxide, silicon(IV) oxide, chromium(III) oxide, manganese(IV) oxide, iron(III) oxide, iron(II,III) oxide, copper(II) oxide and lead(II,III,IV) oxide.

THERMITE PROCESS (ALUMINOTHERMY): A process for extracting a metal from its oxide that involves burning the oxide together with aluminium powder.

THERMO ACIDOPHILES: A group of archaeobacteria that is able to tolerate high temperatures and acidic pH. Examples are the facultatively anaerobic archaea of the genus *Thermoplasma* and the obligately anaerobic archaea of the genera *Acidilobus* and *Caldisphaera*.

THERMOACOUSTIMETRY: A technique in which the characteristics of imposed acoustic waves are measured as a function of temperature after passing through a substance (and/or its reaction product(s)), whilst the substance is subjected to a controlled temperature program.

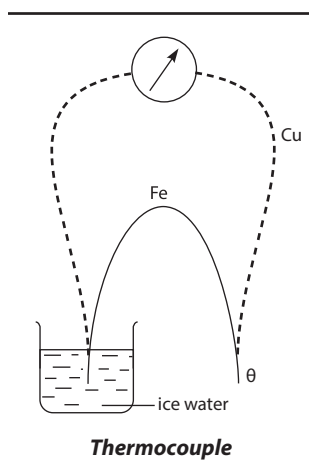
THERMOCHEMICAL EQUATION: A chemical equation that shows either mass and enthalpy relations; or a balanced stoichiometric chemical equation that includes the enthalpy change, ΔH .

THERMOCHEMISTRY: The branch of physical chemistry concerned with heats of chemical reaction, heats of formation of chemical compounds, etc. It involves the measurement, interpretation and analysis of heat changes accompanying chemical reactions and changes in state.

THERMOCHROMISM: A thermally induced transformation of a molecular structure or of a system that produces a spectral change, typically, but not necessarily, of visible colour. Examples are baby bottles (changes to a different colour when cool enough to drink) and kettles.

THERMOCLINE: 1. A layer of water, for example, in a lake, where there is an abrupt change in temperature that separates the warmer surface water from the colder deep water. 2. A steep temperature gradient that exists in the middle zone of a lake and gives rise to thermally induced vertical stratification of water. The thermocline may represent a temperature change of 1°C for every incremental depth of 1 metre of water. 3. A thin but distinct layer in a large body of fluid, for example, water such as an ocean or lake or air such as an atmosphere, in which temperature changes more rapidly with depth than it does in the layers above or below.

THERMOCOUPLE: A device used to measure temperatures accurately. It consists of two dissimilar metal wires joined together at their ends. When the two junctions are kept at different temperatures, a potential difference is generated between the points of contact as a result of one end of the wire being at a fixed temperature and the other end going through varying temperatures. This potential difference is called thermoelectric electromotive force (emf) and it is a measure of the temperature difference between the two points. The magnitude of this emf varies linearly with the temperature difference and a galvanometer can be calibrated to give direct temperature readings, with the known melting points of metals and lead being used as fixed points.



THERMODILATOMETRY: A technique in which a dimension of a substance under negligible load is measured as a function of temperature while the substance is subjected to a controlled temperature program. Linear thermodilatometry and volume thermodilatometry are distinguished on the basis of the dimensions measured.

THERMODYNAMIC CONTROL (EQUILIBRIUM CONTROL): The term characterises conditions that lead to reaction products in a proportion governed by the equilibrium constant for their interconversion and/or for the interconversion of reaction intermediates formed in or after the rate-limiting step.

THERMODYNAMIC TEMPERATURE: The base quantity, T in the system of quantities upon which SI is based, defined by the third law of thermodynamics and not in terms of the properties of any real material. Its theoretically lowest temperature is the null or zero point. At this point, the particle constituents of matter have minimal motion and can become no colder. The SI base unit, kelvin is the fraction $1/273.16$ of the thermodynamic temperature of the triple point of water.

THERMODYNAMICS: The study of the laws that govern the conversion of energy from one form to another, the direction in which heat will flow and the availability of energy to do work or the study of energy conversion between heat and mechanical work and subsequently the macroscopic variables such as temperature, volume and pressure.

THERMOELECTRIC EFFECT (SEE-BECK EFFECT): The generation of electromotive force (emf) in a circuit containing two dissimilar metals or semiconductors when the junctions between the two are kept at different temperatures.

THERMOELECTRICITY: A direct electric current generated by maintaining a temperature difference at the point, where two different materials come into contact, as in a thermocouple or the direct conversion of heat into electrical energy. For a given thermoelectric circuit operating in a given temperature range, the magnitude of the current depends mainly on the temperature difference between the two junction, the greater the temperature difference, the larger the current. Thermoelectric circuits have been used in small thermoelectric generators to furnish power in remote areas and in space probes for radio transmitters and receivers and other devices that require relatively small amounts of electric power. The process is called the Seebeck effect, after the German physicist Thomas J. Seebeck, who discovered the phenomenon in the 1820's.

THERMOELECTROMETRY: A technique in which an electrical characteristic of a substance (and/or its reaction products) is measured as a function of temperature whilst the substance is subjected to a controlled temperature program. The most common measurements are of resistance, conductance or capacitance.

THERMOGENESIS: Heat generation in the body through processes such as respiration, shivering and metabolism. It occurs especially in birds and mammals, animals that maintain their temperature within a certain narrow range, but it is also found in some cold blooded vertebrates and invertebrates. There are three types of thermogenesis: **shivering**, **non-shivering** and **diet induced** thermogenesis. Shivering is the commonest, in which chemical energy is converted into heat rather than mechanical work. The second type is non-shivering thermogenesis, which takes place in fat cells and involves the conversion of stored fat into heat. The third is the diet induced thermogenesis.

THERMOGENIC SOILS: Soils that have been influenced primarily by high soil temperatures as the dominant soil-formation factor; developed in subtropical and equatorial regions.

THERMOGRAPH: A recording thermometer used in meteorology to obtain a continuous record of temperature changes over a period on a graph or the record of this outcome or a record obtained by the technique of thermography.

THERMOGRAPHY: 1. An instrument that shows pattern of heat radiated from a person or an animal's body. It is a medical technique that makes use of the infrared radiation from human skin to detect an area of elevated skin temperature that could be associated with an underlying cancer. The heat radiated from the body is proportional to the local blood flow, thus an area of poor circulation produces less radiation. A tumour on the other hand has an abnormally increased blood supply and is shown on the thermogram. 2. A device that continuously records temperature readings. 3. A process for producing raised lettering, as on stationery or calling cards, by application of a powder that is fused by heat to the fresh ink.

THERMOGRAVIMETRY (TG): A technique in which the mass of a substance (and/or its reaction products) is measured as a function of temperature whilst the substance is subjected to a controlled temperature program.

THERMOHALINE CIRCULATION (GLOBAL OCEAN CONVEYOR): The flow of ocean water caused by changes in temperature and salinity in the deep ocean and wind-driven currents on the surface.

THERMOKARST: Irregular land-surface with small hummocks that results from the melting of ground ice in a region underlain by permafrost.

THERMOLUMINESCENCE: The luminescence produced in a solid when its temperature is raised or the phosphorescence (continuous emission of light without heat) released by some previously irradiated substances when they are heated. It arises when free electrons and holes, trapped in a solid as a result of exposure to ionising radiation, unite and emit photons of light. The process is used by geologists and

archaeologists in thermoluminescent dating. It estimates the age of an object fairly accurately by comparing the luminescence produced by heating a piece of pottery of the object, with the luminescence produced by heating similar materials of known age.

THERMOLYSIS (PROLYSIS): The chemical decomposition of a substance by heat. It is an important process in chemical manufacturing such as thermal cracking of hydrocarbons in the petroleum industry.

THERMOMAGNETOMETRY: A technique in which a magnetic characteristic of a substance (and/or its reaction products) is measured as a function of temperature whilst the substance is subjected to a controlled temperature program.

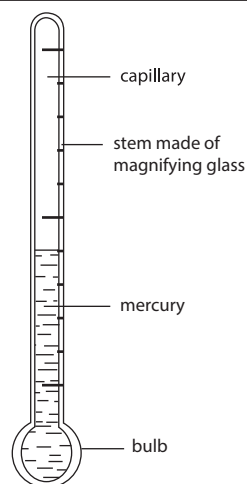
THERMOMECHANICAL MEASUREMENT: A technique in which the deformation of a substance (and/or its reaction products) under nonoscillatory load is measured as a function of temperature whilst the substance is subjected to a controlled temperature program.

THERMOMETER: An instrument for measuring temperature. It consists of a graduated glass tube and a bulb containing mercury or alcohol that rises in the tube when the temperature increases. There are many types designed to suit the degree of accuracy required or the range of temperature to be measured.

THERMOMETRIC ANALYSIS: A method for determining the transformations a substance undergoes while being heated or cooled at an essentially constant rate, for example, freezing point determination.

THERMOMETRIC TITRATION (ENTHALPIMETRIC TITRATION):

An analytical method in which one reactant (the titrant) is added continuously or stepwise to an adiabatic or isoperibol vessel containing another reactant. The enthalpy change(s) of the ensuing reaction(s) causes a temperature change which, when plotted against volume of titrant, may be used to find the titration endpoint(s). It is used for reactions such as neutralisation, redox, precipitation and organic condensation.



Thermometer

T

THERMONEUTRAL ZONE (TNZ): The comfortable temperature range within which an organism need not generate extra heat to keep warm or lose metabolic heat to cool down. In temperate regions the range of temperature is wide, animals adapted to cold environments tend to have broader thermoneutral zones than those living in hot environments.

THERMONUCLEAR REACTION: A nuclear fusion reaction which occurs between various nuclei of the light elements when they are constituents of a gas at very high temperatures. It occurs most readily between energy-rich nuclei of low atomic number (small charge) such as isotopes of hydrogen (deuterium and tritium) and less readily among a few other nuclei of higher atomic number. Thermonuclear reactions, the source of energy generation in the Sun and the stable stars, are utilised in the fusion bomb. **Thermo-nuclear energy** is the energy obtained from nuclear fusion reactions.

THERMONUCLEAR REACTOR (FUSION REACTOR): A reactor in which nuclear fusion takes place with the controlled release of energy. In this kind of reaction, two light atomic nuclei fuse together to form a heavier nucleus and in doing so, release a large amount of energy. These reactors do not yet exist because there are two basic problems in the creation of a self-sustaining thermonuclear reactor. These are heating the reacting nuclides to the huge ignition temperature of about $40 \times 10^6 \text{K}$ for a deuterium-tritium reaction and containing the reacting nuclides for long enough for the fusion energy released to be greater than the energy required to achieve the ignition temperature.

THERMONUCLEAR WEAPON: A weapon in which very high temperatures obtained by means of fission are used to bring about the fusion of light nuclei such as those of hydrogen isotopes, for example, deuterium and tritium with the release of energy.

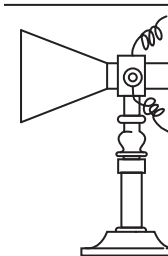
THERMOPARTICULATE ANALYSIS: A thermoanalytical technique in which the release of particulate matter from a substance (and/or its reaction products) is measured as a function of temperature, whilst the substance is subjected to a controlled temperature program.

THERMOPERIODIC RESPONSE: The response of a plant to a diurnal fluctuation in temperature. For example, growth of tomato plants is best when a certain day/night temperature regime is experienced. Meristematic activity and extension growth are often affected by thermoperiodicity.

THERMOPHILE: An organism that can tolerate high temperatures and that grows optimally at temperatures above 45°C . The majority of thermophilic organisms are prokaryotes, such as the archaeobacterial found in hot springs and undersea hydrothermal vents. Some of these thrive at a temperature of 80°C and can survive in temperatures above the boiling point of water (100°C).

THERMOPHILIC: 1. Having affinity for high temperature. 2. Describes an organism that thrives well in high temperatures.

THERMOPILE: A device used to detect and measure the intensity of radiant energy. It is made of a number of thermocouples connected together in series to achieve greater sensitivity. They are used to measure the heat radiation received from the Sun and also as a safety device that ceases to produce an electric current if a pilot light blows out.



Thermopile

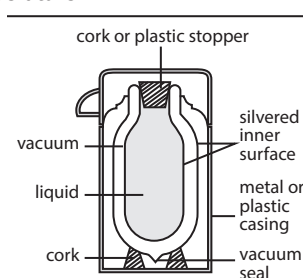
THERMOPLASTIC: A polymer such as nylon, which softens and melts, when it is heated and can be moulded and re-moulded into new shapes. There are more than 40 types of thermoplastics, including acrylic, polypropylene, polycarbonate and polyethylene.

THERMOPTOMETRY: A technique in which an optical characteristic of a substance and/or its reaction product(s) is measured as a function of temperature whilst the substance is subjected to a controlled temperature program. Measurements of total light, light of specific wavelength(s), refractive index and luminescence lead to thermophotometry, thermospectrometry, thermorefractometry and thermoluminescence, respectively; observation under the microscope leads to thermomicroscopy.

THERMORECEPTORS: Sensory receptors which are usually nerve endings in the skin that are stimulated by heat or cold. Skin thermoreceptors (hot and cold receptors) detect changes in environmental temperature.

THERMOREGULATION: The maintenance of a steady body temperature regardless of changes in the environment or the regulation of body temperature by physiological or behavioural means. Some animals, particularly mammals and birds can maintain fairly constant internal temperature, whereas in others, body temperature varies with the environment as temperature.

THERMOS: The trade name for a type of vacuum flask, used to keep beverages hot or cold. The thermos flask has a double-walled silvered glass with a vacuum between the walls. The thermos prevents heat loss by three main methods: conduction, convection and radiation. The cork supports and the cork lid of the flask prevent heat loss by conduction. Heat loss by conduction is also prevented by the vacuum within the double wall of the flask. The smooth shiny surface of the flask reflects back



A thermos flask

the heat that would have been lost through radiation. The vacuum between the double walls of the flask prevents loss by convection.

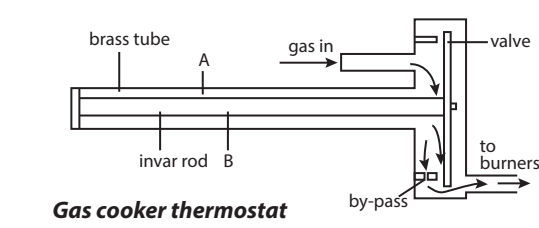
THERMOSEQUENCE: A sequence of related soils that differ, one from the other, primarily as a result of temperature as a soil-formation factor.

THERMOSETTING: Polymers such as Formica, which cannot be softened again once it has been formed. They rather decompose at high temperatures.

THERMOSONIMETRY: A technique in which the sound emitted by a substance (and/or its reaction product(s)) is measured as a function of temperature whilst the substance is subjected to a controlled temperature program.

THERMOSPHERE: Atmospheric shell extending from the top of the mesosphere to outer space. It is a region of more or less steadily increasing temperature with height, starting at 70 or 80 km. It includes the exosphere and most or all of the ionosphere (not the D region).

THERMOSTAT: A device that controls the heating or cooling of a substance in order to maintain it at a constant temperature. It employs a temperature sensor to operate a switch or valve to control electricity or fuel supply. They are used in central heating ovens, car engines, refrigerators, air conditioners, etc. One of the examples of the thermostat is a gas cooker thermostat. Temperatures in gas cookers are regulated by a thermostat which consists of a brass tube A projecting into the oven. The brass tube encloses an invar rod B which is attached to the brass tube at C. A valve is attached to the end of the invar rod in the oven to regulate the flow of gas from the cylinder into the burners. A temperature increase causes the brass tube A to expand and move to the left. The invar rod also moves with it and causes the valve to partially close to reduce the gas inflow.



THERMOTROPIC MESOPHASE: A mesophase formed by heating a solid or cooling an isotropic liquid or by heating or cooling a thermodynamically stable mesophase.

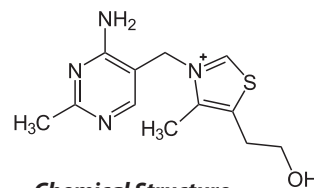
THEROPHYTE: A plant-like form in the Raunkiaer's system of classification. They are annual plants that complete their life in a short period when conditions are favourable and survive harsh conditions as seeds. They are found in deserts and arid areas.

THESIS: A statement or theory put forward and supported by arguments, especially a lengthy written essay submitted, for example, as part of the requirements for a university degree.

THETA FUNCTION: Any of a class of important functions used in topology, number theory and analysis. The basic one from which all others can be derived is the entire function, $\theta_3(z, q) = 1 + 2 \sum_{n=1}^{\infty} q^{n^2} \cos(2\pi n z)$, where $q = e^{\pi i \tau}$ for $\text{Im} \tau > 0$. The Jacobi theta functions are the elliptic analogues of the exponential function and may be used to express the Jacobi elliptic functions.

THETA VACUUM (θ VACUUM): The vacuum state for non-Abelian gauge fields (in the absence of fermion and Higgs fields). In the theta vacuum there are infinite numbers of degenerate states with tunnelling between these states. This means that the theta vacuum is analogous to a Bloch function in a crystal. When a massless fermion field is present, the tunnelling between states is completely suppressed. When there are fermion fields with a small mass, tunnelling is much smaller than for pure gauge fields, but it is not completely suppressed. Theta vacuum helps in understanding the vacuum state in strongly interacting gauge theories, such as quantum chromodynamics.

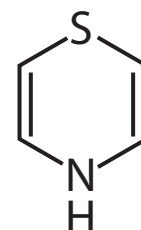
THIAMINE (THIAMIN; ANEURIN; VITAMIN B₁): A white water-soluble crystalline B vitamin complex with the chemical formula $C_{12}H_{17}N_4OS$. It is necessary for carbohydrate metabolism and normal neural activity. Thiamine also plays a vital role in the activities of enzymes, proteins that are involved in bodily processes such as digestion. Good sources



Chemical Structure of Thiamine

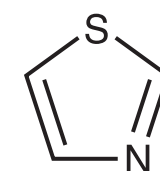
of thiamine include wheat germ, dry beans, peas, enriched cereals, breads, pasta, nuts, eggs and most vegetables. Lean pork is one of the best sources of the vitamin, as are organ meats, such as liver. Deficiency of thiamine causes the disorder beri-beri in human beings.

THIAZINE: A member of a group of heterocyclic compounds that contain sulfur and nitrogen in addition to four carbons in a six-membered ring. It is used in making dyes.



Chemical Structure of Thiazine

THIAZOLE (1, 3 THIAZOLE): A heterocyclic compound containing a five-membered ring with sulfur and nitrogen hetero atoms with the chemical formula C_3SNH_3 . A range of thiazole dyes are manufactured containing the ring system.



Chemical Structure of Thiazole

THICK CLIENTS (HEAVY CLIENTS): Full-featured computers that are connected to a network. Unlike thin clients, which lack hard drives and other features, thick clients are functional whether they are connected to a network or not. **Thin clients** function as regular PCs, but lack hard drives and typically do not have extra I/O ports or other unnecessary features. Since they do not have hard drives, thin clients do not have any software installed on them. Instead, they run programs and access data from a server.

THICK FILAMENT: A bipolar myosin filament that is 10-12 nanometres in width and 1.6 metres long occurring in striated muscle and composed of staggered arrays of myosin molecules. The thick filaments are the major constituents of the A-band of the sarcomere in skeletal muscle; they interact with the thin filaments to bring about muscle contraction.

THIGMOMORPHOGENESIS: A response in plants to chronic mechanical stimulation such as touch by altering their growth patterns. In the wild, these patterns can be evinced by wind, raindrops and rubbing by passing animals. An example is thickening of stems in response to strong winds. Some plants respond to the touch stimulus within seconds, while others show morphogenetic alterations over long periods of time, ranging from days to weeks. Various signalling molecules and phytohormones, including intracellular calcium, jasmonates, ethylene, abscisic acid, auxin, brassinosteroids, nitric oxide and reactive oxygen species, are involved in touch responses.

THIGMOTAXIS: A taxis in which the movement is parallel to lines of stress in the substrate. It is seen, for example, in bacteria of the order Myxobacteriales.

THIGMOTROPISM (HAPTOTROPISM): The growth of an aerial plant organ in response to localised physical contact stimuli. For example, when the tendrils of sweet pea touches a supporting structure, it curves in the direction of the support and coils round it. Climbing plants, such as vines, develop tendrils that coil around supporting objects. Touched cells produce auxin and transport it to untouched cells. Some untouched cells will then elongate faster so cell growth bends around the object.

THIN: 1. A free-flowing consistency similar to that of water, not viscous. 2. Of a photographic negative, lacking contrast or density. 3. Having a small diameter or cross section, for example, thin wire. 4. In logic, to weaken a form of argument, especially by adding to its premises; for example, from A ... B we derive A, C ... B by thinning.

THIN CLIENT (LEAN CLIENT): Low cost system in which the terminals depend on the host computer for software. It has little or no installed software but has access to software that is managed and delivered by network servers that are attached to it. A thin client is an alternative to a full-function client such as a PC.

THIN FILM TRANSISTOR LCD; TFT LCD (ACTIVE MATRIX LCD): A type of colour liquid crystal display commonly used in laptop computers. TFT screens have one or more transistors to control each pixel on the screen, as opposed to the less expensive passive matrix displays where each pixel is not matched with a transistor. TFT LCDs have a faster refresh rate than other types of LCD, reducing the lag between computer and display.

THIN-LAYER CHROMATOGRAPHY (TLC): A type of liquid chromatography that can separate chemical compounds of differing structure based on the rate at which they move through a support under defined conditions. The stationary phase is a thin layer of an absorbing solid prepared by spreading slurry of the solid on a plate and drying it in an oven.

THIN-SKINNED DEFORMATION: Deformation in plate tectonics at a convergent boundary which occurs with shallow thrust faults that only involves cover rocks, typically sedimentary rocks, not deeper basement rocks. **Thick-skinned** deformation involves thrust faults that reach down into the crystalline basement.

THINNING: Pulling out or cutting off some seedlings in a row, plot or flat, for the benefit of those that remain.

THIO-: A prefix denoting the presence of sulfur or a compound that consists of the chemical group SCN bonded by the sulfur atom to a group or an atom other than a hydrogen atom. It is a term used in chemical nomenclature in the names of compounds in which part or all of the oxygen atoms have been replaced by sulfur, especially of a compound in which oxygen has been replaced by divalent sulfur. An example is thioacid.

THIOACETALS: A term including monothioacetals having the structure $R_2C(OR')(SR')$ (subclass monothioketals, $R \neq H$); and dithioacetals having the structure $R_2C(SR')_2$ (subclass dithioketals, $R \neq H, R' \neq H$).

THIOALDEHYDE S-OXIDES: Compounds having the structure $RC(=S=O)H$.

THIOCARBOXYLIC ACIDS: Compounds in which one or both oxygens of a carboxy group have been replaced by divalent sulfur $RC(=O)SH$ or $RC(=S)OH$ monothiocarboxylic acids, $RC(=S)SH$ dithiocarboxylic acids.

THIOCYANATE: A salt or ester of thiocyanic acid used especially of a compound such as thioacid or sodium thiocyanate ($NaSCN$), in which oxygen has been replaced by divalent sulfur.

THIOESTER: Compounds with the functional group $C-S-CO-C$. They are the product of esterification between a carboxylic acid and a thiol. Thioesters are widespread in biochemistry, the best-known derivative being acetyl-CoA.

THIOETHER: A member of a class of organic compounds with the general formula RSR' , where R and R' are alkyl or aryl groups, which are analogous to ethers, in which sulfur takes the place of oxygen. They are named as alkyl or arylsulfides. An example is ethyl methyl sulfide ($C_2H_5SCH_3$).

THIOHEMIACETALS: Compounds of structure $R_2C(SR')OH$ or $R_2C(OR')SH$ (monothiohemiacetals) or $R_2C(SR')SH$ (dithiohemiacetals), $R' \neq H$.

THIOKETONE S-OXIDES: Compounds having the structure $R_2C=S=O$ ($R \neq H$).

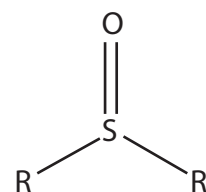
THIOKETONES: Compounds in which the oxygen of a ketone has been replaced by divalent sulfur $R_2C=S$ ($R \neq H$), for example, $CH_3C(=S)CH_2CH_3$ butane-2-thione.

THIOL: A member of a class of unpleasant smelling organic compounds, with the general formula RSH , where R is an alkyl or aryl group analogous to alcohols, in which sulfur takes the place of oxygen in the hydroxyl group. An example is methane thiol, CH_3SH . The $-SH$ is called the thiol or mercapto group.

THIOLATES: Derivatives of thiols, in which a metal (or other cation) replaces the hydrogen attached to sulfur, e.g. $CH_3S^-Na^+$ sodium methanethiolate.

THIONYL CHLORIDE (SULFUR OXYCHLORIDE; SULFUROUS OXYCHLORIDE): A colourless fuming pungent liquid with the chemical formula $SOCl_2$, density 1.638 g/cm^3 (liquid), boiling point 79°C (352 K) and decomposes in water at 140°C (413 K). It is soluble in benzene. It is synthesised industrially by reacting sulfur trioxide and sulfur dichloride: $SO_3 + SCl_2 \rightarrow SOCl_2 + SO_2$. It is used as a chemical intermediate and catalyst and also used in organic synthesis to convert an $-OH$ group to $-Cl$. It is a useful drying agent.

THIONYL GROUP: The group $=SO$, a sulfur atom plus an oxygen atom as occurs in sulfur dichloride oxide and thionyl fluoride. In organic chemistry, the thionyl group is known as a sulfoxide group or sulfinyl group and has the general structure $RS(=O)R'$. It occurs in compounds such as thionyl fluoride. Thionyl chloride ($SOCl_2$) is a common reagent used in organic synthesis to convert carboxylic acids to acyl chlorides.



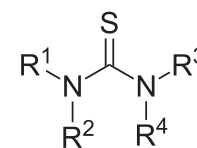
Chemical Structure of Thionyl Group

THIOPHENE: Heterocyclic compound with the chemical formula C_4H_4S . It consists of a flat five-membered ring and is aromatic as indicated by its extensive substitution reactions. Related to thiophene are benzothiophene and dibenzothiophene, containing the thiophene ring fused with one and two benzene rings, respectively.

THIOSULFATE (SODIUM THIOSULFATE): An ion and an acid radical with the chemical formula $S_2O_3^{2-}$. They form salts with alkali metals. One such alkali metal, sodium thiosulfate is used in processing photographic films and prints. They readily decompose in acid solution to give elemental sulfur and hydrogen sulfite ions (HSO_3^-). It rapidly dechlorinates water and is notable for its use to halt bleaching in the paper-making industry. Thiosulfate is also useful in smelting silver ore, in producing leather goods and to set dyes in textiles.

THIOSULFURIC ACID: An unstable acid which readily decomposes to sulfurous acid and sulfur with the chemical formula $H_2S_2O_3$, molar mass 114.14 g/mol and melts at 0°C (273 K). Its salts, thiosulfates are used in photography.

THIOUREA (THIOCARBAMIDE): A colourless crystalline organic compound which is the sulfur analogue of urea with the chemical formula $(NH_2)_2CS$, density 1.405 g/ml and melting point and 182°C (455 K). It is produced from ammonium thiocyanate or by the reaction of hydrogen sulfide with calcium cyanamide in the presence of carbon dioxide. It is used in medicine, as a developer in photography and photocopying and in various organic syntheses.



Chemical Structure of Thiourea

THIRD BODY: A species, other than the reactant itself, which brings about the energisation of a molecule that can undergo a unimolecular process or brings about a combination reaction between atoms or radicals. In the latter case a third body is also called a chaperon.

THIRD CLASS LEVER: Levers in which the effort is applied between the output load on one end and the fulcrum or pivot on the opposite end. Examples include fishing rod, hockey stick, nail clippers, stapler, tong/tweezers, etc.

THIRD GENERATION (3G): A type of wireless mobile telecommunications technology, which is the upgrade of 2G and 2.5G GPRS networks, for faster Internet speed. 3G telecommunication networks support services that provide information transfer rate of at least 0.2 Mbit/s . Later 3G releases, often denoted 3.5G and 3.75G, also

provide mobile broadband access of several Mbit/s to smartphones and mobile modems in laptop computers. 3G technology finds application in wireless voice telephony, mobile Internet access, fixed wireless Internet access, video calls and mobile TV.

THIRD IONISATION ENERGY: 1. The amount of energy required to remove one mole of electrons from one mole of positive ions with a double charge. 2. The minimal energy required to remove to infinity an electron from the atom or molecule isolated in free space and in its ground electronic state. In atomic physics, the unit of ionisation energy is the electronvolt (eV). In chemistry, it is normal to give the value in kJ/mol of the related quantity, called molar ionisation energy, but often called ionisation energy.

THIRD LAW OF THERMODYNAMICS: A law stating that the entropy of a substance approaches zero as its temperature approaches absolute zero.

THISTLE: A common name for several plants with prickly stems and leaves of the family composite flowers. Examples are the Canada thistle, which bears small lilac or white flowers; the musk thistle, which bears purple flowers; and the globe thistle, which bears whitish or blue flower.

THIXOTROPY: The property of certain gels or fluids that are viscous under normal conditions, but less viscous or fluid when shaken, agitated or otherwise stressed. Examples are mayonnaise and toothpaste.

THOMSON EFFECT (KELVIN EFFECT): A phenomenon which is explained as when electric current flows through a conductor, the ends of which are maintained at different temperatures, heat is evolved at a rate approximately proportional to the product of the current and the temperature gradient. If either the current or the temperature gradient is reversed heat is absorbed rather than evolved. It was predicted and subsequently observed by Lord Kelvin in 1851.

THOMSON LAMP: A paradoxical machine devised as a thought-experiment by British philosopher James Thomson (1921-1984) in 1970 to highlight the difficulty in Zeno's paradoxes of deciding whether a supertask, an infinite number of tasks infinite time, has been completed. The lamp is switched on and after a minute off; after a further half minute it comes on again, goes off after another quarter minute and so on, changing its state after each term of a convergent series of time intervals. Since this series of intervals has a sum of two minutes, the process must have terminated after that time, so the lamp must be either on or off; yet it cannot be either on or off, since each time it entered either state it was immediately switched again. This does not show, however, as Thomson argued, that Zeno's paradoxes are unresolved by taking limits, since these paradoxes concern convergent rather than oscillating series of tasks.

THOMSON MODEL (THOMSON'S ATOMIC MODEL): A model which describes the atom as a solid sphere of positive charge with the negative electrons stuck uniformly in it. It was proposed by Joseph John Thomson (1856-1940), the British physicist.

THOMSON SCATTERING: The scattering of electromagnetic radiation by free charged particles, especially electrons, when the photo energy is small compared with the energy equivalent to the rest mass of the charged particles. The electric field of the incident wave accelerates the particle, causing it to emit radiation at the same frequency as the incident wave, scattering the wave. It was first explained by Joseph John Thomson (1856-1940), the British physicist.

THOMSON, SIR JOSEPH JOHN (1856-1940): British physicist, educated at the University of Cambridge, where he became a professor at the Cavendish Laboratory (1884-1918). He is best known for his work on cathode rays, which led to his discovery of the electron in 1897, which he called corpuscles. He also studied the conduction of electricity through gases and was awarded the Nobel Prize for Physics in 1906 for his research into the electrical conductivity of gases. His son, Sir George Thomson (1892-1975), discovered electron diffraction, for which he shared the 1937 Nobel Prize for Physics with Clinton J. Davisson (1881-1958), who independently made the same discovery. He also discovered isotopes and invented mass spectrometry.

THOMSON, SIR WILLIAM, 1ST BARON KELVIN (1824-1907):

British mathematical physicist and engineer, who entered the University of Glasgow at 10, published two papers by 17 and graduated from Cambridge University at 21. As the chair of natural philosophy at University of Glasgow, he did important work in the mathematical analysis of electricity and formulation of the first and second Laws of Thermodynamics and did much to unify the emerging discipline of physics in its modern form. In 1848, he invented the absolute temperature scale named after him. He served as chief consultant for the laying of the first Atlantic cable (1857-58). For his work on the transatlantic telegraph project he was knighted by Queen Victoria, becoming Sir William Thomson. He patented a mirror galvanometer for receiving telegraph signals in 1858. His work in electricity and magnetism led ultimately to James Clerk Maxwell's theory of electromagnetism. He also contributed to the determination of the age of the Earth and the study of hydrodynamics. He had extensive maritime interests and was most noted for his work on the mariner's compass, which had previously been limited in reliability.

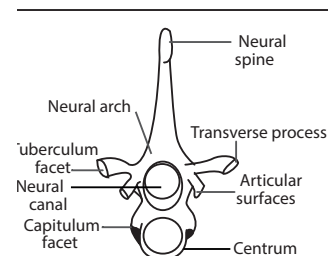
THONNINGIA SANGUINEA: A genus of annual or perennial flowering plant in the family Balanophoraceae, containing only 1 species, *Thonningia sanguinea* with the common name ground pineapple. It is distributed throughout much of tropical southern and western Africa. It is a fleshy herb devoid of chlorophyll and does not photosynthesise, therefore, it is parasitic on roots of trees and shrubs. It forms bulb-like swellings at the points where it attaches to the roots of its host plants. The branching, fleshy, yellowish tuber is horizontal, trailing 10 – 15 cm below soil surface. The leaves are triangular or ovate, forming an involucre around the spadix. The flowers are unisexual or androgynous inflorescences and resembling that of a pineapple or palm tree. It is used prophylactically against bronchial asthma in Ghana and also to treat sexually transmitted diseases. It has also been found to have antioxidative and hepatoprotective actions. In some parts of Africa, it is used to treat diarrhoea, worms, leprosy, skin infections and abscesses, dental caries, gingivitis, fever, malaria, heart disease, rickets and rheumatism.

THORACIC CAVITY (CHEST CAVITY): The space within the chest or thorax bounded by the rib cage and above the abdominal cavity in vertebrates that contains the heart and lungs. It is separated from the abdominal cavity by the diaphragm.

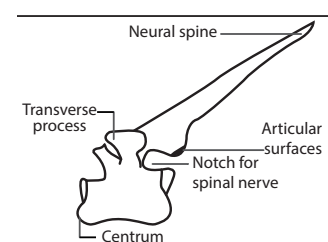
THORACIC DUCT: The key collecting vessel of the lymphatic system. It runs longitudinally in front of the backbone and drains its lymph into the superior vena cava or the major efferent lymph duct into which lymph from most of the peripheral lymph nodes drains. Recirculating lymphocytes that have left the circulation in the lymph node return to the blood through the thoracic duct.

THORACIC VERTEBRAE: The vertebrae of the upper back, between the cervical and the lumbar vertebrae, which articulates with the ribs. In mammals, there are 12-15 thoracic vertebrae and humans possess 12. They are characterised by long neural spine for muscle attachment and articulating facets for attachment of the ribs.

THORAX: The anterior region of the body trunk of animals, containing the heart and the lungs of vertebrates, except fish. In mammals, it is enclosed by the ribcage and is separated from the abdomen by the diaphragm. In insects



Thoracic vertebra of a mammal (Anterior view)



Thoracic vertebra of a mammal (Lateral view)

the thorax bears the wings and legs and it is divided into three parts namely: the prothorax, mesothorax and metathorax, each bearing a pair of legs.

THORIUM: Silvery-white radioactive element and one of the actinides with the symbol Th, atomic number 90, atomic weight 232.038, approximate melting point 1,750 °C (2023 K), approximate boiling point 4,500 °C (4773 K) and approximate relative density 11.7. It has several isotopes, with half-lives of up to 1.39×10^{10} years. Its isotopes have mass numbers from 223 to 234. The most stable isotope, thorium-232, has a half-life of 1.39×10^{10} years, captures slow or thermal neutrons and is used to generate the fissile uranium-233. It occurs in monazite sand in Brazil, USA and India.

THORIUM CARBIDE: A yellowish solid with the chemical formula ThC_2 , which melts above 2630 °C (2903 K) with a molecular weight of 244.0488 and decomposes in water. It is used in nuclear fuel.

THORIUM DIOXIDE (THORIUM(IV) OXIDE): White insoluble powder with the chemical formula ThO_2 , produced mainly as a by-product of lanthanide and uranium production and used as a refractory and in non silica optical glass.

THORIUM FLUORIDE (THORIUM(IV) FLUORIDE; THORIUM TETRAFLUORIDE): A white, hygroscopic toxic powder with the chemical formula ThF_4 , which melts at 1111°C (1384 K). It can be produced by reacting thorium with fluorine gas. It has excellent optical transparency in the range 0.35-12 μm and its radiation is primarily due to alpha particles, which can be easily stopped by a thin cover layer of another material. It is used in making thorium metal, magnesium-thorium alloys and in high-temperature ceramics.

THORIUM NITRATE (THORIUM(IV) NITRATE HYDRATE; THORIUM NITRATE HYDRATE): An explosive white crystal with the chemical formula $\text{Th}(\text{NO}_3)_4 \cdot x\text{H}_2\text{O}$. The anhydrous form decomposes at 500°C (773 K). It is soluble in water and alcohol. It has low chemical toxicity, but alpha emission is expected to constitute a hazard if a fairly large amount is inhaled or ingested. Its dust may irritate eyes and cause diffuse dermatitis. Beta and gamma emission is small. It is a strong oxidiser. It is used in medicine and as an analytical reagent.

THORIUM SERIES: The 4n chain of Th-232 or series of nuclides resulting from the decay of thorium-232 into the stable isotope lead-208. Beginning with naturally occurring thorium-232, this series includes the following elements: actinium, bismuth, lead, polonium, radium and radon. All are present, at least transiently, in any natural thorium-containing sample, whether metal, compound or mineral. The series terminates with lead-208.

THORN: A sharply pointed woody growth projecting from the stem of some trees, bushes and woody plants, usually protective in function. Botanically, in plants such as the honey locust and hawthorn, thorns are distinguished as modified stems. **Spines** as found in the barberry are strong, sharp-pointed, usually woody outgrowth from a stem or leaf. **Prickles**, as found in the rose and blackberry are sharp pointed epidermal outgrowths of the bark.

THOROUGHFARE CHANNELS: 1. Shortcuts within the capillary network that allow blood to bypass a capillary bed. 2. Central channel in the arteriolar system in which flow through the channel is controlled by the sphincter effect of the branch of the arteriole, the metarteriole.

THOUSAND: The fourth digit to the left of the decimal point in the decimal number system or 10^3 ; or the cardinal number equal to 10×100 or 10^3 . A **thousandth** is one part of a thousand equal parts of something or the place value that represents one thousandth of a whole.

THOUSAND-HEADED KALE: A variety of kale grown for feeding livestock, usually in the winter months. It has many branches and small leaves. The dwarf thousand-head produces a large number of new shoots during the winter.

THREAD: 1. Fine cord of spun cotton, silk, flax, wool, etc., especially for sewing and weaving. 2. In computer programming or software, a thread is a portion of a program that is executed in conjunction with the main program that performs another task. For example, a program may have an open thread waiting for a specific event to occur or may have a thread running a separate job, allowing for the main program to perform other tasks. A program is capable of having multiple threads open at once and will either terminate or suspend the thread when the thread's.

THREAD CELL (NEMATOBlast; CNIDOBlast): A type of specialised cell found only in the ectoderm of the Cnidaria. If the sensory projection on the surface of the thread cell is touched, for example, by a prey, the thread is shot out and adheres to the prey, coils round it or injects poison into it. Numerous thread cells on the tentacles of jellyfish produce their sting.

THREAD VEIN: A very slender vein, especially one that is visible through the skin, most commonly occurring on the legs, although other parts of the body may be affected. They are tiny veins just under the skin that have become dilated. With age, changes occur in the skin and the subcutaneous tissue, which become thinner and more fragile, so that blood vessels under the skin become much more obvious. Thread veins do not signify any underlying medical problem.

THREADWORM: Any of various small nematode worms of the genus *Enterobius*, family Oxyuridae that are parasitic on horses, rabbits and other mammals. One species *Enterobius vermicularis*, infests the human intestines and rectum.

THREATENED SPECIES: Species that are likely to become endangered in the foreseeable future throughout all or a significant portion of their range. The International Union for Conservation of Nature (IUCN) treats threatened species not as a single category, but as a group of three categories, depending on the degree to which they are threatened: vulnerable species, endangered species and critically endangered species. A **vulnerable species** is one that is likely to become endangered unless circumstances such as habitat loss or destruction threatening its survival and reproduction improve. An **endangered (EN) species** is one, which has been categorised as likely to become extinct. A **critically endangered (CR) species** is one, which has been categorised by the IUCN as facing a very high risk of extinction in the wild. It is the highest risk category assigned by the IUCN Red List for wild species. Conservation biologists use the IUCN Red List to categorise endangered and critically endangered species. There are currently 2129 animals and 1821 plants with this assessment worldwide.

THREE N PLUS ONE (3N + 1) PROBLEM: The unsolved decision problem of whether the sequences generated by starting with any natural number and repeatedly taking the odd part of $(3n + 1)$ always terminates with 1.

THREE-BODY PROBLEM: The mathematical problem predicting the positions, masses and velocities of three massive bodies, which orbit each other gravitationally, at any point in time, given their present positions, masses and velocities, according to Newton's laws of motion and of universal gravitation. An example would be to completely solve the behaviour of the Sun-Jupiter-Saturn system or that of three mutually orbiting stars.

THREE-DIMENSIONAL (3D): An object which has length, breadth and height. For example, a geometric solid such as a cube or sphere is a three-dimensional figure.

THREE-CIRCLE THEOREM: The theorem which states that a complex function that is analytic in an annulus is such that $\log m(r)$ is a convex function of $\log r$, where $m(r)$ is the maximum modulus of the function on a ring of radius r . This result is so named as it involves circles of three radii.

THREE-POINT CONTACT: The relation between two curves, surfaces, etc., at a point at which the defining equations of the

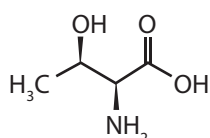
intersection have at least a triple root. In Euclidean geometry this is equivalent to them touching and having the same curvature at that point.

THREE-POINT LINKAGE (THREE-POINT HITCH): A method of joining implements to a tractor. Automatic couplers for three-point-linkage permit implements which is attached rapidly and safely.

THREE-SPACE PROPERTY: A property that is inherited by a space, S , whenever it is possessed both by a subspace, T and by the factor space, S/T . For example, being Noetherian is a three-space property. Such a property of groups is called a **polyproperty**.

THREE-TIMES-A-DAY: A milking system in which cows are milked three times a day. The application of this system can increase milk yields.

THREONINE (2-AMINO-3-HYDROXYBUTANOIC ACID): A colourless crystalline amino acid with the chemical formula $C_4H_9NO_3$. It is obtained from the hydrolysis of protein and is an essential component of human nutrition.



Chemical Structure of Threonine

THRESH: To use a flail, machine or other implement to separate the seeds of a harvested plant from the straw, chaff, husks and other residue. **Thresher** is a harvester of crop with a flail, machine or other implement. **Threshing** is the removal of grain from the plant after harvest.

THRESHOLD (THRESHOLD POTENTIAL): The minimum intensity of a stimulus (usually an electric current) that is necessary to generate an action potential in a neuron.

THRESHOLD FREQUENCY: The minimum frequency of the incident radiation capable of dislodging an electron from the surface of a metal or the frequency of incident radiant energy below which there is no photoemissive effect.

THRESHOLD MOISTURE CONTENT: The minimum moisture condition, measured in terms of either moisture content or moisture stress, at which biological activity just becomes measurable.

THRESHOLD POTENTIAL: The potential an excitable cell membrane must reach for an action potential to be initiated. Threshold potentials are necessary in order to regulate and propagate signalling in both the central nervous system (CNS) and the peripheral nervous system (PNS).

THRIPS: A tiny sucking insect of the order Thysanoptera that feeds on the sap of plants and a major pest of cereals and fruit trees. It has four long thin wings fringed with hairs.

THROAT: 1. The passage that leads from the back of the nose and mouth to the trachea and oesophagus. It includes the pharynx and the larynx, the latter being at the top of the trachea. 2. Any narrowing entry such as the throat of a carburettor. 3. The opening of a tubular corolla or calyx where the tube joins the limb.

THROMBIN: An enzyme that catalyses the conversion of fibrinogen to fibrin. The process is an important step in the clotting of blood.

THROMBOCYTE (PLATELET): A disk-shaped non-nucleated particles found in large quantities in the blood. They play an important role in blood clotting.

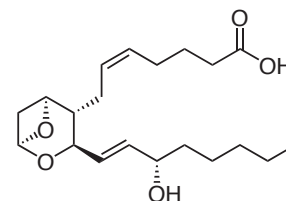
THROMBOPLASTIN: Reactions leading to formation of blood clot by converting prothrombin to thrombin in the early stages of blood clotting.

THROMBOSIS: The formation of a solid mass or a clot of blood in a blood vessel or in the heart that may partially or completely block an artery or vein. The chief causes of thrombosis are injury to the inner lining of a blood vessel; blood disorders that result in thickened blood and an increased tendency toward clotting; and restricted blood flow, caused by such problems as obesity. When thrombosis occurs

in an artery, the tissues that the artery normally supplies with blood are denied of blood-borne nutrients and suffer infarction and die.

THROMBOXANE A_2 (TXA_2):

A co-factor that is released by activated platelets during injury to function together with prostaglandins to promote blood clotting by forming a platelet plug and causing constriction of blood capillaries to reduce bleeding. Its action antagonises that of prostacyclin, which dilates blood vessels and prevents the formation of blood clots.



Chemical Structure of Thromboxane

THROUGH-BOND ELECTRON TRANSFER: Intramolecular electron transfer for which the relevant electronic interaction between the donor and acceptor sites is mediated by through-bond interaction, via the covalent bonds interconnecting these sites, as opposed to through-space interaction.

THROUGH-SPACE ELECTRON TRANSFER: Electron transfer for which the relevant electronic interaction between the donor and acceptor sites is mediated either by direct orbital overlap or by superexchange interaction via intervening molecular entities not covalently bound to the donor or acceptor sites.

THROUGHPUT: 1. In computing, it is the amount of work done by the computer in a given length of time. It depends on a number of parameters that include the processor speed, speed of the peripheral devices and the design of the operating system. 2. The rate of transfer of data or information in a communication network from one location to another in a given amount of time. It is measured in bits per second (bps) or bytes per second (Bps). It is used to measure the performance of hard drives and RAM, as well as Internet and network connections.

THRUM: A heterostylous flower such as primrose, oxlip, cowslip and polyanthus having the stigma on a fairly short style below the anthers and long stamens.

THRUSH (ORAL CANDIDIASIS; ORAL THRUSH): A mucous membrane infection affecting the mouth, tongue, inner cheeks, the roof of the mouth and throat of children and adults with reduced immune function, especially those with diabetes or kidney failure. It is caused by the yeast-like fungus, *Candida albicans* that inhabits the human digestive tract, the vagina and also found on the skin in small amounts. It is characterised by small creamy lesions or eruptions and redness or soreness in the affected area. Those on steroids and prolonged use of antibiotics can be affected. The term thrush is also used as a synonym for **candidiasis** or **vaginal thrush**, an infection of the vagina by the yeast-like fungus, *Candida albicans*.

THRUST: 1. The forward force that propels an aerospace vehicle or marine craft resulting from the stream of air or gases ejected backwards by a propeller or a rocket engine. Thrust is a vector quantity and its magnitude is usually given in newtons (N) in SI units. It is usually calculated as the product of the rate of mass discharge and the velocity of the exhaust gases, relative to the vehicle. 2. A continuous force exerted sideways or downward by one structure or another such as by an arch or an abutment against a wall. 3. A force in the Earth's crust that results in recumbent folding of rock strata.

THRUST GENERATOR: A device that produces motive power. In an electric propulsion system, for example, it is composed of an electric power source and a device which expels a high velocity flow of the propellant.

THUE-SIEGEL-ROTH THEOREM (ROTH'S THEOREM): The culminating theorem of rational approximation that for any algebraic irrational number α and for any $\kappa > 2$, there are only finitely many rational numbers p/q solving $|\alpha - p/q| < Cq^{-\kappa}$.

THULIUM: Silvery-white metallic element of the rare-earth group and one of the Lanthanides with the chemical symbol Tm, atomic number 69, relative atomic mass 168.934, melting point about 1545 °C (about 1818 K), boiling point about 1950 °C (about 2223 K), with a relative density of 9.32. Its radioactive isotope emits gamma-rays and X-rays, which are used in portable radiographic equipment. The stable isotope ^{169}Tm makes up 100% of the naturally occurring element.

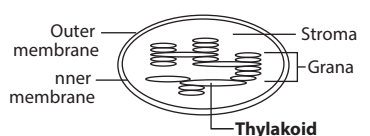
THUMB: 1. The thickest digit of the human hand located next to the forefinger. 2. A short thick digit in some animals, such as in many primates, that is adapted for grasping and corresponds to the human thumb.

THUNDER: The crashing or booming sound which usually follows a flash of lightning or a loud rumbling noise caused by the rapid expansion of air suddenly heated by the electrical discharge of lightning. The sound is produced by lightning bolt, which heats the air around it so quickly (within a few millionths of a second) and to such a high temperature, about 10,000°C (10,273 K) that the air molecules are pushed apart with great force, much like in an explosion. A wave of compressed air (a sound wave) moves out from the lightning bolt.

THUNDERBOLT: 1. A flash of lightning that makes a loud sound of thunder and that can hit someone, an object or structure. 2. In computer science, it is an interface used to connect peripheral devices to a computer using an expansion bus. It is a serial interface that combines existing PCI Express and DisplayPort technology, allowing for longer distance of data transfer and cheaper cables. Thunderbolt was developed by Intel, which also owns the rights to the technology.

THUNDERSTORM: A convective or violent, short-lived atmospheric disturbance storm accompanied by lightning and thunder and a variety of weather conditions especially heavy rain or hail, high winds and sudden temperature changes. They occur when intense heating causes a parcel of moist air to rise, resulting in instability and the development of cumulonimbus cloud. They are most frequent in the tropics but are also common in the middle latitudes.

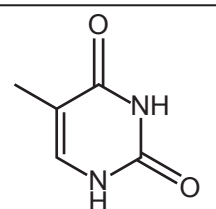
THYLAKOID: Any of the flattened sack-like membranous structures that is stacked on top of one another to form the grana of plant chloroplasts. Chlorophyll and other photosynthetic pigments are situated in the thylakoid membranes, the site for the light-dependent stage of photosynthesis.



Chloroplast showing the thylakoid

THYMIDINE (DEOXYTHYMIDINE; DEOXYRIBOSYLTHYMINE; THYMINE DEOXYRIBOSIDE): A nucleoside made up of one thymine molecule linked to a deoxyribose sugar molecule. In cell biology it is used to synchronise the cells in S phase.

THYMINE (5-METHYLPYRIMIDINE-2,4(1H,3H)-DIONE; 5-METHYLURACIL): Organic compound of the pyrimidine family, often called a base, consisting of a ring containing both nitrogen and carbon atoms and a methyl group. Its chemical formula is $\text{C}_5\text{H}_6\text{N}_2\text{O}_2$, symbol T, melting point 316-317°C (589-590 K) and boiling point 335°C (608 K). It occurs in combined form in many important biological molecules, particularly DNA, where its complementary base is adenine. Its corresponding nucleoside or nucleotide may be prepared from DNA by selective techniques of hydrolysis.



Chemical Structure of Thymine

THYMOCYTE: An immature or developing lymphocyte in the thymus, serving as the forerunner of a T-cell. Progenitor cells enter the thymus from bone marrow and become thymocytes, which proliferate and undergo distinct developmental stages marked by changes in their CD surface proteins.

THYMOL: A colourless crystalline aromatic compound with the chemical formula $\text{C}_{10}\text{H}_{14}\text{O}$, density 0.96 g/cm³, melting point 51 °C (324 K) and boiling point 232 °C (505 K). It is only slightly soluble in water at neutral pH, but it is extremely soluble in alcohols and other organic solvents. It is also soluble in strongly alkaline aqueous solutions due to deprotonation of the phenol. It is obtained from thyme oil and other volatile oils or produced synthetically. It is used in antiseptic mouthwashes, fungicide and as a preservative.

THYMOSIN: A hormone secreted by the thymus gland. Its primary function is to stimulate the production of T-lymphocytes, which are an important part of the immune system. Thymosin also plays a role in the development of B-lymphocytes to plasma cells to produce antibodies.

THYMUS: An organ, present only in vertebrates, that is concerned with the development of lymphatic tissue and hence the antibody producing white blood cells (lymphocytes) and the immune response. It is a bi-lobed organ in the region of the lower neck above and in front of the heart. In humans, it undergoes progressive shrinking after one year of age.

THYRATRON: A thermionic valve that functions as a gas-filled relay. It is a hot-cathode gas tube in which one or more control electrodes initiate but do not limit the anode current except under certain operating conditions. It is now replaced by the silicon controlled rectifier.

THYRISTOR: A type of semiconductor rectifier which uses a small control current to switch a large current on and off. It is silicon-controlled and its anode-cathode current is controlled by a signal applied to a third electrode.

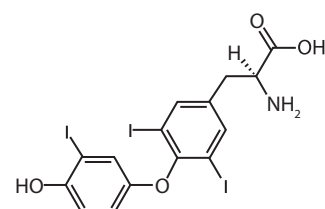
THYROGLOBULIN (TGB): 1. A thyroid protein that is the precursor to iodine-containing hormones. It is typically present in the colloid of thyroid gland follicles. 2. A substance extracted from the thyroid glands of hogs, formerly used as a thyroid hormone supplement to treat hypothyroidism.

THYROID GLAND: An endocrine gland in vertebrates, found in the lower part of the trachea of human beings and other vertebrates. It secretes the thyroid hormones such as thyroxine which controls the speed of chemical reaction (metabolic rates in cells and growth). Excessive action of the thyroid gland can cause Graves's disease, while low activity can cause myxoedema.

THYROID HORMONE: Either of the tyrosine-based hormones, thyroxine (T_4) or triiodothyronine (T_3), which are secreted by the follicular cells of the thyroid gland. These iodine-containing hormones, which are formed in the thyroid from thyroglobulin and then released into the blood stream, help control the rate of all metabolic processes in the body and influence physical development and activity of the nervous system.

THYROID STIMULATING HORMONE (TSH; THYROTROPIN): A hormone secreted by the anterior pituitary gland that controls the synthesis and secretion of the two thyroid hormones of thyroxine and triiodothyronine in the thyroid gland. TSH secretion is controlled by thyrotrophin releasing hormones from the hypothalamus. It is sometimes taken in a drug form to boost metabolism and reduce body weight.

THYROXINE: The principal iodine-containing hormone of the thyroid gland with the chemical formula $\text{C}_{15}\text{H}_{11}\text{I}_4\text{NO}_4$. It is produced by the thyroid gland and increases the rate of cell metabolism and regulates growth. A synthetic form is used to treat hypothyroidism, a deficiency in the production of thyroid hormones.



Chemical Structure of Thyroxine

THYRSE: A compact, cylindrical or ovate inflorescence with no terminal flower on the main axis, but with a terminal flower on each lateral partial inflorescence. An example is found in the lilac.

THYRSULA: A small cyme borne in the leaf axil, as occurs in many members of the Labiatae (Lamiaceae), the mint or deadnettle family with herbs or shrubs often with an aromatic smell. Some examples from this family include mints, thyme, tulsi, spearmint and coleus.

THYSANURA: An order of medium-sized wingless insects traditionally placed in the subclass Apterygota and comprising the bristletails and silverfish. These typically have slender bodies with long threadlike antenna and a segmented tail-like extension to the abdomen bordered on each side by the elongated paired cerci.

Ti PLASMID: A plasmid of a tumour-inducing bacterium that integrates a segment of its DNA into the host chromosome of a plant. It is frequently used as a carrier for genetic engineering in plants.

TIBIA: 1. The larger of the two long bones of the human lower leg and many other vertebrates. It is the main load carrying bone of the lower leg and it articulates with the femur at the knee and the tarsus at the ankle. 2. The fourth segment of an insect's leg, which is attached to the femur.

TICK: A small spider-like parasite that fastens itself on the skin of, for example, dogs and sucks blood. Ticks belong to the suborder Metastigmata in the order Acari, class Arachnida, phylum Arthropoda.

TIDAL ENERGY (TIDAL POWER): The energy that results from the Sun and Moon's gravitational forces, which causes tides. This type of energy can be used to produce electricity and other forms of power by turbines operated by tidal flow. In certain areas such as Canada's Bay of Fundy, the tidal range reaches more than 15 metres, which is a potential for the generation of tidal power. Perhaps the drawback is that power is not continuous and varies with the seasons. The first working modern tidal power plant was built in France in 1961-67 and has 24 power units of 10,000 kilowatts each.

TIDAL FLATS: Coastal areas, especially near an estuary, of nearly flat, barren muddy or marsh land that is periodically submerged by tidal waters. Normally these materials have an excess of soluble salt.

TIDAL FORCE: The ability of one massive object to cause tides on another object nearby. If the tidal force is large enough it can give rise to tidal heating. A tidal force comes about because of the differences in gravitational pull on an object due to a large mass around which the object is moving.

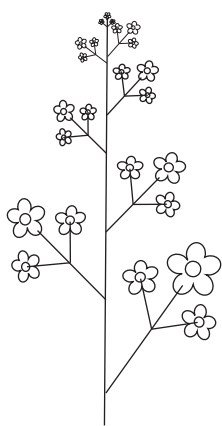
TIDAL RADIUS: The distance from the centre of a planet to the point at which its gravitational pull equals that of the Sun.

TIDAL RANGE: The difference between the water level at high tide and the water level at low tide, expressed in metres.

TIDAL VOLUME: The volume of air taken in or expelled by an animal breathing normally at rest during each cycle of ventilation or breathing. Adults averagely have a tidal volume of 500 cm³.

TIDAL WAVE (TSUNAMI): A large sea wave that is often generated by an underwater earthquake or volcanic eruption. It may be caused by underwater mudslide, earthquake, volcanic eruption or coastal landslide that generates waves over 10 m high, often travelling for hundreds of kilometres with speeds of up to 965 km/h.

TIDE: The regular rise and fall of the water level in the Earth's oceans due to the differences between the gravitational forces between the



Thyrs inflorescence

Earth and another body such as the Moon and Sun or the cyclic rise and fall of sea level as a result of the gravitational pull of the Moon and the Sun. The Moon is more effective than the Sun in causing tides.

TIDE GAUGE: A device fitted with sensors at a coastal location (and some deep sea locations) which continuously measures the level of the sea with respect to the adjacent land. Time-averaging of the sea level so recorded gives the observed relative sea level secular changes. This data is critical for many coastal activities, including safe navigation, sound engineering, and habitat restoration and preservation.

TIE LINE: 1. A horizontal line in a liquid-vapour phase diagram that is perpendicular to an isopleth. The starting point is the first point at which the liquid can coexist with its vapour as the pressure is reduced; the end point is the point beyond which only vapour is present. 2. A communications link between extensions of a private telephone system. 3. A connection between systems, such as electrical power or communications systems.

TIE MOLECULE: A molecule that connects at least two different crystals.

TIETZE'S EXTENSION THEOREM: The theorem that a topological space is normal if and only if every continuous mapping from a closed subset of the space into the unit interval has a continuous extension with range in the unit interval.

TIFF (TAGGED IMAGE FILE FORMAT): In computing, it is a raster graphics file format in which graphical images are stored and transmitted; the images are stored as a bit maps (arrangement of dots).

TIGHT JUNCTION (ZONULA OCCLUDENS): Junction that occurs between cells packed without intercellular spaces, essential for movement of substances across cells to seal off leakages into intestinal lumen from intercellular fluid. They occur in tightly packed cells such as epithelium.

TILAPIA: Any of numerous, mostly freshwater, fish species of the genus Tilapia, family Cichlidae, native to Africa. They are mainly freshwater fish, inhabiting shallow streams, ponds, rivers and lakes and less commonly found living in brackish water. Tilapias have laterally flattened bodies that range from about 10 to 30 cm in length. One species grows to 9 kg. Tilapia species are easy to raise and harvest for food. They grow rapidly, resist disease and readily eat abundant algae and zooplankton.

TILE DRAINAGE: In agriculture, it is a drainage system that removes excess water from soil below the surface. It consists of placing short lengths of pipes end to end at suitable depths and spacings in the soil or subsoil to provide water outlets from the soil. The pipes may be made of concrete, ceramic, fibre, plastic, etc.

TILIA: A genus of about 30 species of deciduous trees that are commonly referred to as lime tree; linden or basswood that is native to the temperate regions of America, Europe and Asia. The trees can grow between 20-40 metres high. The bark is grey, finely fissured or ribbed. The leaves are heart-shaped and alternate. The creamy-white, fragrant flowers are perfect and bisexual and pollinated by insects. The tiny nut-like fruits hang attached to ribbon-like greenish-yellow bracts on the branches which spread numerously from the sturdy trunk. The trees are propagated by grafting and cutting as well as seeds. The lime tree houses lots of aphids and ants due to the rich supply of sap produced by the plant, serving as food for these insects. The lime fruit which contains sour pulp is rich in vitamin C and also used to flavour foods and beverages. The trees are planted ornamentally in parks and gardens to provide shade. The wood is soft and easily worked and used for model building and intricate carving. Some of the species include *Tilia americana* commonly known as American Basswood, is native to North America north of Mexico, used as a source of food for honey bees; and *Tilia platyphyllos*, commonly known as large-leaf linden, large-leaved lime or broad-leaved lime, which is native to central and southern Europe including Great Britain.

TILING (TESSELATION): A collection of smaller shapes that precisely covers a larger shape, without any gaps or overlaps. Usually, the shape to be tiled is a flat plane but other shapes and three-dimensional objects can be tiled, too.

TILL: 1. To prepare land for the growing of crops by ploughing or harrowing. 2. Also called **boulder clay**, it is a deposit of clay, mud, gravel and boulders left by a glacier.

TILLAGE: A combination of mechanical operations with implements such as ploughing and harrowing of land in preparation for growing crops. Tillage aids seed germination by creating a smooth, uniform soil surface for planting. After planting, farmers use tillage to control weeds between crop plants including vegetable, fruit, forest, medicinal and farm crops. **Tillable** is soil that is workable with tools and equipment for growing crops.

TILLER: A shoot that develops from axillary or adventitious buds at the base of a stem, often in response to injury of the main stem. Tillering is seen when the trunk of a tree is lopped, as in the practice of coppicing. It is also characteristic of the growth of grasses, giving the tufted appearance of many species.

TILTH: The physical condition of soil as related to its ease of tillage, fitness as a seedbed, and impedance to seedling emergence and root penetration.

TIMBER: 1. Growing trees with wood suitable for building, carpentry, etc. or wood prepared for use in building, furniture, etc. 2. A large piece of wood used in the framework of a wooden ship or for building.

TIMBERLINE (TREELINE): 1. The highest point on a mountain where trees will grow and above which no tree grows as it is too cold for it to survive. 2. A geographic boundary beyond which trees cannot grow. On the Earth, it is the northernmost or southernmost latitude at which trees can survive.

TIMBRE: The quality of a sound that comes from its tone rather than its volume or pitch. Timbre largely results from a characteristic combination of overtones produced by different instruments.

TIME: The continuum in which events occur in succession from the past to the present and on to the future or a one-dimensional quantity used to sequence events, to quantify the durations of events and the intervals between them. It is used together with other quantities such as space to quantify and measure the motions of objects and other changes. Time is quantified in comparative terms such as longer, shorter, faster, quicker, slower or in numerical terms using units such as seconds, minutes, hours, days. The SI base unit of time is the second. Since 1967, it has been defined as the duration of 9 192 631 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium 133 atom.

TIME COMPLEXITY: A function that describes the amount of time required for a program to run on a computer to perform a particular task. The function is parameterised by the length of the program's input.

TIME CONSTANT: A quantity that gives a measure of the rate of discharge of a capacitor through a circuit consisting of the capacitor and a resistor. It is the time taken for the quantity to fall to $1/e$ of its original value when it varies with time t as e^{-kt} . Generally, it is the time required for an instrument to indicate a given percentage of the final reading resulting from an input signal.

TIME DEPENDENCE OF FUNDAMENTAL CONSTANTS: The possible variation with time in the values of the fundamental constants of nature such as the speed of light. Such a variation has been proposed in various theories but there is no conclusive evidence at present for any such variation.

TIME DILATION (TIME DILATATION): The increase in the time interval between two events, when they occur in a reference frame, which is moving relative to the observer. For example, if two identical clocks are synchronised and placed side by side in an internal frame

of reference they will read the same time for as long as they both remain side by side. If one of the clocks, however, has a velocity relative to the other, which remains beside a stationary observer, the travelling clock will show, to that observer that less time has elapsed than the stationary clock. In general, the travelling clock goes more slowly by a factor $(1-v^2/c^2)$, when measured in a frame of reference travelling at a velocity v relative to another frame of reference. Time dilation becomes noticeable only at speeds approaching that of light and has been accurately confirmed by the apparent increased lifetime of unstable subatomic particles travelling at nearly the speed of light and by the precise timing of atomic clocks carried on airplanes.

TIME DIVISION MULTIPLEXING (TDM): A method of synchronising transmissions in a common medium such that users have the same access at different time slots to the medium.

TIME REVERSAL: The mathematical operation analogous to running a movie film backward, i.e. replacing time t with time $-t$. In other words, a transformation from a given physical system undergoing a given sequence of events to a system in which the exact reverse sequence of states is undergone. The symmetry of time reversal with the symbol T is known as T invariance. As with CP violation, T violation occurs in certain weak interactions, notably kaon decay.

TIME SCALE: 1. Characteristic time for a process to take place. It is numerically defined as the time required for a perturbation in a process to show at least half of its final effect. 2. **Geological time scale** is an internationally developed and agreed division of time into chronological units, whether relative (with units in the correct temporal sequence) or absolute (with numerical ages attached). 3. In astronomy and physics, **dynamical time scale** is the time over which a star responds to deviations from hydrostatic equilibrium. 4. **Nuclear time scale** is an estimate of the lifetime of a star based solely on its rate of fuel consumption. 5. **Thermal time scale** is an estimate of the lifetime of a star once the fuel reserves at its centre are used up. In other words, the thermal time scale is the time in which a star would radiate away all its thermal energy if the nuclear energy production were suddenly turned off. 6. **Planck time** is the time scale beneath which quantum effects are comparable in significance to gravitational effects.

TIME SERIES: In statistics, a sequence of measurement over time usually at regular intervals of a random variable. It may be **continuous**, where observation is made at every instant of time; or **discrete**, where an observation is made at usually regularly spaced intervals.

TIME SHARING: A way of enabling several users to access the same computer at the same time through remote terminals. The computer rapidly switches between user terminals and programs allowing each user to work as if he or she had sole use of the system.

TIME SLICE: In multitasking systems, the period of time a process allows control of the processor, thus enabling the apparent simultaneous execution of two or more processes.

TIME TRAVEL: The act of moving to a different time by moving through space-time. Time travel is theoretically allowed in general relativity theory.

TIME-LAPSE PHOTOGRAPHY: A form of photography used to record a slow process, such as plant growth. A series of single exposures of the object is made on film at predetermined regular intervals. When the film produced is projected at normal cinema speeds the process appears to be taking place at an extremely high rate.

TIMEBASE: A voltage that depends on time and is applied to the electron beam in a cathode-ray tube at regular intervals so that the luminous spot on the screen is deflected in a predetermined manner and usually sweeps it repeatedly and horizontally in a straight line. The period during which the spot returns to its starting point called the **flyback**, is suppressed in devices like television. In a cathode-ray oscilloscope the timebase can be put to more complicated uses.

TIMELINE: A linear representation of important events in a chronological order.

TIMER REVERSAL: An operation representing a transformation from a given physical system undergoing a given sequence of events to a system in which the exact reverse sequence of events take place. The symmetry of time reversal is known as T variance. It has the symbol T.

TIN: A very soft silvery metal, malleable and somewhat ductile element with the symbol Sn that is a member of Group 14 (IVA) of the Periodic Table. Ordinary tin melts at about 232 °C (about 505 K), boils at about 2602 °C (about 2875 K) and has a relative density of 7.31, atomic weight 118.71 and atomic number 50. The most important tin-bearing mineral is cassiterite, SnO_2 . No high-grade deposits of this mineral are known and most of the world's tin ore is obtained from low-grade alluvial deposits. Two allotropic forms of tin exist: white (β) and grey (α) tin. Grey tin is an amorphous powder with a relative density of 5.75. It is obtained when white tin changes into an allotropic form at temperatures below 13 °C (283 K). Tin reacts with both strong acids and strong bases, but it is relatively resistant to solutions that are nearly neutral. The natural element has 21 isotopes and five radioactive isotopes are known. It is used in alloys and to coat thin steel plate in food cans to stop corrosion of the steel by food. Other important uses include as solder alloys, bearing metals, bronzes, pewter, etc.

TIN PLATE: A flexible sheet, iron or steel, electroplated with a thin protective coat of tin, used to make food cans and pots. The steel provides strength and the tin provides the corrosion resistance, ensuring that the food inside it is not contaminated.

TIN(II) CHLORIDE: A white soluble solid occurring in two forms: anhydrous and dihydrate with the chemical formula SnCl_2 . The density for the anhydrous is 3.95 g/cm³ and that for the dihydrate is 2.71 g/cm³ with a melting point of 247 °C (520 K) for the anhydrous and 37 °C (310 K) for the dihydrate. The boiling point is 623 °C (896 K) at which it decomposes. The compound is made by dissolving metallic tin in hydrochloric acid and is partially hydrolysed in solution. It is a reducing agent, used as a catalyst in organic reactions and as an anti sludge agent for oils.

TIN(II) SULFIDE: A grey-black cubic or monoclinic solid, virtually insoluble in water with the chemical formula SnS , relative density 5.22, melting point 882 °C (1155 K), boiling point 1230 °C (1503 K). It has a layer structure similar to that of black phosphorous. Its heat of formation is low and it can be made by heating the elements together. It reacts with hydrochloric acid to give tin(II) chloride and hydrogen sulfide.

TIN(IV) CHLORIDE: A colourless fuming liquid with the chemical formula SnCl_4 , relative density 2.226, melting point 33°C (306 K) and boiling point 114°C (387 K). It dissolves sulfur, phosphorus, bromine and iodine. It is used to coat glass with tin(IV) oxide to make it conductive, as a mordant and in the preparation of inorganic tin and organotin compounds.

TIN(IV) HYDRIDE (STANNANE; TIN TETRAHYDRIDE TIN HYDRIDE STANNANE): An unstable gas used as a reducing agent in organic chemistry with the chemical formula SnH_4 , density 5.4 g dm⁻³ (gas), melting point -146 °C (127 K) and boiling point -52 °C (221.15 K). It is prepared by the reduction of tin chlorides using lithium tetrahydridoaluminate (III). It decomposes slowly at room temperature to give metallic tin and hydrogen and ignites on contact with air.

TIN(IV) OXIDE (TIN DIOXIDE): A white crystalline solid, with the chemical formula SnO_2 , relative density 6.95, melting point of 1127 °C (1400 K) and sublimates between 1800°C (2073 K) and 1900 °C (2173 K). It is used as a pigment and as refractory material.

TIN(IV) SULFIDE (TIN DISULFIDE): A yellow insoluble solid with the chemical formula SnS_2 , density 4.5 g/cm³ (solid) and a melting point of 600 °C (873 K). It is prepared by the reaction of hydrogen sulfide with soluble tin(IV) salt or by the action of heat

on thiostannic acid. It is used as a pigment. It occurs naturally as the rare mineral berndtite.

TINCAL (BORAX; SODIUM BORATE; SODIUM TETRABORATE; DISODIUM TETRABORATE): A soft, light, colourless crystalline mineral with the chemical formula $(\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O})$, density 1.73 g/cm³ (solid), melting point of the anhydrous is 743 °C (1016 K) and boiling point of 1575 °C (1848 K). It is found as soft, whitish crystals or encrustations on the shores of hot springs and in the dry beds of salt lakes in arid regions, where it occurs with other borates, halite and gypsum. It is used as a component of glass and pottery glazes in the ceramics industry, as a solvent for metal-oxide slags in metallurgy, as a flux in welding and soldering and as a fertiliser additive, a soap supplement, a disinfectant, a mouthwash and a water softener.

TINCTURE: A nonvolatile solution prepared by using alcohol as the solvent, for example, iodine solution.

TINNITUS: In medicine, it is the sensation of constant buzzing or ringing in the ears. The phenomenon may originate from prolonged exposure to noisy conditions such as drilling, machinery, loud music or from damage to or disease of the middle or inner ear.

TINSEL FLAGELLUM: A type of flagellum occurring in members of the phylum Oomycota bearing numerous hairs to increase the surface area and power of beating of the flagellum.

TISSUE: A group of cells, not necessarily identical, but from the same origin, that together carries out a specific function. It is a cellular organisational level intermediate between cells and a complete organ. Organs are formed by the functional grouping together of multiple tissues. The four basic types of tissue are nerve, muscle, epidermal and connective. In animals, for example, the nervous tissue is specialised to perceive and transmit stimuli.

TISSUE CULTURE: The growth of the tissues or cells of living organisms in a suitable culture medium or the process by which cells from a plant or animal are removed from the organism and grown under controlled conditions in a sterile medium containing all the necessary nutrients. They are useful for gaining information about factors that control the growth and differentiation of cells. In plants it is a form of vegetative propagation.

TISSUE ENGINEERING: The bioengineering of living tissues such as skin, bone, cartilage, cornea and spinal tissues using culture medium and growth factors. This became a popular method after the successful tissue engineering of thin sheet of collagen gel that was used clinically for generating artificial skin.

TISSUE FLUID (INTERSTITIAL FLUID): A solution that bathes and surrounds the cells of multicellular animals, consisting of water, ions and dissolved gases and substances that are formed when blood is ultra-filtered from the capillaries into the intercellular space. The tissue or interstitial fluid is found in the interstitial spaces, also known as the tissue spaces, surrounding the body cells. On average, a person has about 11 litres of interstitial fluid, providing the cells of the body with nutrients and a means of waste removal. Plasma and tissue fluids are very similar. Plasma, the major component in blood, communicates freely with interstitial fluid through pores and intercellular clefts in capillary endothelium. To prevent a build-up of tissue fluid surrounding the cells in the tissue, the lymphatic system plays a part in the transport of tissue fluid. Tissue fluid can pass into the surrounding lymph vessels and eventually ends up rejoining the blood. Sometimes the removal of tissue fluid does not function correctly and there is a build-up. This causes swelling and can often be seen around the feet and ankles, for example as in elephantiasis. The position of swelling is due to the effects of gravity.

TISSUE PLASMINOGEN ACTIVATOR (tPA): A naturally occurring substance in the body tissues that activates the enzyme

plasmin that helps to dissolve blood clots. Recombinant tPA (r-tPA), produced in bacteria by genetic engineering, has, like streptokinase, been used to dissolve blood clots in the coronary arteries of heart-attack victims to prevent damage to heart muscle and to reduce neurological damage following ischaemic stroke. It has been shown to be more effective than streptokinase when used in conjunction with heparin.

TISSUE RESPIRATION (INTERNAL RESPIRATION; CELL RESPIRATION): The metabolic process by which living cells absorb oxygen and release carbon dioxide or break down carbohydrates, amino acids and fats to produce energy in the form of Adenosine Triphosphate (ATP) or the exchange of gases between cells and tissue fluid which means diffusion of oxygen from pulmonary alveoli to blood and of carbon dioxide from body cells.

TISSUE TYPING: The process of determining the type of histocompatibility antigens on both the recipient and the potential donor's cells or tissues before organ transplantation.

TITANIUM: A hard or strong lightweight silver-grey, light-weight corrosion-resistant metallic element, one of the transition elements with the symbol Ti, atomic number 22, relative density 4.5, atomic weight 47.87, melts at about 1668 °C (1941 K) and boils at about 3287 °C (3560 K). It is found combined in almost all rocks and soils and in plants and animals, natural waters and deep-sea dredgings. Its chief commercial ores are minerals ilmenite (FeTiO₃), rutile (TiO₂) and sphene (CaO·TiO₂·SiO₂). Titanium burns in oxygen at 610 °C (883 K) to form titanium dioxide and it burns in nitrogen at 800 °C (1073 K) to form titanium nitride (TiN). Alloys containing titanium are used for parts for high-speed aircraft, spacecraft, missiles and ships, in electrodes, in chemical, desalination and food-handling equipment and in prostheses.

TITANIUM(IV) CHLORIDE (TITANIUM TETRACHLORIDE; TITANIC CHLORIDE): A colourless, toxic liquid with the chemical formula TiCl₄, boiling point 136 °C (409 K). It is soluble in water and fumes when exposed to moist air. It is used to make titanium and titanium salts, as a dye mordant and polymerisation catalyst and in smoke screens and pigments.

TITANIUM(IV) OXIDE (TITANIA; TITANIUM DIOXIDE): A white inert solid with the chemical formula TiO₂, density 4.23 g/cm³, melting point 1843 °C (2116 K) and a boiling point of 2972 °C (3245 K). It occurs in three crystalline forms: rutile, brookite and anatase. It is used as a white pigment and as a component in various dielectrics for electrical capacitors, paint, sunscreen and food colouring.

TITIUS-BODES' LAW: The coincidental numerical sequence that gives the distance of the planets of the solar system from the Sun in terms of probability. With the exception of Neptune and Pluto, if 4 is added to each number in the series, 0, 3, 6, 12, 24, and the answers divided by 10, the resultant sequence gives the distances of the planets in astronomical units of the first seven planets. It was first announced in 1766 by the German Johann Daniel Titius (1729-1796). It was made popular by Johann Elert Bode (1747-1826). Bode's law also suggested that a planet should be found between Mars and Jupiter, where the asteroid belt was later discovered.

TITLE BAR: A bar located along the top of a window or a dialogue box that displays the name of the window or software program being used. Besides giving a description of the open program or window, the title bar has several other useful features. Some of the functions of the title bar, which may not be supported in all operating systems include: (i) By clicking and holding down the mouse button on the title bar the window can be moved around when not maximised. (ii) Double-clicking the title bar maximises the window or set the window into window mode allowing it to be moved around your screen. (iii) Windows 7 introduced several new advancements that also allow users to click and drag the title bar to any edge of any of the screens to make a window in that portion of the screen.

TITRANT: The solution in a burette that is being added incrementally to the aliquot.

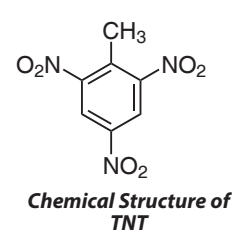
TITRATION: A volumetric analysis used to find the concentration of one compound in a solution by determining how much of it will react with a known amount of another compound in solution; or the process, operation or method of determining the concentration of a substance in solution by adding to it a standard reagent of known concentration in carefully measured amounts until a reaction of definite and known proportion is completed, as shown by a colour change or by electrical measurement and then calculating the unknown concentration. **Titrant** is a substance, such as a solution, of known concentration used in titration. **Titrate** is to measure the concentration of a solution by titration.

TITRATION ERROR: The difference in the amount of titrant or the corresponding difference in the amount of substance being titrated, represented by the expression: end-point value – equivalence-point value.

TITRE: 1. The volume of a solution delivered from a burette at the end point of a titration or the minimum quantity of solution required to complete a reaction in a titration or the concentration of a solution as measured by titration. 2. The number of infectious virus present in a suspension. 3. A measure of the amount of antibody present in a sample of serum or the concentration of an antibody in serum.

TNT (TRINITROTOLUENE; 2,4,6-TRINITROTOLUENE; TRILITE; TOLITE; TRINOL; TROTYL):

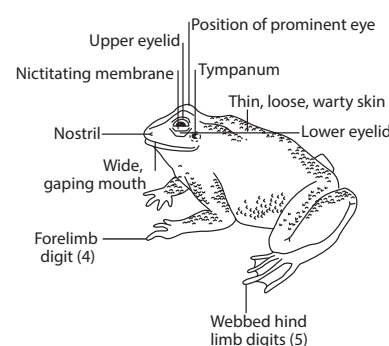
A yellow highly explosive crystalline solid with the chemical formula CH₃C₆H₂(NO₂)₃. It is made by nitrating toluene (methylbenzene). It is an explosive material with convenient handling properties unaffected by ordinary shocks and jarring and must be set off by a detonator. It is sometimes used as a reagent in chemical synthesis and also used to



generate charge transfer salts. The explosive yield of TNT is considered to be the standard measure of strength of bombs and other explosives.

TOAD: A group of froglike amphibians belonging to the order Anura of the class Amphibia that are usually distinguished from frogs by the roughness of the skin, due

to a large number of glandular tubercles and by the shorter hind legs. Toads are usually nocturnal animals, hiding during the day in dark, damp places and hopping about at night in search of insects, grubs, slugs, worms and other invertebrates. They are often brownish or greyish in colour and have warty skin; a flat head; swollen parotid glands on the side of the neck behind the eyes; bright, jewel-like eyes with a transverse pupil; and slightly webbed toes.



Lateral view of toad

TOADSTOOL: The common name for many umbrella-shaped fruiting bodies of fungi or mushrooms of the Agaricaceae family. The term usually refers to those that are inedible or poisonous.

TOBACCO MOSAIC VIRUS (TMV): A rigid rod-shaped RNA-containing virus that causes distortion and blistering of leaves (mosaic disease) in a wide range of plants, especially the tobacco plant. It is transmitted by insects which feed on plant tissue.

TOCOLYTIC DRUG: Drug that prevents premature labour by inhibiting the contractions of the uterus. It is usually used to permit pregnancy to proceed to its natural term to permit the foetus to gain in size and maturity.

TOCOPHEROL (VITAMIN E): One of a group of fat-soluble organic compounds that make up vitamin E. They are present in vegetable oils and leafy greens. Its deficiency leads to several adverse effects on health. An example is sterility in rats and wastage of muscles in animals.

TOEPLITZ MATRIX (DIAGONAL-CONSTANT MATRIX): A matrix in which all the elements are constant along any diagonal that slopes from northwest to southeast or left to right. Given $2n - 1$ numbers a_k , where $k = -n + 1, \dots, -1, 0, 1, \dots, n - 1$, a Toeplitz matrix is a matrix which has constant values along negative-sloping diagonals as shown in the diagram.

$$\begin{bmatrix} a_0 & a_{-1} & a_{-2} & \cdots & a_{-n+1} \\ a_1 & a_0 & a_{-1} & \ddots & \vdots \\ a_2 & a_1 & a_0 & \ddots & a_{-2} \\ \vdots & \ddots & \ddots & \ddots & a_{-1} \\ a_{n-1} & \cdots & a_2 & a_1 & a_0 \end{bmatrix}$$

Toeplitz Matrix

TOG: A unit of measure of thermal insulation used in the textile trade; a light summer suit provides 1.0 tog.

TOGGLE: A term used to describe the process of switching back and forth between settings or between programs. For example, a user may toggle between a software program and a software calculator.

TOGGLE KEY: A term used to describe a computer keyboard key that has one or more functions. The most common toggle key is Caps Lock, which toggles the letter keys between lowercase and uppercase mode. Some keyboards also have other toggle keys, such as Num Lock, Scroll Lock and Insert.

TOKAMAK: A device using a magnetic field to confine plasma in the shape of a torus (doughnut). It was developed in the mid-1960s by Soviet plasma physicists. It consists of a complex system of magnetic fields that confine the plasma of reactive charged particles in a hollow, doughnut-shaped container.

TOKEN: 1. A portion of data that is passed from computer to computer in token-ring networks. When a computer has a token, it is capable of communicating to other computers and devices on the network. It thus avoids collusion (two or more computers trying to communicate at the same time). 2. In programming or batch files a token often refers to a segment or portion of a much larger piece of data. There are five categories of tokens: (i) constants; (ii) identifiers; (iii) operators; (iv) separators; and (v) reserved words. For example, the reserved words "new" and "function" are tokens of the JavaScript language. Operators, such as +, -, * and /, are also tokens of nearly all programming languages. 3. In security systems, a **hard token** is small card that displays an identification code used to log into a network. When the card user enters the correct password, the card displays the current ID needed to log into the network. This adds an extra level of protection to the network because the IDs change every few minutes. Security tokens also come in software versions, called **soft tokens**.

TOKEN RING (802.5): A network in which all computers on the network are connected in a loop. The term token is used to describe a 3 byte frame round the loop; when a computer on the network is able to decode that token, the information is received on that computer. The token ring is used by ARCNET, token bus and FDDI.

TOLERABLE-WINDOWS APPROACH (SAFE-LANDING APPROACHES): These approaches analyse greenhouse gas emissions as they would be constrained by adopting a long-term climate rather than greenhouse gas concentration stabilisation-target (e.g., expressed in terms of temperature or sea level changes or the rate of such changes). The main objective of these approaches is to evaluate the implications of such long-term targets for short- or medium-term "tolerable" ranges of global greenhouse gas emissions.

TOLERANCE: 1. The ability of an organism to withstand extremities of environmental conditions. For example, tilapia can survive in water with very low oxygen concentration but other fishes die in such a condition. Tilapia, therefore, has a high tolerance. 2. The build up to the resistance of drugs or chemicals which occur after prolonged application. 3. The phenomenon by which the cells of the immune system are constrained from mounting an immune response against

itself. 4. In agriculture, it is the ability of a plant to survive and reproduce after application of herbicides or during infestation by plant pathogens. This may either be inherent (natural) or acquired through genetic manipulation.

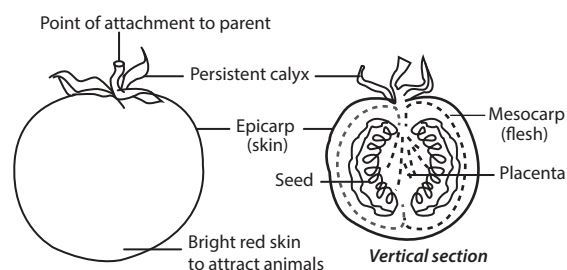
TOLL-LIKE RECEPTOR (TLR): Any of a family of receptor proteins associated with innate immune responses by lymphocytes, macrophages and other cells. They bind certain constituents of invading pathogens such as bacterial lipopolysaccharide or peptidoglycan and activate a signal pathway inside the cell that triggers the release of cytokines and chemokines.

TOLLEN'S REAGENTS: Ammoniacal solution of silver nitrate used as a test for aldehydes and reducing sugars. It is made by adding sodium hydroxide to silver nitrate to give silver(I) oxide, which is dissolved in aqueous ammonia. The sample is warmed with a reagent in a test tube. Aldehydes reduce the complex Ag^+ ion to metallic silver, which deposits as a mirror of silver on the walls of the test-tube, an indication of positive test.

TOLUENE (METHYLBENZENE; PHENYLMETHANE): A colourless liquid aromatic hydrocarbon, derived from benzene by replacement of hydrogen atom by a methyl group. Its chemical formula is $\text{CH}_3\text{C}_6\text{H}_5$, melting point -95°C (178 K) and boiling point 110.8°C (394 K). It is insoluble in water but highly soluble in most organic solvents. It can be obtained from coal tar or made from methylcyclohexane (extracted from crude oil) by catalytic dehydrogenation. Its main uses are as a solvent and as a raw material for producing trinitrotoluene (TNT). It is also used in aviation fuel and other high-octane fuels, in dyestuffs, explosives and as a solvent for gums and lacquers.

TOLUIDINE BLUE: A general purpose alkaline aniline stain commonly used to give an even colour to the specimen and distinguish it from the embedding material. It is frequently used to stain the $1\ \mu\text{m}$ sections cut on ultramicrotomes as an adjunct to transmission electron microscope studies.

TOMATO: Common name for a vine-like herb of the nightshade family *Lycopersicon esculentum*, belonging to the genus *Cycopersicon* and its soft juicy red, yellow or green fruit with bright-red skin and pulpy seedy flesh, eaten cooked or raw as a vegetable. The numerous varieties differ greatly in plant form and fruit type, the latter ranging from a small currant size through cherry, plum and pear forms to the large, nearly round fruits. It is native to South America and grown throughout the world. Technically, tomato is not a vegetable but a fruit; but a ruling in *Nix v. Hedden* by the US Supreme Court in 1893 established that it is a vegetable.



Longitudinal section of tomato fruit

TOMENTOSE: Covered in short, matted or tangled, soft, woolly hairs, such as that of the leaves of hoary mullein (*Verbascum pulverulentum*).

TOMOGRAPHY: Imaging by sections or sectioning, through the use of any kind of penetrating wave (ultrasound, gamma or x rays) of internal structures by focusing on a specific plane within the body to produce a cross-sectional image. It allows the examination of structures that are obscured by overlying organs and soft tissues and do not show up clearly on conventional x-ray images.

TON: The imperial unit of mass, with the symbol *t*. The **long ton**, used in the UK, is 1,016 kg; and the **short ton**, used in the USA, is 907 kg.

TONE: 1. The state of sustained tension in muscles that is necessary for the maintenance of posture or the natural firmness of muscles or of the body generally, when not being flexed. 2. A sound, especially one produced by a machine.

TONG: Any of the various kinds of scissors-like devices consisting of two arms hinged at one end and pressed together at the other end in order to handle an object without touching it. Tongs are designed in many shapes and sizes for specific needs such as at home, in workshops and in laboratories. For example, in a blacksmith's shop, large ones are used to handle hot metals; and various types are also used in science laboratories, for example, beaker tongs are used to handle hot beakers.

TONGUE: The fleshy, movable, muscular organ, attached in most vertebrates to the floor of the mouth, that is the principal organ of taste, an aid in chewing and swallowing and, in humans, an important organ of speech. The upper surface is covered with small projections called papillae, which give it a rough texture. The colour of the tongue, usually pinkish-red but discoloured by various diseases, is an indication of health.

TONNE (METRIC TON): SI unit of mass, which is equal to 1,000 kilograms or approximately the mass of one cubic metre of water at four degrees Celsius. Its symbol is *t*.

TONELLI'S THEOREM: In mathematics, the theorem of measure theory which gives conditions for the order of integration of an iterated integral to be reversed. If (X, Σ, μ) and (Y, T, ν) are sigma-finite measure spaces and F is a non-negative $(\Sigma \times T)$ -measurable function, then $\iint F(x, y) \mu(dx) \nu(dy) = \iint Fd(\mu \times \nu)$.

TONICS: Agents or substances containing properties which tone down the body. Certain herbs can be effective in doing this; such herbs include dandelion, meadowsweet, moringa, turmeric and willow

TONOPLAST (VACUOLE MEMBRANE): The single semi permeable membrane that bounds the vacuole of plant cells and separates the fluid-filled vacuole from the surrounding cytoplasm.

TONSIL: 1. Either of the two small oval masses of tissue, one on either side of the back of the mouth or masses of lymphoid tissue situated at the back of the mouth and throat in higher vertebrates. They are concerned with the production of lymphocytes and, therefore, with defence against infection. 2. A lump of tissue shaped like a tonsil of the mouth, such as either of the two small lumps in the brain tonsils of the cerebellum.

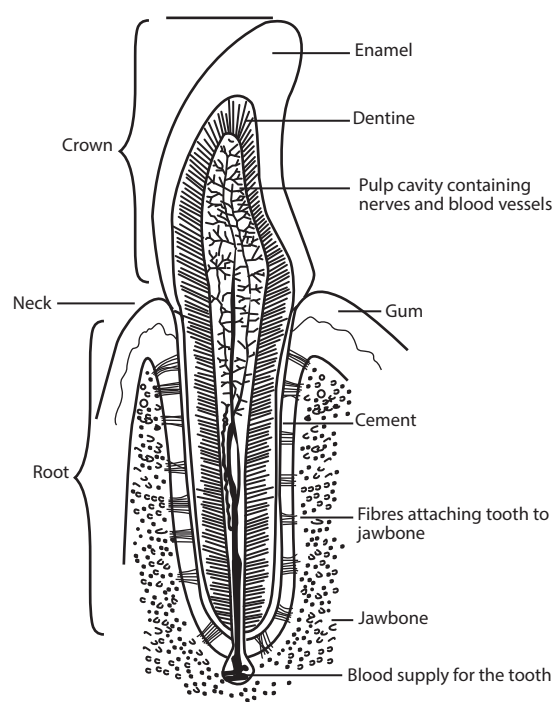
TONSILLITIS: Inflammation of the tonsils due to virus or bacterial infection. Tonsillitis may follow sore throat or other pharyngeal disease or it may be a primary tonsil disease. Sore throat and red swollen tonsils, which may exude pus or cause swallowing difficulty, are common, with lymph nodes at the angle of the jaw are usually tender and swollen.

TOOL: 1. A device or implement, especially one held in the hand, used to carry out a particular function. Examples of **tools** are hammer, scissors, pliers and hoes. 2. In software, tools are accessories that help developers develop software programs more effectively. 3. In hardware, **tools** are what enable a user to install, remove or perform other actions on their computer or their computer hardware.

TOOLBAR: A set of icons or buttons located at the top of a computer screen that is part of a software program's interface or an open window. When it is part of a program's interface, the toolbar typically sits directly under the menu bar. Microsoft Word has a toolbar with icons that allow users to open, save and print documents, as well as change the font, text size and style of the text.

TOOTH: 1. Any of the hard bone-like structures rooted in sockets in the jaws of a vertebrate used basically for biting and chewing food as well as for attacking, grooming, defensive and other functions. Adult humans typically have 32 teeth: 16 in the upper jaw and 16 in

the lower jaw that fit together and work in harmony to chew food. Humans are heterodonts, having teeth of different sizes and shapes that serve different functions, such as tearing and grinding. They include incisors, canines, premolars and molars, each with a specific size, shape and function. Human teeth are made of four distinct types of tissue: enamel, dentin, pulp and cementum. **Enamel**, which is about 0.16 cm thick, forms the outer layer of the tooth above the gum line. It is the hardest substance in the human body and protects the inner layers of the teeth from harmful bacteria and changes in temperature from hot or cold food. **Dentin**, the pulp or core of the tooth is a hard mineral material located beneath the enamel, which surrounds and protects the tooth. Pulp contains blood vessels, which carry oxygen and nutrients to the tooth and nerves. The nerves transmit pain and temperature sensations to the brain. The outer layer of the tooth that lies below the gum line is **cementum**, a bone-like substance that anchors the tooth to the jawbone. 2. A sharp part on an invertebrate such as one of the pointed denticles or ridges on the exoskeleton of an arthropod or the shell of a mollusc made of horny, calcareous or chitinous material and functioning like or resembling a vertebrate tooth. 3. An object with the shape or function of a tooth, for example, one of the jagged indentations along the edge of a saw or leaf or an object with the shape or function of a tooth. 4. A part that sticks out from the edge of a gear wheel or sprocket, designed to interlock with a similar part on another wheel. 5. The roughness of a surface, especially that of paper, that allows paints, glues and other substances to stick to it.



Longitudinal section of an incisor tooth

TOP SOIL: The upper, fertile layer of soil profile composed of rock particles, humus, air, water and living organisms. Plant roots take their nutrients from it.

TOPAZ: A transparent gemstone, which usually crystallises in orthorhombic system with a hardness of 8 on the Mohs scale and a relative density of 3.4 to 3.6. It is found in acidic igneous rocks such as granites. It is predominantly an aluminium silicate mineral with some fluorine in the structure with variable colour. It is a very popular gemstone and most commonly used to make decorative pieces and jewelry.

TOPICAL EFFECT: Consequence of application of a substance to the surface of the body which occurs at the point of application.

TOPOCHEMICAL REACTION (TOPOTACTIC REACTION): A reversible or irreversible reaction that involves the introduction of a guest species into a host structure and that results in significant structural modifications to the host, for example, the breakage of bonds.

TOPOGRAPHY: The study and mapping of the features on the surface of Earth and other observable celestial objects including planets, moons and asteroids. It deals with such issues as relief and contours, distribution of mountains and valleys, patterns of rivers and all other features, natural and artificial producing the landscape. **Topographic map** is a map that shows surface features, or topography of the Earth.

TOPOISOMER: An isomer, typically of large complex molecules that is distinct from other similar isomers by virtue of its topology such as degree of twisting or interlocking of ring structures. The DNA may be distinguished according to the degree of supercoiling of the double helix. DNA topoisomers can be interchanged by enzymes called topoisomerases

TOPOISOMERASE: An enzyme that determines the way in which DNA is packed in living cells by altering the degree of supercoiling of the DNA molecules. They unwind and wind DNA, in order for DNA to control the synthesis of proteins and to facilitate DNA replication. For example, DNA gyrase, a topoisomerase found in *E. coli*, introduces negative supercoiling by breaking both strands of DNA double helix and passing another stretch of the double helix through the gap, which is then sealed.

TOPOLOGICAL DEFECT (TOPOLOGICAL SOLITON): A stable configuration of a system that results from topological distinct solutions to differential equations describing the system. A number of different phenomena are identifiable as topological defects, including screw dislocation in crystals, solitons and skyrmions.

TOPOLOGICAL DIMENSION: An integer that defines the number of coordinates needed to specify a given point of an object or set X . A single point, therefore, a topological dimension equals to zero; a curve has dimension one, a surface has dimension two and so on.

TOPOLOGICAL GROUP: A group, such as the set of all real numbers that constitutes a topological space and in which binary operation and the group's inverse function are continuous functions with respect to the topology. For example, a Hausdorff space with a group operation such that the map $(x, y) \mapsto xy^{-1}$ is continuous, is a topological group.

TOPOLOGICAL INVARIANT (TOPOLOGICAL PROPERTY): A geometrical quantity that remains unchanged by continuous deformations. Charges of magnetic monopoles, analogous quantities in gauge theories and compactness, are examples of topological invariant.

TOPOLOGICAL ISOMORPHISM (CONTINUOUS TRANSFORMATION; HOMEOMORPHISM): A one-to-one transformation between points in two geometric figures or topological spaces that is continuous in both directions. A homeomorphism which also preserves distances is called an **isometry**. Two objects are homeomorphic if they can be deformed into each other by a continuous, invertible mapping.

TOPOLOGICAL SPACE: In mathematics, a set with an associated family of subsets, the open sets, including the whole set and the empty set, that is closed under set union and finite intersection. The family of subsets is usually denoted τ .

TOPOLOGICAL TRANSFORMATION: A transformation that pairs a figure with its image after shrinking, stretching, twisting, bending, or turning inside out without tearing, breaking, and sticking together. Shapes that can be changed into one another by a topological transformation are called **topologically equivalent shapes**. A circle is topologically equivalent to an ellipse and a sphere is equivalent to an ellipsoid. Also, a plate and a bowl are the same topologically equivalent, because the bowl can be flattened into the plate.

TOPOLOGICAL VECTOR SPACE (LINEAR TOPOLOGICAL SPACE): A vector space, such as a normed space that constitutes a topological space over a topological field in which addition and scalar multiplication are continuous functions, where the domains of these functions are endowed with product topologies.

TOPOLOGY: 1. The study of changes in topography that occur over time and, especially, of how such changes taking place in an area affect the history of that area. 2. Also called **algebraic topology**, it is the branch of geometry concerned with the properties of solids that are unchanged by continuous deformation such as twisting or stretching or the study of the properties of geometric figures that are independent of size or shape and are not changed by stretching, bending, knotting or twisting. 3. The family of all open subsets of a mathematical set, including the set itself and the empty set, which is closed under set union and finite intersection and constitute topological space. 4. **Point set topology** is the branch of mathematics that is concerned with the generalisation of the concepts of continuity, limits, etc., to sets other than the real and complex numbers. 5. **Network topologies** are the relationships between parts linked together in a system such as a computer network. The main network topologies are star, bus, ring, hierarchical. 6. The anatomy of a part of the body.

TOPOMERIZATION: The identity reaction leading to exchange of the positions of identical ligands. The indistinguishable molecular entities involved are called **topomers**.

TOPOSEQUENCE: A sequence of related soils with differing profile characteristics caused by topography as a soil-formation factor.

TOPOTACTIC TRANSITION: A transition in which the crystal lattice of the product phase shows one or more crystallographically equivalent orientational relationships to the crystal lattice of the parent phase. An example is transitions in which the anionic array is unchanged during the transition but cation reorganisation occurs.

TOPSOIL (SURFACE SOIL): Also referred to as **A horizon**, it is the upper, outermost layer of soil with a depth of between 7.5 and 25 cm and the layer of soil that is moved in cultivation. It has the highest concentration of organic matter (humus) and microorganisms and is where most of the biological soil activity occurs. Its composition also includes mineral particles, organic matter, water, and air.

TOR: Isolated mass of rock, usually granite, left up standing on a hilltop after the surrounding rock has been broken down.

TORNADO: Extremely violent revolving storm with swirling, funnel-shaped clouds varying in diameter from a few metres to about 2 kilometres that extends to the ground from a cumulonimbus cloud. It is caused by a rising column of warm air propelled by strong wind. Most tornadoes in the Northern Hemisphere create winds that blow anticlockwise around a centre of extremely low atmospheric pressure. In the Southern Hemisphere the winds generally blow clockwise. Peak wind speeds can range from about 120 km/h to almost 500 km/h. The forward motion of a tornado can range from a near standstill to almost 110 km/h. The strongest tornadoes may sweep houses from their foundations, destroy brick buildings, toss cars and buses through the air and even lift railway cars from their tracks. Tornadoes are the most violent of all atmospheric storms.

TORNARIA: The large larva of some acorn worms (Enteropneusta), which is marked by complex bands of cilia.

TOROID: 1. A surface generated by rotating a closed curve for a full turn about a line that is in the same plane as the curve, but does not cross it. The set of all points that the curve crosses in the course of the rotation forms a toroid. 2. A coil of insulated or enameled wire wound on a doughnut-shaped object, such as an O-ring made of powdered iron. It is used as an inductor in electronic circuits, especially at low frequencies where comparatively large inductances are necessary.

TOROSE: Cylindrical with alternate swellings and contractions at intervals. **Torulose** means slightly torose, as occurs in a small fruit which is constricted between the seeds.

TORPOR: In animals, especially in hibernating and aestivating animals, a physiological state that conserves energy typically by reduced metabolism, heart rate, respiration and body temperature

that occurs in varying degrees.

TORQUE (MOMENT; MOMENT OF A FORCE): A measure of the turning effect of a force on an object or a measure of how much a force acting on an object causes that object to rotate. It is the product of a force and its perpendicular distance from a point about which it causes rotation or turning. It can be expressed as: $\tau = r \times F$, where τ is the symbol of torque, r is the perpendicular distance from a point about which it causes rotation or turning, and F is the force that acts on the object to turn. An example is the force generated by an internal-combustion engine to turn a vehicle's drive shaft. The unit of torque is the newton metre (Nm) which is a vector product, unlike the joule which also has newton metre (Nm) as its unit but is a scalar product. A turbine produces a torque on its central rotating shaft.

TORR: A unit of pressure used in high vacuum technology or a unit of pressure, which is equal to about 133.3 pascals or one millimetre of mercury supported in a column.

TORRENT: In computing, it is a file type used by BitTorrent file-sharing protocol. It authorises and points to a remote server that contains the location of different remote hosts with an instance or part of the file to be shared or downloaded. **BitTorrent** is a communication protocol for peer-to-peer file sharing (P2P) which is used for transferring large files, such as digital video files containing TV shows or video clips or digital audio files containing songs over the Internet.

TORRICELLIAN VACUUM: The space or vacuum enclosed above a column of mercury when a tube, closed at one end, is filled with mercury and then inverted in a well of mercury. This space is evacuated except for mercury vapour. The pressure inside the Torricellian vacuum is the vapour pressure of mercury, about 10^{-3} torr.

TORSION: 1. A mechanical strain set up in an object when the object is twisted around its longitudinal axis, typically when one end is fixed; or the state of strain set up in a twisted material or the distortion caused by applying torque in opposite directions to each end of an object or the stress placed on an object that has been twisted. The external twisting forces are resisted by the shear stresses induced in the material of the object. 2. In mathematics, also known as **second curvature**, it is the degree of departure of a curve from its osculating plane. It is the quantity $\tau = -N(s) \cdot B'(s)$, where N and B are respectively the unit principal and binomial vectors to the curve, parametrised by arc length. The quantity $1/\tau$ is the radius of torsion. 3. In zoology, the asymmetrical positioning of the body achieved, through twisting and repositioning of visceral organs in anticlockwise direction through an angle of 180° on the rest of the body during larval development. The phenomenon takes place in the free-swimming larva of gastropods and converts the symmetrical larva into an asymmetrical adult.

TORSION ANGLE (DIHEDRAL ANGLE): 1. An angle formed by the intersection of two planes or the angle between two intersecting planes on a third plane normal to the intersection of the two planes. 2. In chemistry, in a nonlinear chain of atoms A-B-C-D, it is the angle between the plane containing atoms ABC and the plane containing BCD. The torsion angle can have any value from 0° to 180° . If the chain is viewed along the line BC, the torsion angle is positive if the bond AB would have to be rotated in a clockwise sense (less than 180°) to eclipse or align with the bond CD. If the rotation of AB has to be in an anticlockwise sense, the torsion angle would be negative. In proteins, the two torsion angles ϕ and ψ describe the rotation of the polypeptide chain around the two bonds on both sides of the Ca atom.

TORSION BALANCE: 1. An instrument for measuring very weak forces. It consists of a horizontal rod fixed to the end of a vertical wire or fibre or to the centre of a taut horizontal wire. The forces to be measured are applied to the end of the rod. The turning of the rod can be measured by the displacement of a beam of light reflected from a plane mirror attached to it. It can be used to measure the gravitational acceleration at the Earth's surface. 2. Torsion balance may

also refer to a device used in weighing, a type of equal-arm balance.

TORSION ELEMENT: In mathematics, it is an element of an R-module whose product with some non-zero element of the ring is zero. It is equivalent to being nilpotent for the group operation of Z-modules. The set of such elements form the torsion submodule; all other elements are free element. The quotient of a module by its torsion submodule is a torsion free-module.

TORSION MODULE: A module whose elements are all torsion elements.

TORSION SUBMODULE: The set of all the torsion elements of the module.

TORSION-FREE MODULE: In mathematics, it is a module with no non-zero torsion element. It is isomorphic to a submodule of a free module. Torsion-free modules over principal idea domains are free.

TORSIONAL STEREOISOMERS: Stereoisomers that can be interconverted (actually or conceptually) by torsion about a bond axis. This includes E,Z-isomers of alkenes, atropisomers and rotamers.

TORTICOLLIS: A twisting of the neck to one side, causing the head to tilt. It can be a permanent result of a structural condition such as a short neck muscle or temporarily caused by muscle spasm.

TORTOISE: A slow moving four-legged land and fresh water reptile with a hard dome-shaped shell, belonging to the family Testudinidae, closely resembling turtles. The top part of the shell is called **carapace** and the underside is the **plastron** and the two are connected by the bridge. Tortoises range in size from about 10 centimetres to about 1.4 metres long and weigh up to 254 kilogrammes. They are found in many countries, particularly where the weather is warmer for most of the year. It moves at a speed of only 0.2 km/h. Tortoises have a hard outer shell to protect them from predators but the skin on the legs, head and belly of the tortoise is quite soft and can be retracted into the shell to protect itself. Most species of tortoise are herbivorous, eating grasses, weeds, flowers, leafy greens and fruits. They are egg laying, with the female digging nesting burrows to lay between one and thirty eggs at a time, but only a handful of the young tend to survive as they are very vulnerable to attack by all kinds of predators. Tortoises generally have a lifespan similar to the lifespan of humans, although some species of tortoise are known to live over 150 years old. One individual, Harriet, a Galápagos tortoise died at the age of 175 years.

TORTUOSE: In biology, it is an object, which is irregularly twisted or bent. For example, a tortuous aorta is one that is twisted or distorted resulting in narrowing or constriction of the artery, which can cause blood flow to be restricted or blocked. This can lead to health conditions such as high blood pressure; atherosclerosis; or aortic insufficiency (a condition in which the aortic valve of the heart leaks causing blood to flow in the reverse direction during ventricular diastole, from the aorta into the left ventricle).

TORUS (ANCHOR RING): A solid generated by rotating a circle about an external line in its plane, also called an anchor ring. It has the shape of the inner tube of a tyre. If r is the radius of the rotating circle and R the distance between the centre of the circle and the axis of rotation, the volume of the torus is $2\pi^2 R r^2$ and the surface area is $4\pi^2 R r$.

TOTAL CURVATURE: The total curvature of an immersed plane curve is the integral of curvature along a curve taken with respect to arc length. The total curvature of a closed curve is always an integer multiple of 2π , called the index of the curve or turning number – it is the winding number of the unit tangent vector about the origin or equivalently the degree of the map to the unit circle assigning to each point of the curve, the unit velocity vector at that point. This map is similar to the Gauss map for surfaces.

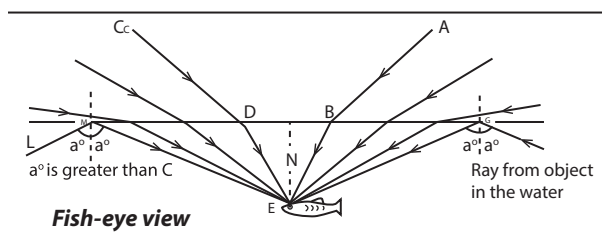
TOTAL DERIVATIVE (TOTAL DIFFERENTIAL): The derivative of a function, f , of several variables with respect to a single parameter in terms of which these variables are expressed. If $z = f(x, y, t)$ with parametric equations $x = u(t)$, $y = v(t)$, then the total derivative

is given by $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$.

TOTAL DIFFERENTIAL EQUATION: A differential equation that contains two or more dependent variables together with their differentials or differential coefficients with respect to a single independent variable which may or may not, enter explicitly into the equation. It is of the form $\sum_{i=1}^n P_i dx_i = 0$, where each P_i is a function of the variables $x_1, x_2, \dots, x_n$. When $n = 2$, the solution of the total equation can be found by means of a solution of the linear differential equation $P_1 + P_2 \frac{dx_2}{dx_1} = 0$. When $n = 3$, this equation is integrable if and only if $\nabla \times V = 0$, where $V = (P_1, P_2, P_3)$.

TOTAL ECLIPSE: A condition in which one celestial body completely obscures or shades another. On Earth, in a total eclipse, the disk of the Sun is fully obscured by the Moon. In partial and annular eclipses only part of the Sun is obscured.

TOTAL INTERNAL REFLECTION: The phenomenon which occurs when light moving from an optically denser medium to a lesser medium strikes the boundary at an angle greater than the critical. If n is the refractive of the medium and c is the critical angle, then $n \sin c = 1$.



TOTAL ION CURRENT: 1. The sum of the separate ion currents carried by the different ions contributing to the spectrum (this is sometimes called the reconstructed ion current) after mass analysis. 2. The sum of all the separate ion currents for ions of the same sign before mass analysis.

TOTAL IONIC EQUATION: Equation for a chemical reaction which relates the predominant form of all species in aqueous solution or in contact with water.

TOTAL MOMENT: The total torque of a system of forces.

TOTAL ORDERING: A relation that orders a set to itself in such a way that every element is related to every other either by the relation or its converse. A relation $\leq$ is a total order on a set S (" $\leq$ totally orders S ") if it has the following properties: (i) reflexivity that $a \leq a$ for all $a \in S$; (ii) antisymmetry ($a \leq b$ and $b \leq a$ implies $a = b$); (iii) transitivity ($a \leq b$ and $b \leq c$ implies $a \leq c$). (iv) totality that for any $a, b \in S$, either $a \leq b$ or $b \leq a$.

TOTAL PROBABILITY: The probability of all possible things happening. It is a law stating that if two events A and B are mutually exclusive then $P(A \cup B) = P(A) + P(B)$. For example, the probability that, a regular six-sided die is rolled, it shows a multiple of 3 is $1/3 = 1/6 + 1/6 = P(\text{shows } 3) + P(\text{shows } 6)$. The generalisation to n mutually exclusive events is the law of total probability: $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$.

TOTAL VARIATION: A measure of the oscillation of the function $h: V_h(a, b) = \sup\{\sum |h(x_{i+1}) - h(x_i)|\}$ over all partition of the interval $[a, b]$, which is finite if and only if the function is of bounded variation on the interval. If the function is decomposed as $f - g$, with f and g monotone increasing, by setting $2f(x) = V_h(a, x) + h(x) - h(a)$, $2g(x) = V_h(a, x) - h(x) + h(a)$, for x between a and b , then the total variation between a and b equals

$f(b) + g(b)$. 2. The measure arising by adding the members of the Jordan decomposition of a signed measure.

TOTALITY: In astronomy, it is the period during a total eclipse of the Sun in which the view of the Sun's surface from a point on the Earth is totally obscured by the Moon. It lasts up to a maximum of 7.67 minutes.

TOTALLY BOUNDED: Of a set in a metric space, having the property that it can be enclosed in the union of a finite number of ϵ -balls round elements of the set; such that for every $\epsilon > 0$ there is a finite epsilon-net, that is, there is a finite set F in the space such that each point of the given set is within epsilon of F . For example, in the reals, the interval $(0, 1)$ is totally bounded, but the rationals are not. A set in a metric space is compact if and only if it is both complete and totally bounded. A metric space is compact if and only if it is complete and totally bounded.

TOTALLY DISCONNECTED: Of a topological space, having the property that every two distinct points can be separated; it has no non-trivial connected subsets. Examples are the rationals and Cantor set. In a totally disconnected space the only connected subsets are the empty set and the one-point sets.

TOTALLY FINITE: Of a measure, finite and such that the space itself is of finite measure.

TOTALLY SIGMA-FINITE: Of a measure, sigma-finite and such that the space itself is of sigma-finite measure. An example of a sigma-finite but not totally sigma-finite measure is provided by counting measure on the ring of countable sets of an uncountable set.

TOTATIVE: An integer less than and relatively prime to a given positive integer n . The number of such totatives is the totient of n . For example, the totatives of 21 are 1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19 and 20.

TOTIENT FUNCTION (EULER PHI FUNCTION; EULER'S TOTIENT FUNCTION): The number theoretic function, written $\phi(n)$, that counts the number of distinct positive integers less than or equal to n that are relatively prime to n . ϕ is multiplicative and so determined by its value on prime powers, for which $\phi(p^{k+1}) = p^k(p-1)$.

TOTIPOTENT: Capable of differentiating into all cell types, as of embryonic stem cells, at a stage before they develop into any or all adult cell types specialised to form, for example, tissues as skin, marrow and muscle, given the appropriate stimulation as compared with a differentiated cell whose fate is determined. For example, the zygote and early embryonic cells are totipotent since they can differentiate into any cell type during development.

TOUCH: The tactile sensation produced by specialised nerve endings in the skin enabling the organism to detect objects or substances in contact with the body. It is the sense that enables the texture of objects and substances to be perceived. Touch receptors are concentrated in the tips of the finger in humans. Touch sensitivity varies in different parts of the body, depending on the number of end organs present in any one area. The tip of the tongue, lips and fingertips are three of the most sensitive areas, the back and parts of the limbs the least so. The sense of touch is very closely related to the other four sensations received by the skin: pain, pressure, heat and cold. There is a specific kind of sensory receptor for each of the five so-called cutaneous senses. For example, light-touch receptors convey only the sensation that an object is in contact with the body, while pressure receptors convey the force or degree, of contact. The blind, for example, learn to read by the Braille system by making use of the sensitivity to touch with the fingertips.

TOUCHPAD (GLIDEPOINT; PRESSURE SENSITIVE TABLET; TRACKPAD): An input device, which allows users to move the mouse cursor without the need of an external mouse. It is found on the majority of portable computers and now also available with some external keyboards.

TOUCHSCREEN: A display screen which enables a computer to react to the touch of a finger and do not require external input devices, such

as mice and keyboards. Some touchscreens require a proprietary pen for input, though most modern touchscreens detect human touch. Touchscreens are commonly used in devices such as Smartphones and Tablet Computers.

TOUR: A Hamiltonian circuit of a graph: the circuit visits every vertex exactly once.

TOURETTE'S SYNDROME: A rare nervous system condition characterised by multiple tics and vocal phenomena such as grunting, snarling and obscene speech. It is named after French physician Georges Gilles de la Tourette (1859-1904).

TOURMALINE: A hard, variously coloured, trigonal or elongated crystalline mineral composed of complex cyclosilicates containing boron. It has a hardness of 7 to 7.5 on Mohs scale and relative density of 2.98 to 3.20. Due to its piezoelectric properties, it is used in polarisers and some pressure gauges. It is also used as a gemstone due to its double refraction.

TOURNAMENT: In mathematics, it is a directed graph (digraph) containing a single edge that connects every pair of vertices. This corresponds to a graph of winners and losers in a round-robin tournament.

TOWBAR: A strong bar which connects the back of a car or tractor, to which another vehicle can be attached to be pulled along.

TOWER SILO: A tall circular tower used for storing forage crops for feeding cattle.

TOWN GAS (COAL GAS; ILLUMINATION GAS): Any manufactured gas supplied as a fuel gas to domestic and industrial users. Examples include coal gas, substitute natural gas and natural gas.

TOXAEMIA: 1. A condition produced by the presence of bacterial toxins in the blood, usually with tissue or organ damage, fever and severe intestinal upset. 2. Also known as **pre-eclampsia** or **preeclampsia**, it is a condition in pregnancy, characterised by abrupt and sharp rise in blood pressure, albuminuria and oedema or swelling of the hands, feet and face.

TOXIC: Poisonous and capable of causing injury or death especially by chemical means. In other words, it is a substance capable of causing death or malfunction of a system when it is ingested, inhaled or absorbed into the body in relatively small amounts.

TOXIC SHOCK SYNDROME: A rare condition caused by toxins produced by *Staphylococcus aureus* (staph) bacteria, by toxins produced by group A streptococcus (strep) bacteria. It is characterised by rapid onset of fever, vomiting and sometimes low blood pressure, which can lead to death. It is seen most often in young women using tampons during menstruation.

TOXIC WASTE (HAZARDOUS WASTE): Waste material that can cause death or injury to living organisms. It is usually the product of industry or commerce, but comes also from residential use, agriculture, the military, medical facilities, radioactive sources and light industry, such as dry cleaning establishments.

TOXICITY: 1. Measurement of how poisonous or toxic a chemical is. 2. The quality or condition of being toxic.

TOXICODYNAMICS: The study of toxic actions on living systems, including the reactions with and binding to cell constituents and the biochemical and physiological consequences of these actions.

TOXICOGENOMICS: The study of the toxic effects of drugs or other substances on patterns of gene expression, in particular cells or tissues. It is thus a synthesis of toxicology and genomics.

TOXICOKINETICS: The study of the process of the uptake of potentially toxic substances by the body, the biotransformation they undergo, the distribution of the substances and their metabolites in the tissues and the elimination of the substances and their metabolites from the body. Both the amounts and the concentrations of the substances and their metabolites are studied. The term has essentially the same meaning as pharmacokinetics, but the latter term should be restricted to the study of pharmaceutical substances.

TOXICOLOGY: The science of the study of poisons. It is the study of the nature, effects and detection of poisons and the treatment of poisoning. A scientist who specialises in the study of poisons is called a **toxicologist**.

TOXIN: A poison produced by a living organism, especially a bacterium or any chemical molecule that can damage the living body or a substance that accumulates in the body and causes it harm.

TOXINOLOGY: Scientific discipline involving the study of the chemistry, biochemistry, pharmacology and toxicology of toxins.

TOXOID: An inactivated bacterial toxin which is strong enough to induce the formation of antibodies and immunity. Toxoids are used for medical purposes such as vaccination or immunisation against certain infections such as diphtheria, tetanus and botulism.

TOXOPLASMOSIS: An infectious disease caused by the one-celled protozoan parasite *Toxoplasma gondii*, which attacks ewes and causes pregnant animals to abort. Although most individuals do not experience any symptoms, the disease can be very serious and even fatal, in individuals with weakened immune systems.

TRABANT (SATELLITE): A spherical body seen attached to one end of a chromosome arm by a narrow filament at mitotic metaphase. Often a pair of trabants is apparent. Usually trabants are only found on one or two chromosomes but occasionally more are found. The number and position of trabants are one of the features of the karyotype. The trabants stalk functions as a nucleolar organiser.

TRABECULA: A bar-like structure extending across a lumen or lacuna. For example, in some plants extensions of the secondary wall thickening form trabeculae across the lumina of xylem tracheids.

TRACE: 1. In mathematics, also called **piercing point**, it is the point at which a line intersects a coordinate plane. 2. The projection of a curve on a given plane, which on the horizontal plane of a standard helix forms a circle. 3. The trace of an algebraic number is the sum of the conjugates of the given number. For example, the trace of $\sqrt{2} + 1$

is $(\sqrt{2} + 1) + (1 - \sqrt{2}) = 2$. 4. The trace of a matrix is the sum of the terms along the principal diagonal. 5. In computer science, a method of checking that a computer programme is functioning correctly by causing the changing values of all the variables involved to be displayed while the program is running. 6. Inherently an immeasurable non-zero amount within the measurement method and scheme being used, for example, visible moisture less than 0.5 mm within a rain gauge read to the nearest millimetre. 7. The line appearing on the face of a cathode-ray tube when the visible dot repeatedly sweeps across the face of the tube as a result of deflections of the electrons.

TRACE ELEMENT: 1. Any element having an average concentration of less than about 100 parts per million atoms (ppma) or less than 110 µg/g. 2. Also known as **micronutrient**, it is any chemical element necessary in minute quantities for the health of a plant or animal or an element that is required in minute amounts for normal growth and development and the functioning of vital enzyme systems. Examples are iodine and manganese. 3. A chemical element present in minute, but detectable amounts in something, such as a metal or ore.

TRACE FOSSIL: Evidence that life existed in the past. Examples of such evidence include footprints, burrows, tracks, eggshells, nests and tooth marks.

TRACEABILITY: 1. The concept that each stage in the supply chain from farm to consumer can be traced so that the quality of food can be guaranteed. 2. It refers to the completeness of the information about every step in a process chain or the ability to chronologically interrelate uniquely identifiable entities in a way that is verifiable.

TRACER: A foreign substance mixed with or attached to a given substance to enable the distribution or location of the latter to be determined subsequently. There are several types of tracers: (i) **physical tracer** is one that is attached by physical means to the object being traced; (ii) **chemical tracer** is a chemical with properties similar to those of the substance being traced with which it is mixed homogeneously; (iii) **isotopic tracer** is a unique isotope, either radioactive or an enriched, uncommon stable isotope, of the element to be traced; (iv) **radioactive tracer** is a physical or chemical tracer

having radioactivity as its distinctive property which allows detection at small concentrations and hence after large transport distances. The composition of aerosols in the troposphere has been used as a qualitative tracer of air masses. The elemental analyses (determined by neutron activation, X-ray fluorescence, etc.) of the aerosols transported from various sources or source regions sometimes have characteristic patterns which are used to define qualitatively the origin of tropospheric aerosols collected in other geographical regions.

TRACEROUTE: A computer network tool that records the route through the Internet between a computer and a specified destination computer both for understanding where problems are in the network and for getting a detailed sense of the Internet itself. It also calculates and displays the amount of time each hop took.

TRACHEA: 1. Also called windpipe, it is a tube in air-breathing vertebrates leading from the larynx and carrying air to the lungs, through the bronchi, bronchioles and alveoli. It is a tube supported by incomplete rings of cartilage that conducts air from the throat to the bronchi. 2. A tubular part of water-conducting plant tissue that provides mechanical support and transport of water and nutrients.

TRACHEAE: A tube in insects and related air-breathing invertebrate animals (terrestrial arthropods especially) through which air is drawn into the body by the pumping action of the abdominal muscles.

TRACHEARY ELEMENT: A water-conducting cell in the xylem. The secondary cell walls of the tracheary elements show various types of thickening. Annular and spiral patterns of thickening allow further extension of the tracheary element and tend to be found in protoxylem. The other forms of thickening are seen in metaxylem. Although bryophytes do not possess xylem, some do possess water-conducting cells known as hydroids.

TRACHEID: A type of elongated cell occurring within the water conducting tissue (xylem) of ferns, conifers and related plants or a cell in the trachea of conifers and other gymnosperm plants, with bands of lignin thickening the cell walls and adding structural support. Water flows from one tracheid to another through regions or pits in the cell walls, which are not thickened.

TRACHEOLES: A fine channel that branches off from an insect's trachea and carries oxygen to its tissues.

TRACHEOPHYTE: Traditional classification of plants with vascular system, conspicuous tissue system and waterproof cuticles; now reclassified into Filicinophyta, Coniferophyta and Angiospermophyta.

TRACHEOSTOMY (TRACHEOTOMY): A surgically made hole through the front of the neck and into the trachea (windpipe) to help with breathing. It is usually temporary, although some are long term or even permanent.

TRACHOMA: An infectious disease caused by the *Chlamydia trachomatis* bacteria. It causes acute or chronic conjunctivitis or corneal inflammation with secondary blood-vessel extension over the cornea resulting in loss of translucency. Eyelid deformity with secondary corneal damage is also common. It is transmitted by direct or indirect contact with eye, nose or throat secretions of an individual with trachoma or indirectly via flies or other insects carrying those secretions on their legs or bodies. It is the leading infectious cause of blindness in the world, but totally preventable and curable.

TRACING: 1. A copy of something made by putting a sheet of translucent paper called **tracing paper** on top of it and drawing the visible outlines on this paper. 2. A graphic record made by an instrument such as a seismograph. 3. Labelling a substance, usually a radioactive/stable isotope or dye that can be traced to identify a compound as it undergoes a chemical reaction or assimilation.

TRACK: 1. Part of the magnetic structure created on a disk surface during disk formatting so that data can be stored on it. 2. The path of an ionising particle as revealed by a track detector.

TRACK DETECTOR: A nuclear detector which makes the paths of ionising particles visible, either directly such as cloud chamber or after suitable treatment (photographic emulsion, polymers).

TRACK LAYER: A heavy-duty caterpillar tractor, used mainly for earthmoving and drainage work.

TRACKBALL: An input device used to enter motion data into computers or other electronic devices. It serves the same purpose as a mouse, but is designed with a moveable ball on the top, which can be rolled in any direction. A trackball requires less arm and wrist motion than a regular mouse takes and therefore, is often less stressful for the user to use.

TRACKER: The part of a high-energy gamma-ray detector that is used to determine the trajectory of the incoming gamma-ray. For a silicon-strip detector-based tracker, the trajectories of electron-positron pairs are recorded. These pairs are produced by the converter.

TRACKING: In computer science, it is synchronising the movement of an on-screen pointer or cursor with that of an input device such as a mouse.

TRACKPOINT (POINTING STICK): A small nob located in the middle of some keyboards. It is a controlling device that works like joystick. It is located between the "G", "H" and "B" keys.

TRACTOR: A powerful self-powered vehicle commonly having large rear-wheels or caterpillar tracks used for pulling farm machinery such as ploughs for tilling the land and loads. It has a hydraulic lift for raising and lowering implements. They are widely used in agriculture, building construction and road construction and for specialised service in industrial plants. Examples of farm implements attached to the tractor are disc plough, mouldboard plough, disc ridge and spring tide harrow as shown in the diagrams.

TRACTRIX: In mathematics, it is an involute of a catenary described by a point initially on the vertex. Its equation in standard form is $x = \operatorname{arccosh}(a/y) \pm \sqrt{(a^2 - y^2)}$.

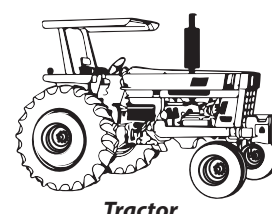
TRADE WINDS: Persistent warm moist winds that blow westward from the high-pressure zones at about latitude 30°N and S toward the doldrums (inter-tropical convergence zone) at the equator. They are thus northeasterlies in the Northern Hemisphere and southeasterlies in the Southern Hemisphere. They are stronger and displaced toward the equator in winter.

TRAFFIC (NETWORK TRAFFIC; DATA TRAFFIC): In computer science this is the load of data or instructions carried by a network or some section of a network. Light traffic speeds up the response time of the instruction or data transfer. A longer response time shows that the device is experiencing much load. The Internet is able to report on the response time on the whole internet by looking at the various points on the internet.

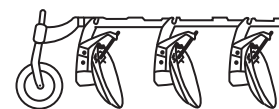
TRAFFICABILITY: The ability of soil to take machinery or stock or all types of traffic without significant soil damage. It is related to the soil water content.

TRAIL (EULERIAN WALK): A walk (an alternating sequence of edges and vertices) in a graph in which all edges are distinct.

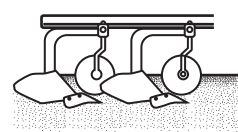
TRAILED IMPLEMENTS: Implements such as harrows which are pulled by a tractor.



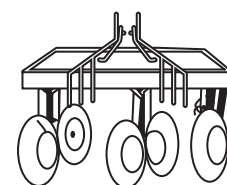
Tractor



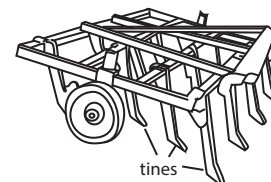
(i) Disc plough



(ii) mouldboard plough



(iii) Disc ridge



(iv) Spring tine harrow

Examples of farm implements attached to a tractor

TRAILER: 1. In telecommunications, it is the ending portion of a data transmission that commonly contains error-correction information that helps verify the information was sent properly. 2. A short movie clip, often used to show a short demonstration or introduction for computer games. 3. A machine used for transporting things and usually pulled by another vehicle.

TRAILING PLANTS: Any plant whose shoots consists of a stem and leaves lie on the ground. Examples are ivy, sweet potato vine, creeping jenny, ivy leaved geranium and trailing fuchsia.

TRAIT: 1. One of the alternative forms of character, for example, tall, short, red, blue and anger and may be inherited or acquired. 2. A genetically determined characteristic or condition. Examples are blue eyes or being taller than average. A phenotypic trait is an obvious and observable trait, which is the expression of genes in an observable way. An example of a phenotypic trait is hair colour.

TRAJECTORY: 1. The path that a projectile makes through space under the action of given forces such as thrust, wind and gravity. 2. A surface or curve that intersects all of a family of surfaces or curves at a constant angle. 3. In mathematics, also called **orbit**, it is the curve $\{y(t) : t \in I\}$, where y is the solution of a system of ordinary differential equation and I is the interval of existence of y ; this is a curve in phase space.

TRAMA: The central tissue in the gill of a fungus, consisting of loosely packed hyphae.

TRAMLINE: A path left clear for the wheels of tractors to drive over. They are used as guide marks for spraying and when applying fertiliser so that damage to crops is kept to a minimum.

TRANCE: A mental state characterised by half-consciousness, loss of perceptions of time and space and loss of response to external stimuli. An example is the state of hypnosis.

TRANQUILLISER (TRANQUILLIZER): Any drug such as benzodiazepines, barbiturates, anti-depressants and beta-blockers used for reducing anxiety or tension. Most tranquilisers can be potentially addictive.

TRANS: A prefix designating certain groups of atoms on opposite sides of the longitudinal axis of a double bond or of the plane of a ring in a molecule.

TRANS ARRANGEMENT (REPULSION): The situation in which an individual is heterozygous for two linked genes (either two units of function or two mutations in the same functional unit) and the recessive (or mutant) allele of one gene is on the same chromosome as the dominant (or normal) allele of the other gene. The homologous chromosome will thus possess the dominant allele of the first gene and the recessive allele of the second gene. If the recessive alleles are in the same functional unit then their effects will show in the phenotype because there is no unchanged functional unit to mask them. If they are in different functional units, then the phenotype will be normal.

TRANS EFFECT: An effect in the substitution of inorganic square-planar complexes, in which certain ligands in the original complex are able to direct the incoming ligand into the trans position. In other words, it is the effect of the ligands on the rate of substitution of the group trans to itself. The proportion of each product depends on the trans ligand. A typical situation occurs in the synthesis of cisplatin.

TRANSACTINIDE ELEMENTS: Elements above lawrencium in the periodic table with an atomic number greater than 103. Transactinides are radioactive and have only been obtained synthetically in laboratories.

TRANSACTION FILE: A collection of transaction records used in updating the master files system.

TRANSACTION PROCESSING: A system that trades day-to-day transactions such as inventory production output orders and sales and creates a database for them.

TRANSAMINASE: An enzyme that catalyses the transfer of an amino group from an α -amino acid to an α -ketoacid, especially α -ketoglutaric acid. It plays an important role in the synthesis of amino acid.

TRANSAMINATION: The formation of one amino acid from another by a biological reaction in amino acid metabolism in which an amine group is transferred from an amino acid to a keto acid to form a new

amino acid; or the transfer of an amino group from one molecule to another without the intermediate formation of ammonia. The coenzyme required for this reaction is pyridoxal phosphate.

TRANSANNULAR STRAIN: In medium-sized ring compounds, the strain due to repulsive nonbonded interactions between substituents or hydrogen atoms attached to non-adjacent ring atoms.

TRANSCIEVER: A hardware device that is used to transmit and receive information. For example, a user using a cordless Bluetooth keyboard or mouse may have a transceiver to send and receive information from the wireless device.

TRANSCellular PATHWAY: The route through cells. Substances moving across epithelia are mostly forced to pass through the epithelial cells rather than passing through the cells. This means that both the mucosal surface and the serosal surface of the cells can act as gate keepers, keeping out certain substances, while transporting others from one side to another often against a concentration gradient.

TRANSCENDENTAL FUNCTION: A function that cannot be expressed algebraically. For example, $\sin x$ cannot be expressed in algebraic terms and hence, if $f(x) = \sin x$, $f(x)$ is a transcendental function.

TRANSCENDENTAL NUMBER: Irrational number that is not the root of a polynomial equation or a number that is not algebraic or that cannot be expressed as an integer. An example is a non-repeating decimal such as π . A transcendental number is also nonalgebraic. Examples are $\log x$, a^x and $\sin x$.

TRANSCRIPTION: The process by which the information for the synthesis of a protein in living cells is transferred from the DNA strand on which it is carried to the messenger RNA strand involved in the actual synthesis.

TRANSCRIPTION FACTOR: A collection of proteins that can control gene activity by increasing or decreasing the binding of RNA polymerases to the DNA molecules during transcription. This is achieved by the ability of the transcription factors to bind to the DNA molecules. Transcription factors contain finger domains which are often in repeated sequence called multi finger loops. General transcription is active in transportation of many genes.

TRANSCRIPTOME: The full unique sequenced RNA molecules with coded genetic information transcripts that an individual produces. The type and the relative abundance of different transcripts, i.e. the messenger RNA's can be obtained by analysing cell content using complementary DNA probe particularly in the short form of DNA microarrays. Such analysis provides a good idea about the expression pattern of the cell's gene.

TRANSCRIPTOMIC: The study of the RNA transcripts of a cell, tissue or organism, with emphasis on determining how the transcriptome and hence pattern of gene expression, changes with respect to various factors such as type of tissue, stage of development and hormones.

TRANSDETERMINATION: A switch in lineage commitment in a stem or progenitor cell to a closely related cell type.

TRANSDIFFERENTIATION: The conversion of one cell type to another. This occurs when a differentiated cell switches to another differentiated lineage without an intermediate stage of dedifferentiation.

TRANSDUCER: 1. A device that converts one form of energy into another, especially in electrical or electronic circuits. Examples are microphones, photoelectric cells and automobile horns. 2. A biological entity that converts energy in one form to another, for example, the rods and cones of the eye or the hair cells of the ear.

TRANSDUCIN: A G-protein that acts as a second messenger in the response to light by vertebrate photoreceptors in the retina. When a photon of light strikes a molecule of rhodopsin (the visual pigment in the rod cells of the retina of the eye), it activates transducin. This causes a chain of events within the rod cell that amplifies the signal and evokes the response.

TRANSDUCTION: 1. The transfer of genetic materials from one bacterium cell to another by means of a transducing bacteriophage. When the phage is grown on the first host, a fragment of the host DNA can be incorporated into the phage particles. This foreign DNA can be transferred to the second host upon infection with progeny phage from the first experiment. 2. The conversion of stimuli detected in the receptor cells into electric impulse which are transported by the nervous system, as it occurs, when the ear converts sound waves into nerve impulses. 3. Any mechanism by which binding of a signal molecule to a cell-surface receptor triggers a response inside the cell. This often involves second messengers.

TRANSECT: 1. A straight line across an expanse of ground along which ecological measurements are taken continuously or at regular intervals. 2. To divide something by running or cutting across it.

TRANSFECTION: The procedure by which foreign DNA introduced during genetic engineering is picked up and replicated into daughter cells.

TRANSFER CELL: A type of plant cell associated with the vascular tissue specialised for the short-distance high-volume transport of photosynthates or sugars. The cells are characterised by the elaboration of an unligified, secondary cell wall to form finger-like projections or wall ingrowths which protrude into the cytoplasm of the cell. These ingrowths are enveloped by plasma membrane, forming a wall-membrane apparatus which increases the surface-volume ratio of the cell.

TRANSFER COEFFICIENT: A quantity used in the analysis of processes in electrochemistry. It is the fraction of the potential difference at the surface of an electrode that assists charge difference in one direction but discourages it in the other direction. Its value lies between 0 and 1, frequently being about $\frac{1}{2}$. Its symbol is α .

TRANSFER ELLIPSE: The path followed by a body moving from one elliptical orbit to another.

TRANSFER LAYER: A layer in the OSI model responsible for managing end-to-end or host-to-host communication sessions. An example is Transmission Control Protocol (TCP).

TRANSFER LENS (RELAY LENS): A lens used to transfer a light beam to another place within an optical system without changing the image. In a Maksutov telescope, for example, a transfer lens is used to make the focal point accessible by bringing it outside the instrument.

TRANSFER LINE: A line provided to carry the sample to be analysed from the sample point to the analytical unit without altering the composition of the sample.

TRANSFER RATE: In computer science, it is the speed at which information is moved between one or more locations. It is commonly measured in bits per second (bps) or bytes per second (Bps).

TRANSFER RNA (tRNA): Single-stranded RNA molecules that bind to amino acids and deliver them to the proper codon on messenger RNA or one of a class of RNA molecules that transport amino acids to ribosomes for incorporation into a polypeptide undergoing synthesis.

TRANSFERABILITY: In chemistry, it is the assumption that a chemical property that is associated with an atom or a functional group in a molecule will have a similar (but not identical) value in a variety of different circumstances. Examples of transferable properties include electronegativity, nucleophilicity, NMRchemical shift, etc.

TRANSFERASE: Enzymes that catalyse reactions in which a group is transferred from one compound to another. Groups that are transferred are C, aldehydic or ketonic residues, acyl, glycosyl, alkyl, nitrogenous, phosphorus and sulfur containing groups.

TRANSFERRIN: A beta globulin protein that stimulates the transport of iron to the liver for storage or to the bone marrow for synthesis of haemoglobin by binding to the protein.

TRANSFINITE: Having cardinality that is or indexed by a sequence whose ordinal number is, a transfinite number.

TRANSFINITE INDUCTION: An extension of mathematical induction to well-ordered sets, for example, to sets of ordinal numbers or cardinal numbers that is equivalent to the axiom of choice and that has the form of complete induction in that one proves that whenever some property $P(\alpha)$ holds for all ordinals $\alpha < \beta$ then $P(\beta)$ holds; one may then conclude that $P(\alpha)$ holds for all α .

TRANSFINITE NUMBER: A cardinal or ordinal number used in the comparison of infinite sets. The transfinite sequence of cardinal numbers start with the smallest, $\aleph_0$ (Aleph-null), followed by the ω (omega). The set of rationals and the set of reals have different transfinite cardinality.

TRANSFORM PLATE BOUNDARY: Region where two tectonic plates slide sideways past each other. As these plates move past one another, the two plates interact and can create cracks or faults within the surrounding area and may cause earthquakes.

TRANSFORMATION: 1. The mathematical conversion of an equation, expression or function into another equivalent entity, for example, by the substitution of one set of variables with another. 2. A process that changes the position of a shape that include **rigid transformation**, which does not change the shape or size of the figure; and **non-rigid transformation**, which changes the size but not the shape of the figure. There are four types of transformations: reflection, rotation, translation and enlargement. 3. The change of one type of atom to another, resulting from a nuclear reaction. 4. A phenomenon in which an external genetic material of a bacterium is used to cause a permanent change in an organism, which is applied in genetic engineering of organisms. This process can be used to develop biological weapons by converting avirulent strains into virulent forms or transforming normal cells into cancer inducing cells (malignant cells). 5. The conversion of normal cells into malignant cells which can be brought about by the action of carcinogen or oncogenic viruses.

TRANSFORMATION GROUP: A group of transformation on a set with composition as the binary operation.

TRANSFORMATION HYPOTHESIS: A hypothetical explanation of how the sporophyte came to acquire a more prominent form in the life cycle of most vascular plants. It postulates that early vascular plants had gametophyte and sporophyte generations that were very similar in appearance, with prominent upright branching forms. Over time, the gametophytes became smaller and simpler, whereas the sporophyte got progressively more elaborate in form. Eventually, the gametophytes became retained within the sporophyte, the situation in modern seed plants. Hence, the modern sporophyte evolved through the transformation of an existing form.

TRANSMUTATION (INDUCED TRANSMUTATION): A nuclear reaction caused by the bombardment of a nucleus with subatomic particles or a process by which nuclei of an element can be changed to form nuclei of a different element often by the bombardment with neutrons.

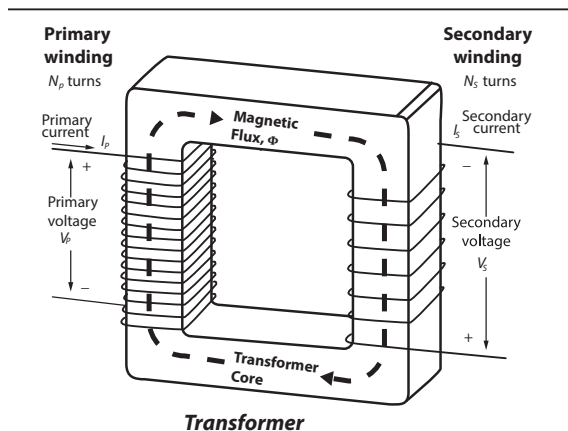
TRANSFORMATION OF THE PLANE: An invertible mapping of the plane to itself.

TRANSFORMATION PROBLEM (CONJUGACY PROBLEM): The problem of finding an algorithm that decides whether the elements represented by two words x and y in G are conjugate elements of G . There exists a finitely presented semigroup for which this problem is insoluble. The problem is to determine whether there exists an element z of G such that $y = xzx^{-1}$.

TRANSFORMATION RULES: In logic, it is the set of rules that specify in purely syntactic terms the methods by which theorems may be derived from the axioms of a formal system; its rules of inference.

TRANSFORMER: A device for transferring electrical energy from one alternating current circuit to another with an increase or decrease change in voltage, current, phase or impedance but without a change of frequency. This is done by applying the principle of magnetic induction between coils to convert voltage and/or current levels. An electrical transformer normally consists of a ferromagnetic core and

two or more coils called windings. One coil, called the primary, is connected to a source of alternating current that produces a constantly varying magnetic field around the coil. The varying magnetic field, in turn, produces an alternating current in the other coil, called the secondary, which is connected to the load. A changing current in the primary winding creates an alternating magnetic field in the core. In turn, the core multiplies this field and couples the most of the flux through the secondary transformer windings. This in turn induces alternating voltage (or emf) in each of the secondary coils. The ratio of the number of turns in the primary coil to the number of turns in the secondary coil, called the turns ratio, determines the ratio of the voltages in the two coils. For example, if there is one turn in the primary and ten turns in the secondary coil, the voltage in the secondary coil will be 10 times that in the primary. Such a transformer is called a **step-up transformer**. If there are ten turns in the primary coil and one turn in the secondary, the voltage in the secondary will be one-tenth that in the primary. This kind of transformer is called a **step-down transformer**. The ratio of the electric current strength or amperage, in the two coils is in inverse proportion to the ratio of the voltages, therefore, the electrical power is the same in both coils. The impedance or resistance to the flow of an alternating current of the primary coil depends on the impedance of the secondary circuit and the turns ratio. With the proper turns ratio, the transformer can, in effect, match the impedances of the two circuits. This is important in some electronic systems because they permit the maximum amount of electric power to be delivered from one component to another, creating efficient performance. In an autotransformer, there is only one coil and both circuits are connected to it. They are connected at different points, so that one circuit contains a larger portion of the coil (that is, has more turns) than the other. Transformers are important in the distribution of electric power. They raise the voltage of the electricity generated at a power plant to the high levels such as 155,000 to 765,000 V needed to transmit the electricity efficiently. Other transformers reduce the voltage substations and where the electricity is used, for example, to 220/240 V for domestic or commercial use. Many household devices, for example, televisions contain transformers to raise or lower house-current voltage as needed.



TRANSFORMING FACTOR: The name given by Frederick Griffith (1879–1941), a British bacteriologist to the unknown material leading to transformation, which was later found to be DNA. He showed that *Streptococcus pneumoniae*, implicated in many cases of lobar pneumonia, could transform from one strain into a different strain. The observation was attributed to an unidentified transforming principle or transforming factor. This was later identified as DNA.

TRANSFUSION: Intravenous delivery of blood or blood products into a patient's circulation to make for deficiency due to disease, injury or surgical intervention.

TRANSFUSION TISSUE: The tissue immediately surrounding at least part of the vascular bundles in the leaves of gymnosperms. Transfusion

tissue is composed of tracheids (transfusion tracheids) with conspicuous bordered pits and parenchyma cells (transfusion parenchyma) containing tannin-like substances and sometimes starch. The main function of the transfusion tissue is believed to be the transport of materials between the vascular bundles and the mesophyll.

TRANSGENE: A gene that is taken from one organism and is inserted into the gene line of another organism so that it is replicated as part of the genome and present in all the recipient's cells. The resultant organism is transgenic.

TRANSGENESIS: The process of transferring cloned genetic material from one organism to another.

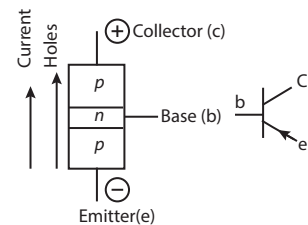
TRANSGENIC: An animal or plant that contains genes from a different species or an organism whose genomes incorporate and express from another species, created by genetic engineering, using suitable vectors to insert the desired foreign genes into the fertilised egg or early embryo of the host. For example, the gene for rat growth hormone can be inserted into a fertilised mouse egg to produce mice with cells that produce rat growth hormone.

TRANSIENT: 1. In electronics, it is the brief malfunction or oscillation in a circuit that lasts a short duration, caused by a sudden rise in the current or electromotive force (emf). 2. **Transient memory** is memory that temporarily holds data while it is being used and when the computer is on. 3. In astronomy, it is a source such as a gamma-ray burst that appears temporarily and then disappears.

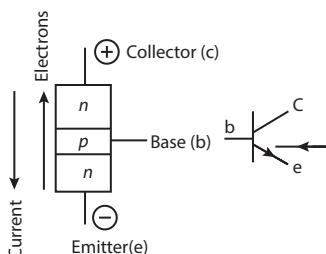
TRANSIENT CLIMATE RESPONSE (TCR): The globally averaged surface air temperature increase, averaged over a 20-year period, centred at the time of CO₂ doubling, i.e., at year 70 in a 1% per year carbon dioxide increase experiment with a global coupled climate model. It is a measure of the strength and rapidity of the surface temperature response to greenhouse gas forcing.

TRANSIENT SPECIES (CHEMICAL SPECIES): A short-lived intermediate in a chemical reaction. It can be defined only in relation to a time scale fixed by the experimental conditions and the limitations of the technique employed in the detection of the intermediate. The term is a relative one. Transient species are sometimes also said to be 'metastable'.

TRANSISTOR: An electronic component made of semiconductor materials with three or more electrodes that can regulate a current passing through it or a semiconductor device, capable of amplification in addition to rectification. It can be used as a switch or an amplifier. It is the basic unit in radio, television and computer circuits. One type of transistor is the Bipolar Junction Transistor (BJT). It is a semiconductor device with three doped semiconductor regions, which are the base, collector and emitter, separated by two p-n junctions. The p-n junction between the base and the emitter has a barrier voltage (V_0) of about 0.6 V, which is an important parameter of a BJT. Unlike the Field Effect Transistor (FET), in which current is produced only by one type of charge carrier (electrons or holes), in BJT, current is produced by both types of charge carriers (electrons and holes), hence the name bipolar. There are two types of BJT: npn and pnp. In an NPN transistor, a thin and



(fig. i) A p-n-p bipolar junction transistor



(fig. ii) An n-p-n bipolar junction transistor

Examples of Transistor

are two types of BJT: npn and pnp. In an NPN transistor, a thin and

lightly doped P-type material is sandwiched between two thicker N-type materials; while in a PNP transistor, a thin and lightly doped N-type material is sandwiched between two thicker P-type materials. The main current that is controlled is called the collector current and goes from collector to emitter or from emitter to collector, for the PNP or NPN transistor, respectively. The small controlling current called the base current that controls the main current, called the collector current goes from base to emitter or from emitter to base, for the PNP or NPN transistor, respectively. The base current is the only current that goes through the base wire of the transistor; and the collector current is the only current that goes through the collector wire. It is the amount of current flowing in the base circuit that controls the amount of current flowing in the collector circuit. The larger collector current is, therefore, proportional to the base current according to the relationship or more precisely it is proportional to the base-emitter voltage. This results in current amplification.

TRANSITION: 1. A change of a system from one quantum state to another. 2. In genetics, it is a substitution mutation in which a purine is replaced by another purine, for example, $A \rightarrow G$ or a pyrimidine with another pyrimidine such as $T \rightarrow C$. 3. The stage of childbirth that occurs before the expulsive phase of labour, characterised by strong uterine contractions and nearly full cervical dilation.

TRANSITION ELEMENT (TRANSITION METAL): A metallic element that has a partially filled d-orbital in the neutral state or one of its ionic states of an element that forms at least one ion with a partially filled d sub-shell or any chemical element with valence electrons in two shells instead of only one. This structure gives metallic elements their outstanding ability to form ions containing more than one atom (complex ions or coordination compounds), with a central atom or ion (often of a transition metal) surrounded by ligands in a regular arrangement. Examples of such metallic elements include manganese, iron and gold.

TRANSITION INTERVAL: The range of concentration over which the eye is able to perceive change in hue, colour intensity, fluorescence or other property of a visual indicator arising from the varying ratio of the two relevant forms of the indicator.

TRANSITION MATRIX: A matrix used to describe the transition probabilities of moving from one state to another in a Markov chain. Each of its entries is a nonnegative real number representing a probability.

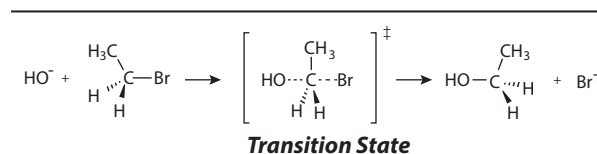
TRANSITION POINT: 1. Also known as **transition temperature**, it is the temperature at which one crystalline form of a substance changes from one state of aggregation to another, called a **first-order transition**; or the temperature of culmination of a gradual change, such as the lambda point or Curie point (a second-order transition). 2. A point at which the constants of a circuit change in such a way as to cause reflection of a wave being propagated along the circuit.

TRANSITION REGION: 1. A thin and very irregular layer of the Sun's atmosphere, a few hundred kilometres deep that lies between the chromosphere and the corona. Within, the temperature climbs steeply from about 20,000 K in the chromosphere to over 1,000,000 K at the base of the corona. 2. The part of the plant body between the stem and the root in which there are intermediate arrangements of the tissues. Vascular transition takes place in this region, ensuring continuity of vascular tissues throughout the plant. 3. An intercalary meristem located between the blade-like portion and the narrow stem-like portion of the plant body in the Laminariales (an order of brown algae).

TRANSITION SPECIES: Any intermediate species in an elementary reaction. An activated complex is an example of a transition species.

TRANSITION STATE: Transition state of a chemical reaction is a particular configuration defined on the state corresponding to the highest energy along the reaction coordinate, which is most likely to go on to form products; or the association of atoms of the highest

energy formed during a chemical reaction. It can be considered as a short-lived intermediate that breaks down to give products. In the substitution of bromine in bromoethane with a hydroxyl group via SN_2 mechanism, the transition state occurs prior to the formation of the product is shown in the diagram below:



TRANSITIVE: 1. Of a relation, having the property that if applied from A to B and from B to C, also applies from A to C; or relating to a relationship between three elements such that if the relationship holds between the first and second elements and between the second and third elements, it necessarily holds between the first and third elements. Examples of transitive relationships are equality of numbers and divisibility of integers. 2. Of a graph, having the property that the adjacency relation on the vertices is transitive. In other words, an **edge-transitive graph** is a graph such that any two edges are equivalent under some element of its automorphism group. 3. Of an action of a group on a non-empty set, having the property that the set has precisely one orbit.

TRANSITIVE CLOSURE: In mathematics, it is a relation R that is transitive on a given set X and contains a given relation on the set. In other words, a relation R on a set X is transitive if, for all x, y, z in X, whenever xRy and yRz then xRz . It can be denoted as $x < y$ and $y < z$ then $x < z$. Examples of transitive relations include the equality relation on any set and the "less than or equal" relation on any linearly ordered set. Set A is transitive, if and only if whenever $x \in A$ and $y \in x$, then $y \in A$ or, equivalently, whenever $x \in A$ and x is not an urelement, then x is a subset of A. Similarly, a class M is transitive if every element of M is a subset of M.

TRANSITIVE SET: In mathematics, it is a set of which every member is also a subset. In other words, a set A is transitive, if and only if whenever $x \in A$ and $y \in x$, $y \in A$ or, equivalently, whenever $x \in A$ and x is not an urelement, then x is a subset of A. Similarly, a class M is transitive if every element of M is a subset of M.

TRANSITIVITY PROPERTY: The transitivity property of equality states that, if $a = b$ and $b = c$, then $a = c$. All real and complex numbers obey this property. The transitive property of inequality states that, if $a > b$ and $b > c$, then $a > c$. Real numbers obey this property, but complex numbers do not.

TRANSLATE: 1. In Euclidean geometry, to move a figure or body laterally without rotation, dilation or angular displacement. 2. The image of some figure or body under a translation. 3. In topological groups, the function derived from a given function f on a group G when the argument is multiplied (on the left or right) by a given element a of G, that is the left translate of f by a , denoted L and the right translate, $R_a f$ or f_a are respectively defined by $L_a f(x) = f(ax)$ and $R_a f(x) = f(xa)$ for all x in G.

TRANSLATION: 1. The process in living cells in which the genetic information encoded in messenger RNA in the form of a sequence of base triplets is translated into a sequence of amino acids in a polypeptide chain during protein synthesis or the process by which information in messenger RNA directs the sequence of amino acids assembled by a ribosome during protein synthesis. 2. The motion of a body in which all the points in the body follow parallel paths or the movement of a body in a straight line so that every point on the body follows a parallel path and no rotation takes place. 3. Movement by the same amount and in the same direction, by every point in a line or in a plane figure. It is a geometrical way of representing a vector and referred to isometric transformation.

TRANSLATIONAL EQUILIBRIUM: An object is in translational equilibrium if the net force acting on it is zero: sum of all forces = 0. For example, when an object has zero overall acceleration, it is either not moving or moving at a constant velocity.

TRANSLATOR: 1. In botany, it refers to the structure that connects between the pollinia of adjacent anthers in the Asclepiadaceae. 2. In computer technology, this is a software package that codes from one language to another different language.

TRANSLOCATION: 1. The movement of minerals, chemical compounds and other soluble materials within a plant. There are two main processes: (i) the uptake of soluble minerals from the soil and their passage upwards from the root to the water conducting vessels. (ii) the transfer of organic compounds synthesised by the leaves, to all metabolising organs. 2. The transfer of part of a chromosome to a new position on the same or on a different chromosome with resultant rearrangement of the genes; or a type of chromosome alteration in which a section of one chromosome breaks off and become joined to another chromosome leading to a loss of genetic information from the previous chromosome. 3. The capture, transport and release or introduction of species, habitats or other ecological materials such as soil from one location to another.

TRANSLOCATED HERBICIDES: Chemical compounds that are absorbed in one part of the plant, and conducted to all parts of the plant via the xylem and phloem. Its effects on the plant may take up to two weeks.

TRANSLUCENT: Of or referring to a substance through which light can pass but through which objects cannot be seen. In this situation, the passage of radiation occurs, but not without some scattering or diffusion. For example, a thin sheet of paper is translucent and oil on paper produces translucence. Also, frosted glass allows light to pass through it but an object cannot be seen clearly through it due to the fact that the light rays are scattered.

TRANSMEMBRANE DOMAIN: A functional region of protein that spans the phospholipid bilayer of a biological membrane such as the plasma membrane of a cell. Integral membrane proteins typically comprise two or more such domains alternating with intracellular and extracellular domains arranged on either side of the membrane.

TRANSMISSION: 1. In telecommunications, it is the process of sending, propagating and receiving an analogue or digital information signal over a physical point-to-point or point-to-multipoint transmission medium, either wired, optical fibre or wireless. Transmission technologies and schemes typically refer to physical layer protocol duties such as modulation, demodulation, line coding, equalisation, error control, bit synchronisation and multiplexing. Transmission of a digital message or of a digitised analogue signal, is known as **data transmission** or **digital communication**. 2. The mechanical system, including shafts and gears, by which power is transmitted from an engine to the point, where it is to be used or a set of gears and the protective casing that covers this in a vehicle or engine. 3. The ability of a material to let incoming radiation pass completely through it. 4. The one way transport of nerve impulse from one neuron to another across a synapse. 5. The spread of infection from person to another.

TRANSMISSION CONTROL PROTOCOL (TCP): A protocol within the TCP/IP protocol suite, developed for the internet to get data from one network device to another to ensure that data is not lost in transmission. TCP ensures the reliability of data transmission across Internet connected networks and ensures that all data arrive accurately and at the other end. It takes care of keeping track of the individual units of data, called packets that a message is divided into for efficient routing through the Internet. For example, when an HTML file is sent from a Web server, the Transmission Control Protocol (TCP) program layer in that server divides the file into one or more packets, numbers the packets and then forwards them individually to

the IP program layer. The task of **Internet Protocol (IP)** is to take care of handling the actual delivery of the data by moving of data packets between nodes using an address such as 166.70.10.23 of a computer or other network device on a network using IP or TCP/IP. TCP/IP software is built into all major operating systems, such as Unix, Windows and the Mac OS.

TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL: Two protocols, TCP and IP that are used together to allow computers to communicate over the public Internet, many wide area networks (WANs) and most local area networks (LANs). TCP ensures the reliability of data transmission across Internet connected networks and takes care of keeping track of the individual units of data, called packets that a message is divided into for efficient routing through the Internet. For example, when an HTML file is sent from a Web server, the Transmission Control Protocol (TCP) program layer in that server divides the file into one or more packets, numbers the packets and then forwards them individually to the IP program layer. IP handles the actual delivery of the data by moving data packets between nodes using an address such as 166.70.10.23 of a computer or other network device on a network using IP or TCP/IP. TCP/IP software is built into all major operating systems, such as Unix, Windows and the Mac OS.

TRANSMISSION ELECTRON ENERGY LOSS SPECTROSCOPY (TEELS): Any technique in which an electron transparent specimen is bombarded with a finely focused electron beam and the energy distribution of the transmitted electrons is measured. This energy spectrum contains features corresponding to discrete losses of energy of the transmitted electrons due to excitation of electronic or plasmon states and provides information on the identity (and in some cases) chemical bonding of the elements in the sample.

TRANSMISSION ELECTRON MICROSCOPE (TEM): A microscopy technique in which a magnified image of a sample is produced by transmitting a beam of electrons through an ultra thin specimen, interacting with the specimen as it passes through. The electrons are accelerated by means of an electron gun, using voltages in the range 40-100 kV and the apparatus is evacuated to minimise electron scatter by air. An image is formed from the interaction of the electrons transmitted through the specimen; the image is magnified and focused onto an imaging device, such as a fluorescent screen, on a layer of photographic film or to be detected by a sensor such as a CCD camera.

TRANSMISSION HIGH ENERGY ELECTRON DIFFRACTION (THEED): Any technique which is based on the diffraction of high energy electrons ($E_0 = 10 - 200 \text{ ke V}$) in crystalline materials and evaluation of the angular distribution of the transmitted electrons. The diffraction pattern represents an image of the reciprocal lattice and therefore contains information about crystal structure.

TRANSMIT: To make heat, sound, light or other radiation pass or spread through space or a medium.

TRANSMITTANCE (TRANSMISSION COEFFICIENT): 1. The ratio of some form of radiation transmitted through a surface to the energy falling on it or the ability of a material to let incoming radiation pass completely through it. It is measured as the ratio of incident radiation to transmitted radiation. The reciprocal of the transmittance is the opacity. 2. The fraction of incident light at a specified wavelength that passes through a sample.

TRANSMITTER: 1. An electronic equipment used to generate and broadcast radio frequency electromagnetic waves and which modulates it so that it carries a meaningful signal and sends it out from an antenna for communication purposes. 2. The part of a telephone system that converts sound into electrical energy.

TRANSLATOR: In computing, a programming language processor that converts a computer program written in one language into another. Translators include the following: (i) **Compilers**, used to translate a program written in a high-level language into machine

code (object code); (ii) **Interpreter** programs that read, translate and execute one statement at a time from a high-level language program; and (iii) **Assemblers** that are used to translate a program written in a low-level assembly language into a machine code (object code) file so it can be used and executed by the computer.

TRANSMITTING TISSUE (CONDUCTING TISSUE): The specialised thin-walled tissue that constitutes the central part of the style in some angiosperms, through which the pollen tube grows down to the funiculus. It may be involved in self-incompatibility systems.

TRANSMUTATION: The conversion of one chemical element into another. It may occur naturally or can be induced. An induced nuclear reaction is caused by the bombardment of nuclei with particles or the transformation of the atom of one chemical element into the atom of another by disintegration or nuclei bombardment. For example, plutonium is obtained through the neutron bombardment of uranium. Natural processes of transmutation include radioactive decay, nuclear fission, and nuclear fusion.

TRANSONIC: Flight in the range between the onset of compressibility effects (around Mach number of 0.7) and the establishment of fully supersonic flight conditions (around Mach number of 1.4). While the aircraft itself may be travelling less than the speed of sound, the air going around the aircraft exceeds the speed of sound at some locations on the aircraft. In the regions where the local airspeed is near or greater than the speed of sound, compressibility effects are encountered and the air density may vary because of local shock waves, expansions or flow choking.

TRANSPARENT: 1. Of a material or substance, allowing light to pass through, so that objects may be seen clearly. For example, water and glass are transparent. 2. Permeable to electromagnetic radiation of specified frequencies, as to visible light or radio waves. 3. In computer science, it refers to a hardware device that has no effect on the data that passes through it or a software program that works so efficiently that the user is unaffected or unaware that it is even working.

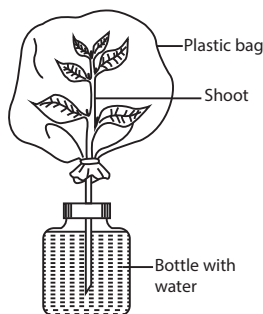
TRANSPIRATION: The loss of water from a plant, especially through stomata to the atmosphere in form of water vapour. It occurs mainly from the leaves through the pores (stomata), whose primary function is gas exchange.

TRANSPIRATION STREAM: The flow of water from the roots to the leaves via the xylem vessels that is caused by transpiration at the leaf surface.

TRANSPLANT: 1. The transfer of a tissue or organ from one human being to another or from one part of the body to another (skin grafting). 2. To uproot plants and replant them in another place.

TRANSPONDER: A radio transmitter or receiver that automatically generates or transmits a reply signals when the correct interrogation is received. An example of a transponder is a radar beacon mounted on an aircraft or a missile in which a receiver is tuned to the incoming radar frequency and a transmitter that radiates the received signal at a higher intensity.

TRANSPORT: In chemistry, it is the action of moving materials within the analytical instrument. Transport can involve any of several means including **pressure flow**, where the materials are moved by fluid pressure, either continuously or discontinuously; **centrifugal flow**, where the materials are moved by centrifugal force; **spontaneous motion**, where the materials are moved due to their intrinsic properties, for example, diffusion, capillarity; and **discrete transport**, where materials are enclosed within a moving container.



An experiment to show that transpiration occurs in plants

TRANSPORT COEFFICIENTS: Quantities that characterise transport in a system. Electrical and thermal conductivity are examples of transport coefficients. An inverse transport coefficient gives a measure of resistance to flow in a system and resistivity is an example. Calculation of these coefficients from first principles is one of the main purposes of non-equilibrium statistical mechanics.

TRANSPORT NUMBER: The fraction of the total charge (current) carried by a particular type of ion in the conduction of electricity through electrolytes. Its symbol is t .

TRANSPORT PROTEIN: A protein whose primary function is to transport a substance from one part of the cell to another, from one cell to another or from one tissue to another. This can be achieved by forming pores or channels through which particular ions or molecules can pass or binding the substance on one face of the membrane, then changing shape so that the substance is carried by the protein through the membrane to be released at the appropriate place.

TRANSPORT PROTOCOL: A communications protocol responsible for establishing a connection and ensuring that all data has arrived safely. Often, the term transport protocol implies transport services, which includes the lower-level data link protocol that moves packets from one node to another.

TRANSPORT THEORY: The theory or phenomenon involving the transfer of matter or heat, focusing on the calculation of transport coefficients and inverse transport coefficients, such as viscosity and conductivity. Transport theory uses approximate methods, including the kinetic theory of gases and kinetic equations, such as the Boltzmann equation.

TRANSPPOSE: To move a term from one side of an algebraic equation to another, reversing its sign to maintain equality. It is often referred to as **changing the subject of a formula**.

TRANSPPOSE OF A MATRIX: A matrix formed by turning all the rows of a given matrix into columns or the other way round. For a matrix A , the transpose of A is given by A^T . For example, the transpose

of the matrix $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ is given by $A^T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

TRANSPOSON (TRANSPOSABLE GENETIC ELEMENT): A mobile DNA element that can be inserted at many different sites in a genome, by moving from place to place or by replicating copies of itself that insert elsewhere in the genome. They can disrupt gene expressions or cause deletions and inversions and hence affect both the genotype and phenotype of the organisms in questions. They account for a significant proportion of the repetitive DNA in eukaryotes.

TRANSPOSON MUTAGENESIS (TRANSPPOSITION MUTAGENESIS): A biological process that allows genes to be transferred to a host organism's chromosome, interrupting or modifying the function of an extant gene on the chromosome and causing mutation.

TRANSURANIC ELEMENT: Elements above uranium in the periodic table with an atomic number greater than 92. They are mostly unstable or radioactive. More than 20 such elements have been identified. These elements consist of more than 100 radioactive isotopes, which are characterised by radioactive instability. These radioisotopes are produced artificially by bombarding heavy atoms either with neutrons produced in nuclear reactors or with charged particles accelerated to high energy in particle accelerators. The first 10 transuranium elements, together with actinium, thorium, protactinium and uranium, constitute the actinide elements, which are chemically analogous to the rare earth elements. They are neptunium, plutonium, americium, curium, berkelium, californium, einsteinium, fermium, mendelevium and nobelium.

TRANSVERSAL: A line that intersects or cuts two or more lines.

TRANSVERSALITY CONDITIONS: Orthogonality conditions that must hold at the boundary of a problem. A curve satisfying a transversality condition is called a **transversal**.

TRANSVERSE ACCELERATION: The inertial force produced by acceleration acting across the body, perpendicular to the long axis of the body, as in a chest-to-back direction.

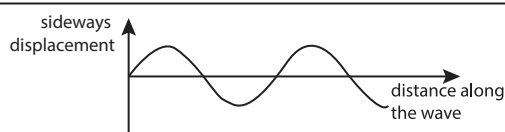
TRANSVERSE AXIS: The axis of a hyperbola that passes through the two foci.

TRANSVERSE COMPONENT: The component of a vector that is perpendicular to the radius vector. In polar coordinates, the transverse component of velocity is $T(dr/dt)$ and the transverse component of acceleration is $r \frac{d^2\theta}{dt^2} + 2 \frac{dr}{dt} \frac{d\theta}{dt}$.

TRANSVERSE PROCESS: The long projections that extend laterally from each neural arch of a vertebra.

TRANSVERSE TUBULES (T TUBULES): A series of the unfolding of the plasma membranes of muscle fibres that enclose each myofibril at the Z line. The T tubules transfer action potentials from the sarcolemma to the sarcoplasmic reticulum, to release calcium ions into the cytosol, resulting in the contraction of the muscle fibre.

TRANSVERSE WAVES: A wave in which the displacement of the medium's particles is at right angles to the direction of travel of the wave motion. It makes the medium through which it travels to vibrate in a direction at right angles to the direction of its movement.



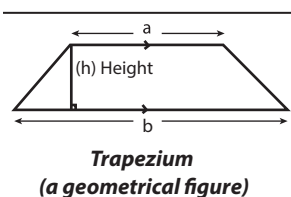
Transverse wave (eg light wave)

TRANSVERSION: A mutation caused by the substitution of a purine base for a pyrimidine base (or vice versa) in the DNA. It can be caused by ionising radiation and alkylating agents and can lead to a missense mutation.

TRANSVERSUS: A thin sheet of muscle that lies deep to the internal oblique, with its fibres running horizontally forward.

TRAP CROP: Any crop planted in or around another crop to attract pests away from the more valuable crop. Trap crops are usually heavily treated with pesticides, burnt or ploughed in.

TRAPEZIUM: 1. In geometry, it is a four-sided plane figure or quadrilateral that has two parallel sides. The parallel sides of a trapezium are called bases and the nonparallel sides are called legs. The area of a trapezium is given by $A = \frac{1}{2}(a + b)h$, where a and b are the lengths of the parallel sides and h is the perpendicular height between the parallel sides. 2. In human anatomy, it is a carpal bone in the wrist, characterised by a deep groove on its palmar surface. It is situated at the radial side of the carpus, between the scaphoid and the first metacarpal bone of the thumb. It is homologous with the first distal carpal of reptiles and amphibians.



TRAPEZIUM RULE (TRAPEZOIDAL RULE): A rule for finding the approximate area under a curve by dividing it into pairs of trapezium-shaped sections, forming vertical columns of equal widths with bases lying on the horizontal axis. Generally, the trapezium rule for finding the approximate area under the curve $f(x)$ between the ordinates a and b is given by:

$$\int_a^b f(x)dx = \frac{1}{2}h [y_0 + 2y_1 + \dots + 2y_{n-2} + y_{n-1}], \text{ where } h = \frac{b-a}{n-1}, \text{ where } n \text{ is the number of ordinates.}$$

TRAPEZIUS: A large, diamond-shaped muscle that extends from the back of the skull down to the lower part of the spine in the chest and across the width of the shoulders. Its function is to help support the neck and spine. It is also involved in moving the arm.

TRAPEZOID: A quadrilateral with no parallel sides or a closed plane figure bounded by four line segments or sides, two of which are parallel and two of which are nonparallel. The parallel sides of a trapezoid are called bases and the nonparallel sides are called legs. In an isosceles trapezoid the legs are of equal length. The median of a trapezoid is the line segment connecting the midpoints of the legs; it is parallel to the bases and equal to half the sum of their lengths. The altitude of a trapezoid is the perpendicular distance between the bases. The area of a trapezoid is equal to half the product of the altitude and the sum of the bases (the product of the altitude and the median).

TRAPPING: The interception of a reactive molecule or reaction intermediate so that it is removed from the system or converted into a more stable form for study or identification.

TRASFERRIN: A plasma protein that can bind to iron and transport it to the liver for storage or to the cells of the bone marrow, where it is used for the formation of haemoglobin. One transferrin can bind to two iron ions.

TRASH: In a Macintosh computer, Trash serves the same purpose as the Windows Recycle Bin, where deleted files or folders can be temporarily located. Items can be moved to the Trash by selecting them and dragging them to the Trash icon.

TRAVEL GRAPH: A graph in which distance is plotted against time.

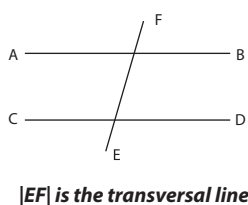
TRAVELLING WAVE: A wave in which energy is transported from one part of a medium to another, in contrast to a standing wave. Travelling waves are observed when a wave is not confined to a given space along the medium. The most commonly observed travelling wave is an ocean wave. As a wave is observed travelling through a medium, a crest is seen moving along from particle to particle. This crest is followed by a trough that is in turn followed by the next crest, which can be observed as a distinct wave pattern (in the form of a sine wave) travelling through the medium. This sine wave pattern continues to move in uninterrupted fashion until it encounters another wave along the medium or until it encounters a boundary with another medium.

TRAVELLING-WAVE ACCELERATOR (LINEAR ACCELERATOR; LINEAR PARTICLE ACCELERATOR, LINAC): A type of particle accelerator that imparts a series of relatively small increases in energy to subatomic particles such as protons and electrons, as they pass through a sequence of alternating electric fields set up in a linear structure. The small accelerations add together to give the particles a greater energy than could be achieved by the voltage used in one section alone. It consists of electrodes and gaps arranged in a straight line. It is proportioned so that when electrical potentials are varied with the proper amplitude and frequency, particles passing through the waveguide receive successive increments of energy and are therefore accelerated. It is used in radiotherapy, in physics research and in the production of radionuclides. In radiotherapy, lines of accelerated x-ray radiation are shaped to target a tumour or surgery site, limiting radiation exposure to healthy tissue while killing cancer cells. A linear accelerator, for example, is used during whole breast radiation, accelerated partial breast radiation and Intensity-Modulated Radiation Therapy (IMRT).

TRAVERS, MORRIS WILLIAM (1872-1961): British chemist and the founding director of the Indian Institute of Science. He worked along with Sir William Ramsay in the discovery of xenon, neon and krypton in 1898. His work on several of the rare gases earned him the name "Rare gas Travers" in scientific circles.

TRAVERSABLE: In mathematics, of a network, forming an Eulerian chain; a network is traversable if each arc can be traced exactly once by beginning at some point without lifting one's pencil from the page. A network is a collection of points, called vertices and a collection of lines, called arcs, connecting these points.

TRAWLER: A diesel engine powered boat that is used in trawling for fish. It is from 12 metres in length to about 90 metres, which tow nets, called **trawls**, for fish or dredge for shellfish. Conventional trawlers are side trawlers, towing their nets from one side of the vessel. Most new boats, however, are stern trawlers and tow and haul over a stern ramp. Shrimp trawlers tow their nets from booms swung out on each side of the boat. In all cases, powerful winches wind in the cables leading to the nets, which in turn are held open by hydrofoils, called trawl boards or doors.

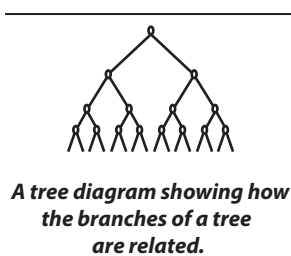


TREAD: The part of the surface of a wheel or tyre or the rubber on its circumference that comes in contact with a rail or road or the depth of grooves on the surface of a tyre.

TREATED SOLUTION: The test solution that has been subjected to reaction or separation procedures prior to measurement of some property.

TREE: 1. A perennial woody plant with a single straight bole or trunk. In most species the trunk produces secondary limbs called branches. Trees provide many valuable products, especially wood, used in construction, for making furniture, building boats, as fuel, wood pulp for papermaking, etc. The fruits and nuts of some trees are edible. In addition, trees take in carbon dioxide and release oxygen during photosynthesis. The root systems help retain water and soil, preventing floods and erosion. Trees and forests provide habitats for a wide variety of animals and they decorate both natural and altered landscapes. Growth rings in the trunk indicate the age of most trees. 2. In mathematics, also known as **tree diagram**, it is a graph that has a unique path from any vertex to any other vertex travelling along the edges or a flow diagram which has no closed paths. If one vertex is distinguished as the root or origin, it is called a **rooted tree**; if not it is a **free tree**; in a **labelled tree**, a unique element is associated with each node. A finite set of node-disjoint trees comprises a **forest**. Also the term defines a compact connected set in which every pair of points is joined by a unique rectifiable path. 3. In computing, it is a data structure conceptualised as an inverted tree with the root at the top. It is used in referring to the hierarchical organisation of files and folders.

TREE DIAGRAM: A graph that has a unique path from any vertex to any other vertex travelling along the edges, or a flow diagram which has no closed paths. Factor trees and probability trees are tree diagrams used, respectively, to factor numbers and to represent probability situations in which there is a series of events.



TREE OF LIFE (*Guaiacum officinale*):

An evergreen tree with high drought tolerance which grows up to a height of about 15 metres at maturity with a trunk diameter of 60 centimetres. It is a slow growing species of lowlands dry forest, woodland and thickets. It is native to the Caribbean and from Panama to Guyana, but can thrive in any well-drained deep soil and tropical forests anywhere. The tree trunk is often crooked with furrowed bark and has umbrella-like canopy. The leaves are compound, sessile, obovate and with pinnate venation and rounded at the apex. The leaflets are entire with an oval shape and sometimes blunt-pointed

at both base and apex. The plant produces subtle fragrant light-blue flowers with hairy petals having ten stamens; and are solitary or produced in a few-flowered clusters. The fruits produced by the plant are succulent, heart-shaped and yellow in colour. Its wood is considered the heaviest and densest of trees and rapidly sinks to the bottom when placed in water. It had been used in commercial shipbuilding for centuries. The resin in the tree is used in treating rheumatism, gout and syphilis.

TREE TOPOLOGY (EXTENDED STAR TOPOLOGY): A type of physical topology with a combination of both the star and the linear bus topologies. It has multiple access points connected to the linear bus, while the nodes are connected to their respective access points.

TREE-AND-BRANCH FILING SYSTEM: In computer science, it is a filing system where all files are stored within directories, like folders in a filing cabinet. These directories may in turn be stored within further directories, called sub directories. The **root directory** is the top-level directory of a file system that contains all the other directories and sub directories. All file systems have a root directory but are labelled differently.

TREELINE (TIMBERLINE): 1. The highest point on a mountain where trees will grow and above which no tree grows as it is too cold for it to survive. 2. A geographic boundary beyond which trees cannot grow. On the Earth, it is the northernmost or southernmost latitude at which trees can survive.

TREMATODA: A class of acoelomate parasitic flatworms of the phylum Platyhelminthes comprising the flukes. They have bilateral symmetry with suckers and well-developed hooks to anchor to their hosts.

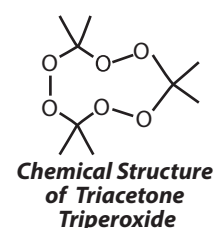
TREMOR: 1. A minor earthquake or a quivering or vibration caused by slippage of the Earth's crust at a fault, especially before or after a major earthquake. 2. A fluctuation in a sound or light. 3. A slight shaking or trembling movement of a part of the body.

TRI-: A prefix used to denote three. For example, **triangulate** or **trigonus** is three-angled; **triacamorous** means having three locales; and **triasristate** means three-awned.

TREND: In statistics, it is a functional relationship between observed data and an independent variable, usually time.

TREND LINE: A straight or curved line on graph showing observations made over time. It may be drawn visually by connecting the actual data points or by using statistical techniques.

TRIACETONE TRIPEROXIDE, TATP (ACETONE PEROXIDE): An organic peroxide with the chemical formula $C_9H_{18}O_6$, relative density 1.18, melting point 91°C (364 K) and boiling point $97\text{--}160^\circ\text{C}$ (370–433 K). It is a highly explosive white crystalline substance very sensitive to heat and shock. It is derived from acetone. Acetone peroxides are common and unwanted by-products of oxidation reactions, such as those used in phenol syntheses. Due to their explosivity, they are hazardous. Numerous methods such as shifting the pH to more alkaline, adjusting the reaction temperature or adding a soluble copper(II) compound are used to reduce their production. Acetone peroxide and benzoyl peroxide are used as flour bleaching agents to bleach and mature flour.



TRIACHASium: A cymose inflorescence with three branches.

TRIACHOCARPOUS: Having hairy fruits.

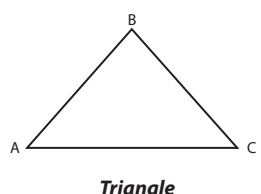
TRIAD: A group of three; used to describe the three-flowered umbels found in some Loranthaceae (mistletoe families of the sandalwood order Santalales).

TRIADELPHOUS: Having stamens arranged into three groups. **Triandrous** means having three stamens.

TRIAL: In statistics, a single experiment or observation. It is done as a test or experiment or as a process or means by which the properties of quantities are tested by experiment.

TRIAL AND ERROR: A method of solving a problem by making a sequence of attempts or trying various theories until error is sufficiently reduced or eliminated.

TRIANGLE: 1. A three-sided plane figure, the sum of whose interior angles is 180 degrees or a two-dimensional geometric figure formed by three sides and three angles. It is a fundamental figure of plane geometry, since it is the polygon with the fewest sides and any other polygon can be subdivided into triangles. A triangle is **scalene** if none of its sides and angles are equal; **isosceles** if two sides are equal; and **equilateral** if all three sides are equal. 2. Any thin, flat, three-sided instrument used as a drawing and drafting guide to rule straight lines at specific angles or for determining the angles of ruled lines. Typically the instruments have angles of 90°, 45°, 60° and 30°. 3. A metal bar bent into the shape of a triangle with one angle open and used as percussion instruments.



TRIANGLE INEQUALITY: 1. The theorem that the sum of any two sides of a triangle is greater than the third. In other words, any side of a triangle must be shorter than the other two sides added together. 2. The requirement that $d(x, y) + d(y, z)$, (for a binary real-valued function d or that $|x| + |y| \geq |x + y|$).

TRIANGLE OF FORCES: A triangle with two sides representing forces acting on a particle, while the third represents the combined effect of these forces. It is a method of calculating the force produced by two other forces (the resultant), based on the fact that if two forces acting at a point can be represented in both magnitude and direction by the sides AB and BC of the ABC, then the side AC will represent the resultant in both magnitude and direction. Also, if three forces acting at a point can be represented in magnitude and direction by the sides of a triangle taken in order, the forces are in equilibrium. The polygon of forces will close and the sum of the forces will be zero.

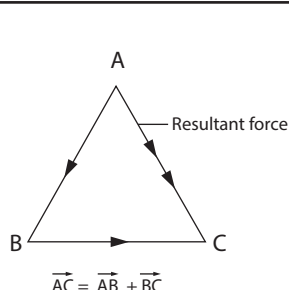


Fig I:
**Triangle of
forces and vectors**

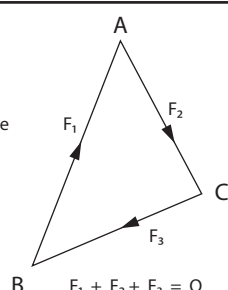


Fig II:
Forces in equilibrium

TRIANGLE OF REFERENCE: A set of three linearly independent points in two dimensional algebraic geometry that are chosen together with a unit point in order to determine a system of homogeneous coordinates for the geometry.

TRIANGLE OF VELOCITIES: 1. Triangle of vectors, in which each vector denotes a velocity or a triangle that can be drawn when the magnitude and direction of three component velocities of a body at rest are represented by the sides of a triangle. 2. The fundamental triangle associated with dead-reckoning, composed of the following vectors: heading and true airspeed, track and groundspeed and wind speed and wind direction.

TRIANGULABLE: Of a topological space, able to be mapped by a (simplicial) triangulation, as is the sphere.

TRIANGULAR: 1. Shaped like or relating to, a triangle. 2. Of a three-dimensional figure, having a triangular base; an example is a tetrahedron.

TRIANGULAR MATRIX: A matrix having only zero entries below or above the main diagonal. It is an **upper triangular** if the zero entries are below the main diagonal and a **lower triangular** if they are above.

TRIANGULAR NUMBER: A number given by the expression $(n+1)\left(\frac{n}{2}\right)$, where $n = 1, 2, 3, \dots$ or one obtained by adding the first n

whole numbers. For example, the 5th triangular number is 15 because $15 = 1 + 2 + 3 + 4 + 5$. It is also a number that can make a triangular dot pattern; for example, 1, 3, 6, 10 and 15 are triangular numbers.

TRIANGULATE: 1. To calculate trigonometrically, especially by triangulation. 2. To divide, an area or region, into triangles or simplices.

TRIANGULATION: 1. A technique used in surveying and navigation to determine distances, using the properties of a triangle. It involves the division of a large area into adjacent triangles for survey purposes using trigonometric relationships to calculate the dimensions of an area bounded by each triangle. One side baseline and the angles to the third point of each adjacent triangle are measured and the lengths of the other sides are calculated from these measurements.

2. A navigation technique that uses the trigonometric properties of triangles to determine a location or course by means of compass bearings from two points, a known distance apart. Space-age global positioning systems enable people to triangulate their locations relative to the known positions of Earth-orbiting satellites. 3. The process of determining the location of a point by measuring angles to it from known points at either end of a fixed baseline, rather than measuring distances to the point directly. The point can then be fixed as the third point of a triangle with one known side and two known angles. 4. Also known as **simplicial triangulation**, it is a homeomorphism of a topological space on to a polyhedron comprising the points of a simplicial complex.

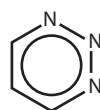
TRIANGULO-: An affix used in names to denote three atoms bound into a triangle.

TRIASSIC: The earliest period of the Mesozoic era. It began about 248 million years ago following the last period of the Palaeozoic era (Permian) and succeeded by the Jurassic. During that period, marine animals diversified and molluscs were the dominant invertebrates.

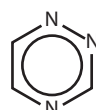
TRIATOMIC MOLECULE: A molecule which is created from three atoms. Examples are carbon dioxide (CO₂) and water (H₂O).

TRIAxIAL: Having three axes.

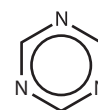
TRIAZINE: A group of organic compounds based on a heterocyclic ring containing three carbon atoms and three nitrogen atoms (C₃H₃N₃). The triazine structure is a heterocyclic ring, analogous to the six-membered benzene ring but with three carbons replaced by nitrogens. The three isomers of triazine are distinguished from each other by the positions of their nitrogen atoms and are referred to as 1,2,3-triazine, 1,2,4-triazine and 1,3,5-triazine. Other aromatic nitrogen heterocycles are pyridines with 1 ring nitrogen atom, diazines with 2 nitrogen atoms in the ring and tetrazines with 4 ring nitrogen atoms. Triazines have weaker bases than pyridine. They include several commonly used agricultural herbicides such as simazine and atrazine. Triazine herbicides work by inhibiting electron transport of photosynthesis.



1,2,3-triazine



1,2,4-triazine



1,3,5-triazine
or
s-triazine

TRIBASIC: 1. An acid containing three replaceable hydrogen atoms and capable of reacting with three hydroxyl ions per molecule, for

example, phosphoric acid (H_3PO_4). 2. A compound such as calcium phosphate that contains three univalent metal atoms or groups in each molecule.

TRIBE: In botany, it is a category used in the classification of plants and animals made up of several similar or closely related genera within a family; or a division in the scientific classification of animals and plants, between a subfamily and a genus. In the case of grasses, examples include the Paniceae, Bambuseae, Aveneae and the Oryzeae.

TRIBO ELECTRICITY (TRIBOELECTRIC EFFECT): Static electricity produced as a result of friction, such as rubbing when certain materials become electrically charged after they come into contact with another different material and are then separated. For example, rubbing glass with fur can build up triboelectricity. The polarity and strength of the charges produced differ according to the materials, surface roughness, temperature, strain and other properties.

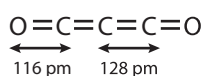
TRIBOLOGY: The science and technology of interacting surfaces in relative motion, including the study of friction, lubrication and wear. Many manifestations of tribology are beneficial and make modern life possible. Many others are serious nuisances and careful design is necessary to overcome problems caused by excessive friction or wear. Friction uses up or wastes, a substantial amount of the energy generated by humans. In addition, a great deal of productive capacity is devoted to replacing objects that have worn out.

TRIBOLUMINESCENCE: The luminescence caused by friction. Some crystalline substances emit light when they are pulled apart, ripped, scratched, crushed or rubbed as a result of static electric charges generated by the friction.

TRIBROMOMETHANE (BROMOFORM; TRIBROMOMETHANE): A colourless liquid haloform having the chemical formula CHBr_3 , density 2.889 g/cm^3 at 15°C , melting point of 8°C (281 K) and boiling point of 149°C (422 K). It is produced naturally by phytoplankton and seaweeds in the ocean and this is thought to be the predominant source to the environment. It can be prepared by the haloform reaction using acetone and sodium hypobromite, by the electrolysis of potassium bromide in ethanol or by treating chloroform with aluminium bromide. However, locally significant amounts of bromoform enter the environment formed as disinfection by-products known as the trihalomethanes, when chlorine is added to drinking water to kill bacteria. It is somewhat soluble in water and readily evaporates into the air. It is an animal carcinogen and can be absorbed into the body by inhalation and through the skin. The substance is irritating to the respiratory tract, the eyes and the skin and may cause effects on the central nervous system and liver, resulting in impaired functions. In the past, it was used as a solvent, sedative and flame retardant, but now it is mainly used as a laboratory reagent.

TRIBUTARY: A stream or river that flows into a lake, a larger stream/river or other body of water.

TRICARBON DIOXIDE (CARBON SUBOXIDE; PROPA-1,2-DIENE-1,3-DIONE): A colourless gas with an unpleasant odour, which is an oxide of carbon with the chemical formula C_3O_2 , melting point -111.3°C (161.9 K) and boiling point 6.8°C (280 K). It is the acid anhydride of malonic acid, from which it can be prepared by dehydration using phosphorous(V) oxide. Tricarbon dioxide is used in the preparation of malonates and as an auxiliary to improve the dye affinity of furs.



Chemical Structure
of Tricarbon Dioxide

TRICARBOXYLIC ACID CYCLE (TCA CYCLE; KREBS CYCLE; CITRIC ACID CYCLE; SZENT-GYÖRGYI-KREBS CYCLE): The cyclical biochemical process whereby acetate is completely oxidised to carbon dioxide and water and electrons are transferred to NAD^+ and FAD. The TCA cycle is localised to the mitochondria in eukaryotic cells and to the plasma membrane in prokaryotic.

TRICARINATE: Having three ridges or keels.

TRICARPELLARY: Having three carpels.

TRICEPS: A muscle that runs parallel to the humerus and works antagonistically to the biceps causing the arms to extend.

TRICHINA WORM: A parasitic nematode worm *Trichinella spiralis*, found in mammals such as humans cats, dogs, pigs and rats. The adults live in the gut and the larvae migrate to the muscles. If consumed in undercooked meat they cause trichinosis in humans.

TRICHINOSIS: A disease caused by the parasitic nematode worm *Trichinella spiralis*, found in flesh-eating mammals such as humans, cats, dogs, pigs and rats. It is caused by eating raw or undercooked pork or other animalsb that are infected with the larvae of a parasitic worm. Early symptoms include diarrhoea, abdominal pain, fatigue, nausea and vomiting; which may develop into high fever, muscle pain and tenderness, swelling of the eyelids or face, weakness, headache and sensitivity to light.

TRICHLOROETHANAL (CHLORAL): Oily colourless liquid with the chemical formula $\text{C}_2\text{HCl}_3\text{O}$, density 1.511 g/cm^3 at 20°C , melting point -57.5°C and boiling point 97.8°C (370.8 K). It has a characteristic pungent smell. It is produced by reacting chlorine with ethanol or acetaldehyde. It is used in making DDT.

TRICHLOROETHANE (1,1,1-TRICHLOROETHANE; METHYL CHLOROFORM): A volatile colourless non-flammable liquid with a sweet odour and the chemical formula $\text{C}_2\text{H}_3\text{Cl}_3$, density 1.32 g/cm^3 , melting point -33°C (240 K) and boiling point 74°C (347 K). It is used as solvents for adhesives, pesticides and lubricants and in industrial cleaning solutions.

TRICHLOROETHENE (TRICHLOROETHYLENE; 1,1,2-TRICHLOROETHENE; 1,1-DICHLORO-2-CHLOROETHYLENE; 1-CHLORO-2,2-DICHLOROETHYLENE; ACETYLENE TRICHLORIDE): A colourless, toxic and non-flammable liquid with the chemical formula C_2HCl_3 , density 1.46 g/cm^3 (20°C), melting point -73°C (200 K) and boiling point 87.2°C (360 K). It has a smell resembling that of chloroform (trichloromethane). It is widely used as a solvent in dry cleaning and degreasing. It is also used to extract oils from nuts and fruit, as an anaesthetic and fire extinguisher.

TRICHLOROMETHANE (CHLOROFORM; FORMYL TRICHLORIDE; METHANE TRICHLORIDE; METHYL TRICHLORIDE; METHENYL TRICHLORIDE): A colourless volatile sweet-smelling liquid haloform with the chemical formula CHCl_3 , relative density 1.48, melting point -63.5°C (209.5 K) and boiling point 61.7°C (334.7 K). It can be made by chlorination of methane followed by separation of the mixture of products or by the haloform reaction. It is an effective anaesthetic but can cause liver damage and it has now been replaced by other halogenated hydrocarbons. It is used as a solvent and raw material for making other compounds.

TRICHOCYST: An organ resembling a thread that protrudes and can be ejected from a minute cavity on the surface of some protozoans, especially ciliates. This organ is capable of stinging or grasping.

TRICHOMES: Hair-like extensions from the epidermis of a plant that provide shade and protection for the plant.

TRICHOTOMY: The property of the real numbers or more generally of any strong total ordering that for any two real numbers a and b , exactly one of the following must be true: $a < b$, $a = b$ or $a > b$.

TRICHROMATIC: Involving the combination of the three primary colours to produce the other colours or relating to standard colour vision, which is able to perceive red, green and blue.

TRICLINIC CRYSTAL SYSTEM: One of the seven crystal systems, described by three basis vectors. In the triclinic system, the crystal is described by vectors of unequal length, as in the orthorhombic system. In addition, all three vectors are not mutually orthogonal.

TRICLISIA: A genus of plant in the Menispermaceae family, comprising about 20 species. Some of the species include *Triclisia dictyophylla* Diels, which occurs in the lowland dense rain-forest from Liberia eastwards to the Cameroons, the Democratic Republic of Congo and Angola; *Triclisia lanceolata*, found in Cameroon and the Democratic Republic of the Congo; *Triclisia macrophylla*, found in Cameroon, Equatorial Guinea and Sierra Leone; *Triclisia macrocarpa* (Baill.) Diels, occurring in Madagascar; and *Triclisia saculeuxii* (Pierre)

Diels, which occurs from Congo eastwards to Kenya and south to Angola and Mozambique. *Triclisia dictyophylla* is a liana that can grow up to 30 m long and 10 cm in stem diameter. Young branchlets are hairy and the old ones are glabrous. The bark is grey and vertically striped. The leaves alternate, simple and entire, very large and slightly heart-shaped or broadly ovate, shortly acuminate at the apex, leathery and glabrous. Flowers are dioecious, occurring in panicles mostly on old wood below the leaves. The oblong-ovate fruit is a large drupe in an enormous bunch, bright orange-red when. All *Triclisia* species are known for the treatment of ailments such as diarrhoea, venereal diseases, swellings, oedema of the legs, gout, fabrifuges, malnutrition, debility, pulmonary and stomach troubles, etc.

TRICOLPATE: Of or referring to a pollen grain having three colpi, as is commonly found amongst most dicotyledonous species.

TRICUSPID VALVE (RIGHT ATRIOVENTRICULAR VALVE): A valve consisting of three flaps, situated between the right atrium and the right ventricle of the mammalian heart. When the right ventricle contracts, forcing blood into the pulmonary artery, the tricuspid valve closes the aperture to the atrium thereby preventing any backflow of blood. The valve reopens to allow blood to flow from the atrium into the ventricle.

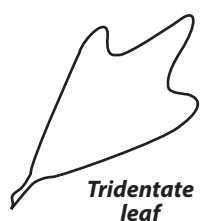
TRICYCLIC: 1. In chemistry, chemical compounds that contain three interconnected rings of atoms, for example, phenothiazine. 2. Having three whorls.

TRIDAX (COATBUTTONS; *Tridax procumbens*): A perennial dicotyledonous herb in the daisy family (Asteraceae or Compositae) with a creeping stem that can reach 75 centimetres long. It is native to Central America and occurs in many tropical and subtropical countries in fallow fields, agricultural fields and disturbed wastelands. It has simple, serrated leaf margins with arrowhead appearance. Its daisy-like yellow-centred white flowers have three serrated ray florets. It produces fruits called achene, which is hard, covered with stiff hairs and feathery with plume-like white pappus at one end. The calyx is fused to form an involucre bearing hair-like structures. The plant produces numerous achenes up to 1500 per plant and it is listed as one of the noxious weeds in the world. *Tridax* is best known for its antiviral, anti-inflammatory, anti-oxidant and antibiotic efficacies.

TRIDENT OF NEWTON: A cubic plane curve with an ordinary double point in the real projective plane that cuts the axis in either one or three points: $x = 0$, $y = 1$, $z = 0$ with an equation of the form $xy = ax^3 + bx^2 + cx + d$.

TRIDENTATE: In botany, of a leaf that has three tooth-like pointed or rounded projections.

TRIDIAGONAL MATRIX: A matrix having zero entries except on the main diagonal and the rows immediately above and below it. In other words, it has nonzero entries only on the main diagonal, the first diagonal below this and the first diagonal above



Tridentate leaf

the main diagonal. An example is the 4 x 4 matrix:

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ 3 & 4 & 1 & 0 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 3 & 3 \end{bmatrix}$$

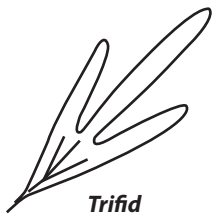
TRIDIGITATE: Divided into three finger-like lobes or divisions.

TRIDYNAMOUS: With stamens arranged in two groups of three, one group often longer than the other.

TRIFID: Partly or fully split into three divisions or lobes.

TRIFOLIATE: Of a compound leaf having three leaflets arising from the same point, such as the clovers (*Trifolium*) and wood sorrels (*Oxalis*). **Trifoliolate** means having three leaflets.

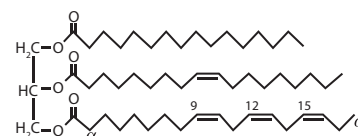
TRIFURCATE: Three-forked; divided into three terminal, long lobes.



Trifid

TRIGEMINAL NERVE: Each of the fifth and largest pair of cranial nerves, which has a wide distribution and the main nerve of sensation for the face. Its motor fibres supply the four muscles of mastication (chewing) and four other muscles: tensor palati, tensor tympani, mylo-hyoid and anterior belly of the digastric. Its sensory fibres supply: (i) part of the dura mater; (ii) the eyeball and other contents of the orbit; (iii) the skin of face and anterior half of the head; (iv) the mucous membranes of the air sinuses, nose, mouth and anterior two-thirds of the tongue; (v) the teeth; and (vi) the mandibular joint. Problems with the sensory part of the trigeminal nerve result in pain or loss of sensation in the face; and problems with the motor root of the trigeminal nerve result in deviation of the jaw toward the affected side and trouble chewing.

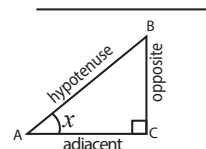
TRIGLYCERIDE (TRIACYLGLYCEROL): An ester of glycerol in which all the three hydroxyl groups are esterified with fatty acid. They are the major constituents of fats and oils and provide a concentrated food energy store in living organisms as well as cooking fats and oils, margarine, etc. Their physical and chemical properties depend on the nature of their constituents.



Chemical Structure of Triglyceride

TRIGONOMETRIC FUNCTIONS (CIRCULAR FUNCTIONS; CYCLOMETRIC FUNCTIONS):

Functions defined in terms of angle and sides of a right-angled triangle and widely used in the solution of many mathematical problems; or any of the six functions: sine, cosine, tangent, cotangent, secant and cosecant that represent ratios of sides of right angle triangle. For example, for acute angle x in a right triangle ABC, the trigonometric functions are as follows:



Trigonometric/cyclometric functions of a right-angled triangle

$$\sin x = \frac{\text{opposite}}{\text{hypotenuse}}, \cos x = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan x = \frac{\text{opposite}}{\text{adjacent}}, \cot x = \frac{\text{adjacent}}{\text{opposite}}$$

$$\sec x = \frac{\text{hypotenuse}}{\text{adjacent}}, \csc x = \frac{\text{hypotenuse}}{\text{opposite}}$$

TRIGONOMETRIC SERIES: The trigonometric series for $\cos x$ and $\sin x$ are given by $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$ and

$$\sin x = 1 - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

TRIGONOMETRIC TABLES: Tables showing the values of the trigonometric functions for various angular values between 0° and 90° enabling their values for any argument to be inferred by means of the trigonometric reduction formulae. They have now been superseded by computers, electronic or digital calculators.

TRIGONOMETRICAL RATIO: Mathematical relationship between the angles and sides of a right-angled triangle. These ratios: **sine**, **cosine** and **tangent** are based on properties of right triangles. The function values depend on the measure of the angle. The functions are expressed in the following ways:

$$(i) \text{ sine } x = \frac{\text{side opposite } x}{\text{hypotenuse}};$$

$$(ii) \cosine x = \frac{\text{side adjacent } x}{\text{hypotenuse}};$$

$$(iii) \tan gent x = \frac{\text{side opposite } x}{\text{side adjacent } x}.$$

The reciprocal ratios are as follows:

$$(i) \text{Cosecant} = \frac{\text{hypotenuse}}{\text{opposite}};$$

$$(ii) \text{Secant} = \frac{\text{hypotenuse}}{\text{adjacent}};$$

$$(ii) \text{Cotangent} = \frac{\text{adjacent}}{\text{opposite}}.$$

TRIGONOMETRY: A branch of mathematics that solves problems relating to plane and spherical triangles. It is the relationship between the angles and sides of right angled triangles. Trigonometrical relationships are ratios and therefore have no units. Its principles are based on the fixed proportions of size for a particular angle in a right-angled triangle. The science of trigonometry is based on certain ratios called trigonometric functions.

TRIHEDRAL: 1. Having or formed by three plane faces meeting at a point. 2. A figure formed by the intersection of three lines in different planes.

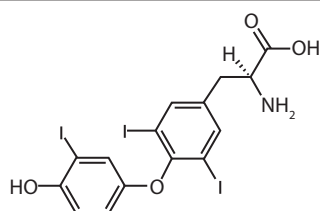
TRIHEDRON: In differential geometry, it is a set of three mutually perpendicular vectors directed from a single point or a figure formed by three planes meeting at a point.

TRIHYBRID: An organism that is heterozygous for three genes, for example, AaBbCc. Trihybrid crosses, especially of the type AaBbCc $\times$ aabbcc, are commonly used devices in chromosome mapping. By comparing the frequencies of the phenotypes, the chromosomal arrangements of the three genes A/a, B/b and C/c can be deduced. This is technically easier than setting up three dihybrid crosses AaBb $\times$ aabb, AaCc $\times$ aacc and BbCc $\times$ bbcc, which would only achieve the same result.

TRIHYDRATE: A crystalline hydrate that contains three moles of water per mole of compound. Aluminium trihydrate is a common primary ingredient present in most solid surface material and accounts for as much as 70% of the total product. Examples of trihydrate are glycerol and the anti-infection drug amoxicillin trihydrate.

TRIHYDRIC: A term used to describe an alcohol that contains three hydroxyl groups per molecule. An example is glycerol.

TRIIODOMETHANE (IODOFORM): A yellow, volatile, sweet-smelling solid with the chemical formula CHI_3 , density 4.008 g mL^{-1} and melting point 121°C (394.25 K). It is synthesised by the reaction of iodine and sodium hydroxide with a methyl ketone, acetaldehyde, ethanol and certain secondary alcohols (CH_3CHROH , where R is an alkyl or aryl group). It is occasionally used as a disinfectant.



Chemical Structure
of Triiodothyronine

TRIIODOTHYRONINE (T_3): A thyroid hormone with the chemical formula $\text{C}_{15}\text{H}_{12}\text{I}_3\text{NO}_4$, similar to thyroxine but more potent, used in the treatment of hypothyroidism. Production of T_3 and its prohormone thyroxine (T_4) is activated by thyroid-stimulating hormone (TSH), which is released from the pituitary gland.

TRIIRON TETROXIDE (FERROSFERRIC OXIDE): A black magnetic oxide with the chemical formula Fe_3O_4 and relative density 5.2. It is formed when iron is heated in steam and also occurs naturally as the mineral magnetite. The oxide dissolves in acids to give a mixture of iron(II) and iron(III) salts.

TRILATERAL: Having three sides.

TRILEMMA: Classically valid argument one of whose premises is the disjunction of three statements that can be expressed as a choice among three unfavourable options, one of which must be chosen or as a choice among three favourable options, only two of which are possible at the same time.

TRILINEAR: Consisting of, bounded by or relating to three lines. Trilinear coordinates $x:y:z$ of a point relative to a given triangle describe the relative directed distances from the three sidelines of the triangle. Trilinear coordinates are examples of homogeneous coordinates.

TRILLION: A number, which is equal to 1 followed by 12 zeros, written as 1 000 000 000 000 (10^{12}).

TRILOBITE: An extinct marine arthropod with a flat oval body and a dorsal exoskeleton divided into three vertical sections. It belongs to the class Trilobita.

TRIM: In statistics, it is to discard extreme observations in a sample; the trimmed is regarded as characterising the 'typical' values of the variable.

TRIMEROPHYTES: A group of extinct land plants that flourished during the latter part of the Devonian period. It is believed to be the ancestral group from which both the ferns and seed plants evolved.

TRIMEROUS (3-MEROUS): A term used to describe flowers in which the parts of each whorl are inserted in threes or multiples of three. This arrangement is characteristic of many monocotyledons.

TRIMETHYLALUMINIUM, TMA (ALUMINIUM TRIMETHYL): A colourless liquid with the chemical formula $\text{Al}_2(\text{CH}_3)_3$, density 0.752 g/mL , melting point 15°C (288 K) and boiling point 125°C (398 K). It ignites in air and reacts with water to give aluminium hydroxide and methane, usually with extreme vigour. It is prepared via a two-step process as follows: $2\text{Al} + 6\text{CH}_3\text{Cl} + 6\text{Na} \rightarrow \text{Al}_2(\text{CH}_3)_6 + 6\text{NaCl}$. TMA is mainly used for the production of methylaluminoxane, an activator for Ziegler-Natta catalysts for olefin polymerisation. TMA is also employed as a methylation agent; used for the preparation of Tebbe's reagent, which is used for the methylation of esters and ketones. It is also used in semiconductor fabrication to grow thin film, high-k dielectrics such as Al_2O_3 via the processes of Chemical Vapour Deposition or Atomic Layer Deposition.

TRIMETHYLAMINE: An organic compound with the chemical formula $\text{N}(\text{CH}_3)_3$. It is a colourless, hygroscopic and flammable tertiary amine with a strong fishy odour in low concentrations and an ammonia-like odour at higher concentrations. It acts as an osmolyte in the cells of the kidney. This compound can also be converted to trimethylamine oxide, which is excreted by many marine fish giving them their characteristic odour.

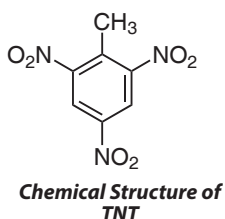
TRIMONOECIOUS: Having male, female and bisexual flowers on the same plant.

TRIMORPHIC: 1. In biology, it refers to having or occurring in three differing forms. 2. In chemistry, it refers to crystallising in three distinct forms.

TRIMORPHIC NUMBER: A number n is called **trimorphic** if n^3 ends in n . An example is 49, since $49^3 = 117649$. Not all trimorphic numbers are automorphic. A number n is called **tri-automorphic** if $3n^2$ ends in n ; for example 667 is tri-automorphic because $3 \times 667^2 = 1334667$, ends in 667. A number n whose square ends in n is called **automorphic**. For example, 5 is automorphic, because $5^2 = 25$, which ends in 5.

TRINERVED (TRINERVATE): Three-nerved, with the nerves all arising from near the base, as shown in the diagram. a trinerved or three-nerved leaf, has

TRINITROTOLUENE (TNT; 2,4,6-TRINITROTOLUENE; TRILITE; TOLITE; TRINOL; TROTYL): A yellow highly explosive crystalline solid with the chemical formula $\text{CH}_3\text{C}_6\text{H}_2(\text{NO}_2)_3$. It is made by nitrating toluene (methylbenzene). It is an explosive material with convenient handling properties unaffected by ordinary shocks and jarring and must be set off by a detonator. It is sometimes used as a reagent in chemical synthesis and also used to generate charge transfer salts. The explosive yield of TNT is considered to be the standard measure of strength of bombs and other explosives.



TRINOMIAL: 1. Consisting of three mathematical expressions or terms or a polynomial with only three terms. Examples are $3a + 2b - 5c$ and $2x^2 + 5x - 3$. 2. Consisting of or relating to three taxonomic names, denoting the genus, species and subspecies or variety of an organism. An example is *Brassica oleracea botrytis*, the cauliflower.

TRIODE: A thermionic valve with three electrodes or an electronic amplification device having three active electrodes. The heated cathode produces electrons which flow to the anode after passing through the negatively biased control grid. Small voltage fluctuations superimposed on the grid bias lead to large fluctuations in the anode current. The triode is thus the first electronic device capable of amplification. It is used in radio-frequency transmitters, producing more than 1kW in power.

TRIOECIOUS: Having male, female and bisexual flowers on different plants.

TRIOLE (TRIHYDRIC ALCOHOL): An alcohol which has three hydroxyl groups per molecule; or a chemical compound such as glycerol containing three hydroxyl groups (-OH functional groups). An example of triol on benzene is benzenetriol.

TRIOSE: A sugar molecule or monosaccharide or simple sugar which has three carbon atoms. There are only three possible trioses: **L-Glyceraldehyde** and **D-Glyceraldehyde**, both aldotrioses because the carbonyl group is at the end of the chain; and **dihydroxyacetone**, which is a ketotriose because the carbonyl group is in the middle of the chain. Trioses are important in cellular respiration. During glycolysis, Fructose-1,6-diphosphate is broken down into glyceraldehyde-3-phosphate and dihydroxyacetone phosphate. Lactic acid and pyruvic acid are later derived from these molecules.

TRIOXIDANE (DIHYDROGEN TRIOXIDE; DIHYDROXY ETHER; HYDROGEN TRIOXIDE; μ -TRIOXIDANEDIIDODIHYDROGEN): A relatively stable inorganic compound of hydrogen and oxygen with the molecular formula H_2O_3 present in the atmosphere and possibly also in living tissue. It was first synthesised in 1994 and is produced in a reaction similar to that used for the commercial production of hydrogen peroxide (H_2O_2) but ozone (O_3) is used instead of oxygen. Hydrogen trioxide is stable at low temperatures but begins to decompose slowly at -40°C forming the high-energy form of oxygen, singlet oxygen.

TRIOXOCARBONATE(IV) ACID: The International Union of Pure and Applied Chemistry's (IUPAC) name for **carbonic acid**. It is a weak carbonic acid and an inorganic compound with the formula H_2CO_3 . It is also a name sometimes given to solutions of carbon dioxide in water, because such solutions contain small amounts of H_2CO_3 . Carbonic acid forms two kinds of salts: the carbonates and the bicarbonates. It is an intermediate step in the transport of CO_2 out of the body via respiratory gas exchange. The hydration reaction of CO_2 is generally very slow in the absence of a catalyst, but red blood cells contain carbonic anhydrase, which both increases the reaction rate and dissociates a hydrogen ion (H^+) from the resulting carbonic acid,



Tripartite leaf

leaving bicarbonate (HCO_3^-) dissolved in the blood plasma. This catalysed reaction is reversed in the lungs, where it converts the bicarbonate back into CO_2 and allows it to be expelled. This equilibration plays an important role as a buffer in mammalian blood.

TRIPALMATE: Palmately compound three times.

TRIPARTITE: 1. Divided into three parts. 2. In botany, of a leaf which is divided into three parts down to the base, but not wholly separate.

TRIPLE: The pickled rumen or the first three stomachs of ruminants such as cattle and sheep. It is cleaned and treated with lime and marketed for consumption as food. It contains a large amount of connective tissue which forms gelatin on boiling. It is a good source of calcium, protein, niacin, iron, zinc and copper.

TRIPETALOUS: Having three petals.

TRIPHENYLMETHYL RADICAL: An unusual aromatic compound with the chemical formula $(\text{C}_6\text{H}_5)_3\text{C}$. It can exist as an anion, cation or free radical. It can be prepared by homolysis of triphenylmethyl chloride by a metal like silver or zinc in benzene or diethyl ether.

TRIPHYLLOUS: Having three leaves.

TRIPINNATE: Of a bipinnate leaf, with the secondary leaflets further subdivided, as occurs in certain ferns.

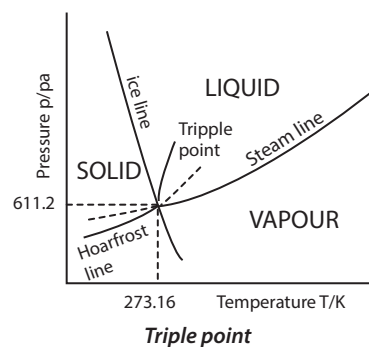
TRIPINNATIFID: Thrice pinnately cleft; said of a pinnatifid leaf when its segments are pinnatifid and the subdivisions of these also are pinnatifid.

TRIPLE: A set sequence with three members.

TRIPLE BEAM BALANCE: An instrument used to measure the mass of an object accurately with a reading error of ± 0.05 gram. It consists of a beam which is the largest size supported on a fulcrum. On one side of this beam, is a pan on which the object to be measured is placed, while on the other side, the beam is split into three parallel beams, each carrying a known weight and together culminating into a pointer pointing to a fixed scale. The weights are slid on their respective beams until zero reading is obtained and a state of balance is achieved.

TRIPLE BLIND STUDY: An extension of the double blind study in which the identities of those enrolled in the study are unknown to the research participant, the individuals who administer the treatment or intervention, and the researchers who analyse the outcomes.

TRIPLE POINT: The temperature and pressure at which the vapour, liquid and solid phases of a substance are in equilibrium or the particular temperature and pressure at which three different phases of one substance can coexist in equilibrium. The triple point for water occurs at 273.16K and 611.2Pa, as shown in the diagram. This value forms the basis of the definition of the Kelvin and the thermodynamic temperature scale.



TRIPLE PRODUCT: A scalar product or a vector product each with three components. A scalar triple product of three vectors a , b and c written as $[a, b, c]$ is obtained by multiplying the three vectors a , b and c in the manner $a \cdot (b \times c)$. The result is a scalar product because,

just like the dot product, it evaluates to a single number. A vector triple product is, however, obtained by multiplying three vectors a , b and c in the manner $a \times (b \times c)$; the result is a vector. It is also equal to $(a \cdot c)b - (a \cdot b)c$. Note that $a \times (b \times c)$ is not equal to $(a \times b) \times c$.

TRIPLE RESPONSE: 1. The vascular changes in the skin in response to mild mechanical injury, producing redness, heat and swelling, which are three of the cardinal signs of inflammation. 2. In botany, it is the three characteristic morphological changes on an etiolated seedling (a yellowed seedling, which lacks chlorophyll and is elongated and spindly as a consequence of being grown in the absence of light) in response to the presence of ethylene. Its effects on etiolated seedling development include the following: (i) increased radial growth of the root and stem, thereby increasing in thickness; (ii) reduced root and stem elongation; and (iii) and reduced abnormal horizontal stem growth.

TRIPLE VACCINE: Vaccine for infants to protect them against diphtheria, tetanus and whooping cough. The vaccine contains a mixture of diphtheria toxoid, tetanus toxoid and pertussis vaccine.

TRIPLE VECTOR PRODUCT (VECTOR TRIPLE PRODUCT): The product of three vectors in three-space, written in the form

$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$; this is not equal to $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$.

TRIPLET: 1. An atomic state in which two spin angular momenta of electrons combine to give a total nonzero spin. A triplet state usually has a lower energy than a singlet state because of the effect of correlations between electrons. 2. Each of three children or animals that are delivered by the same mother during one birth.

TRIPLET CODE: A set of three-nitrogenous base sequences that specify the amino acids for polypeptide chains in nucleic acids. Each triplet, a group of three bases, codes for a specific amino acid. The triplet of bases on the DNA and mRNA is known as a codon; and the triplet of bases on the tRNA is known as an anti-codon.

TRIPLET STATE: A state having a total electron spin quantum number of 1.

TRIPLET-TRIPLET ANNIHILATION: It corresponds to the energy transfer process, whereby two excited molecules in the lowest triplet state yield, upon close proximity, a ground state molecule and an excited molecule in the singlet state

TRIPLET-TRIPLET ENERGY TRANSFER (TTET): Energy transfer from an electronically excited triplet donor to produce an electronically excited acceptor in its triplet state.

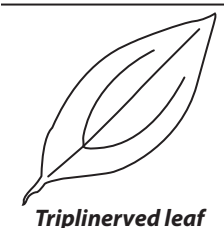
TRIPLET-TRIPLET TRANSITIONS: Electronic transitions in which both the initial and final states are triplet states.

TRIPLINERVED: Three-nerved, with the two lateral nerves arising from the midnerve above the base, as shown in the diagram.

TRIPLOBLASTIC: An animal having a body composed of three embryonic cell layers: the ectoderm, mesoderm and endoderm. Most multicellular animals are triploblastic

except the Porifera, Cnidaria and Ctenophora. They do not exist in porifera, cnidaria and ctenophora. In animals the early embryo acquires fundamental cell layers through the process of gastrulation. These cell layers are called **germ layers**, giving rise to different organs of body during organogenesis.

TRIPLOID: Possessing three representations of each chromosome. It refers to a nucleus cell or organism that has three times a haploid number of chromosomes. Triploid organisms are normally sterile as they lack homologous chromosomes, preventing pairing during meiosis.



TRIPOD: A frame or stand with three legs that are usually collapsible and used for supporting things such as a camera, compass, theodolite, stool, table or other equipment.

TRIPRISMO-: An affix used in names to denote six atoms bound into a triangular prism.

TRIPROTIC ACID: Each unit of acid that yields three protons upon ionisation or an acid that has three ionisable hydrogen atoms in each molecule. An example is orthophosphoric acid.

TRIPTAN: The general term for a family of drugs used for the treatment of migraine attacks.

TRIPTEROUS: Having three wings or winglike appendages.

TRIQUETROUS: Three-edged; with three protruding angles.

TRIRADIATE SCAR: In botany, it is a scar on the surface of a spore that marks the point at which it was joined to the other three spores making up the tetrad. Triradiate scars are seen when the tetrad shows tetrahedral symmetry. It is conspicuous because the spore wall is fairly thin at this point, while the rest of the wall is thickened with cutin.

TRISECT: To divide into three equal parts.

TRISECTING THE ANGLE: The problem of how to construct an angle that is equal to one-third of a given arbitrary angle using only straight-edge and compasses. It is possible with the aid of a protractor or using the limaçon of Pascal or a trisectrix.

TRISECTRIX OF MACLAURIN: A curve which makes angle trisections possible, defined as the locus of the point of intersection of two lines, each rotating at a uniform rate about separate points, so that the ratio of the rates of rotation is 1:3 and the lines initially coincide with the line between the two points. Its equation is $x^3 + xy^2 + ay^2 - 3ax^2 = 0$, which is symmetric around the x -axis, contains the origin and has a vertical asymptote at $x = -a$.

TRISMUS (LOCKJAW): A symptom of tetanus in which spasm of the jaw muscles or muscles of mastication occurs and also inability to open the jaws.

TRISOCTAHEDRON: A solid figure such as a crystal having 24 identical triangular faces, in groups of three, each group corresponding to one face of an octahedron.

TRISODIUM PHOSPHATE(V) (SODIUM ORTHOPHOSPHATE): A colourless crystalline compound with the chemical formula Na_3PO_4 . It is soluble in water and insoluble in ethanol. It is known both as the decahydrate and the dodecahydrate. It may be prepared by boiling sodium carbonate with the stoichiometric amount of phosphoric acid and subsequently adding sodium hydroxide to the disodium formed.

TRISOMY: The condition of a nucleus cell or organism in which one of the pairs of homologous chromosomes has gained an additional chromosome, resulting in a chromosome number of $2n+1$. The additional copy of a chromosome means that there are three copies, not two in a diploid organism. An example is Trisomy 21 in Down's syndrome. It is the cause of a number of human gene abnormalities including the Down's syndrome, Patau's syndrome and Edward's syndrome.

TRISPERMOUS: In botany, three-seeded or containing three seeds.

TRISTICHIOUS: In botany, arranged in three rows on a stem. It is a form of alternate leaf arrangement in which successive leaves arise 120° around the circumference of the stem giving three vertical rows of leaves.

TRISTYLOUS: With three styles.

TRISULCATE: With three furrows or grooves.

TRITIATED COMPOUND: A compound in which a hydrogen atom in each molecule of the compound has been replaced by a tritium atom. It is a process of replacing a stable atom in a compound with a radioisotope of the same element to enable its path through a biological or mechanical system to be traced by the radiation it emits. In some cases a different stable isotope is used and the path is detected by means of a mass spectrometer.

TRITIUM: A radioactive isotope of hydrogen containing two neutrons and one proton in its nucleus. Its symbol is T or ^3H . It occurs naturally in trace amounts; and is a beta (β) - emitter. It has atomic mass of 3 and a half-life of 12.3 years. Although rare in nature, it can be produced artificially and is used in tracers and hydrogen bombs.

TRITON: The largest of Neptune's moons, which orbits Neptune at an average distance of 354,800 km. It rotates once every 5.877 Earth days and completes one orbit in the same amount of time, which means that one side of the moon faces Neptune at all times, while the other faces out into space. With a radius of 1,353 km, it is one of the ten largest moons in the solar system. Based on the moon's density, scientists estimate that Triton is about one-quarter ice and three-quarters rock.

TRIVALENT: In botany, it is the temporary association of three homologous chromosomes in a triploid or trisomic organism. Such associations are observed between mid prophase and late metaphase of the first meiotic division.

TRIVIAL: In mathematics, a trivial solution is a solution of a system of equations that sets the value of all the variables at zero. For example, the equation $x + 5y = 0$ has the trivial solution $x = 0, y = 0$. Nonzero solutions or examples are considered nontrivial.

TRIVIAL FILE TRANSFER PROTOCOL (TFTP): A simple file transfer protocol similar to FTP that is used to for transferring files between network devices or load configuration files to network devices such as routers.

TRIVIAL NAME: 1. A common or vernacular name having no part used in a systematic sense. For example, the trivial name for *Pan troglodytes* is chimpanzee. 2. A common, historic or convenient name for a substance, derived often from the source in which the substance was discovered, but unsystematic and not used in modern official nomenclature. For example, the trivial name for salicylic acid is aspirin. 3. Also known as **specific epithet**, it is the Latin name that follows a capitalised genus name in binomial nomenclature and serves to distinguish a species from others in the same genus. For example, in the name *Nicotiana tabacum*, commonly known as tobacco, the trivial name or specific epithet is *tabacum*.

TRIVIAL RING (ZERO RING): A ring, denoted $\{0\}$ or 0 consisting of only a single element with the ring operations multiplication ($\cdot$) and addition ($+$) defined so that $0 + 0 = 0$ and $0 \cdot 0 = 0$. It is a commutative ring with an identity.

TRIVIAL SUBGROUP: The subgroup of a given group that consists of only the identity element.

TROCHANTER: 1. The second segment of an insect's leg, between the coxa and the femur. 2. Any of several bony knobs on the femur of vertebrates to which muscles are attached.

TROCHLEAR NERVE: The fourth cranial nerve and the only one that arises from the back of the brain stem. Each of the two trochlear nerves controls just one of the small muscles (ocular muscles) which moves each eyeball and is purely motor. It follows the longest course within the skull of any of the cranial nerves. Paralysis results in rotation of the eyeball upward and outward and, therefore, in double vision.

TROCHOID: The curve formed by the path of a point on the extension of a radius of a circle as it rolls along a curve or line, derived from the cycloid. It is also the curve formed by the path of a point on a perpendicular to a straight line as the straight line rolls along the convex side of a base curve, derived from the involute.

TROCHOPHORE: The pelagic planktonic larvae of polychaete worms, some molluscs and certain other invertebrates. It is top-shaped and usually has two bands of cilia encircling the body.

TROJAN (TROJAN HORSE): In computer science, it is a non-replicating program or utility, which is usually malicious but contained inside apparently harmless and useful programs or data, but in

reality is performing background tasks such as giving access to the computer or sending personal information to other computers. For example, a Trojan horse might appear to be a computer game, but when it is opened, the program starts writing over certain parts of the hard drive, corrupting data. Trojan horses are one of the most common methods a criminal uses to infect a user's computer and collect personal information.

TROMINO: A three-square polyomino or one of the two plane figures that can be formed by joining three unit squares along their sides.

TRONA: A mineral form of sodium sesquicarbonate with the chemical formula $\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}$. It consists of a mixed hydrated sodium carbonate and sodium hydrogen carbonate. It is used as a source of sodium compounds.

TROPANE ALKALOIDS: A group of alkaloids derived from the amino acid ornithine. They include hyoscyamine and its stereoisomer atropine, both solanaceous alkaloids. These are prepared from *Datura stramonium* (thorn apple) and *Atropa belladonna* (deadly nightshade), respectively. The cocaine alkaloids, derived from the leaves of *Erythroxylum coca* and *E. novagranatense* (coca), are also examples.

TROPHIC LEVEL: The position that an organism occupies in a food chain or an individual layer on the pyramid of numbers which represent types of organisms living at parallel levels on food chains. It is a stage in a food chain that reflects the number of times energy has been transferred through feeding. For example, plants are eaten by animals that are in turn eaten by predators. Plants and plant-eating animals occupy the first two levels, followed by carnivores, usually to a maximum of six levels. Many animals feed at different trophic levels.

TROPHOBLAST: A thin outer layer of ectoderm that encloses the embryo of mammals attaching the fertilised ovum to the wall of the womb and absorbing nutrients. They are formed during the first stage of pregnancy and are the first cells to differentiate from the fertilised egg.

TROPIC HORMONE: Hormone made by one gland that causes another gland to secrete a hormone; or hormones produced and secreted by the anterior pituitary that target endocrine glands. They include thyroid-stimulating hormone (TSH or thyrotrophin), adrenocorticotrophic hormone (ACTH or corticotropin), luteinising hormone (LH) and follicle-stimulating hormone (FSH).

TROPICAL CYCLONE: A cyclone that develops over tropical or subtropical waters and at least 5° of latitude away from the equator, in which sea temperatures are above 27°C (300K). This is a disturbance about 650 km across, spinning about a central area of very low pressure, with winds over 140km per hour. The violent winds are accompanied by towering clouds, some 4000m high and by torrential rain, with about 150mm frequently falling within a few hours.

TROPICAL DEPRESSION: Strong low-pressure system formed in low latitudes. It forms when a low pressure area is accompanied by thunderstorms that produce a circular wind flow with maximum sustained winds below 62 km/h. Tropical depression can cause flooding rainfall along its path, especially if the system is slow-moving.

TROPICAL RAIN FOREST: The most complex of all communities, located near the equator with temperatures of between 21°C and 30°C high humidity and annual rainfall of between 200-1000 centimetres. Tropical rainforests grow in climatic conditions that are least limiting to plant growth and the growth and productivity (total amount of organic matter produced per unit area per unit time) of tropical rainforests is higher than that of other vegetation, provided that other factors such as soil fertility or consumption by herbivorous animals are not extremely low or high. It also contains more species of plants and animals than all other terrestrial biomes combined. The Amazon rain forest is the world's largest tropical rain forest. It is home to around 40,000 plant species, nearly 1,300 bird species, 3,000 types of fish, 427 species of mammals, and 2.5 million different insects.

TROPICAL STORM: An extreme low pressure system with sustained wind range of 63 to 118 km but less than hurricane intensity. Tropical storms are also known as hurricanes in the Atlantic, tropical cyclones in South East Asia and typhoons in the Pacific regions.

TROPICS (TROPICAL ZONE): The area between the tropics of Cancer and Capricorn, defined by the parallel of latitudes. It is located between the Tropic of Cancer in the Northern Hemisphere at approximately 23° 27' North and the Tropic of Capricorn in the Southern Hemisphere at 23° 27' South. The tropics include all the areas on the Earth where the Sun reaches a point directly overhead and a point directly underneath at least once during the solar year. Outside the tropics, the Sun never reaches a point directly overhead or a point directly underneath at any time during the year.

TROPISM: The directional growth of plant organs in response to an external stimulus, such as light, touch or gravity. Growth towards the stimulus is a positive tropism and growth away from the stimulus is negative tropism. The different types of tropism include geotropism, hydrotropism, phototropism orthotropism, thigmotropism and plagiotropism.

TROPOMYOSIN: An actin-binding protein, found in muscles whose molecules consist of two elongated strands that run along the length of the filament. When the muscle is at rest, the tropomyosin molecules cover the binding sites of the actin molecule, where interaction with myosin occurs. When the muscle contracts, the tropomyosin is displaced by another protein, called troponin, which allows the interaction of actin with myosin.

TROPONIN: A complex three polypeptide chain that is found at regular intervals along the length of an actin filament. It binds to calcium ions during muscle contraction and hence displaces tropomyosin and exposes the binding site on the actin filament. This enables the interaction of actin and myosin to take place.

TROPOPAUSE: The region of the atmosphere which joins the troposphere and stratosphere and where the decreasing temperature with altitude, characteristic of the troposphere ceases and the temperature increase with height which is characteristic of the stratosphere begins. Its height is about 15-17 km over the tropics and 10 km nearer the poles, but it varies seasonally and with changes in weather

TROPOPHYTE: A plant that is adapted to survive in a climate, where there are alternating wet and dry seasons, by having a resting phase during the dry season. For example, many trees in monsoon forests shed their leaves in the dry season.

TROPOSPHERE: The lowest level of the Earth's atmosphere, extending from the Earth's surface to a height of about 15 km. It contains about 80% of the total mass of air and practically all of the atmosphere's water vapour. Within the troposphere, temperature falls with increasing height, although temperature inversions can occur.

TROPYL RADICALS: The delocalised radicals derived formally by abstraction of one hydrogen from the CH₂ group of cyclohepta-1,3,5-triene and substitution derivatives thereof.

TROPYLUMION: The positive ion having a ring of seven carbon atoms with the chemical formula C₇H₇⁺. The ion is symmetrical and has characteristic properties of aromatic compounds.

TROUGH: 1. A long open, usually shallow, box for animals to feed or drink from. 2. A hollow between two waves. It is the point on the medium that exhibits the maximum amount of negative or downward displacement from the rest position. 3. A region of lower atmospheric pressure between two regions of higher pressure. 4. A long open box in which a baker kneads dough for bread, etc.

TROWEL (HAND TROWELL): A small hand-held implement or tool, often with a handle and a flat blade tapering to a tip that is used in construction to apply, shape and smoothen mortar or plaster. Some are designed as a shovel with curved tapering, pointed blade and used as a hand tool for digging the soil and lifting loose materials.

TRUE: 1. One of the truth-values assigned to a sentence in any two-valued logic. Alternatively, the unique designated truth-value in a many-valued logic. 2. Of a proposition in a mathematical theory, validly deducible from the axioms of the theory.

TRUE-BREEDING: An organism having certain biological traits which are passed on to all subsequent generations when bred with another true breeding organism for the same traits. This occurs when self-fertilisation gives rise to the same traits in all offspring, generation after generation, which is interpreted as equivalent to homozygous.

TRUE NORTH: The direction from any point along a meridian towards Geographic North Pole along an imaginary line at an angle of 90° to the equator. It is calculated by using an imaginary line through the Earth rather than by using a compass.

TRUFFLES (TUBERALES): An order of the Discomycetes in which the ascocarps are formed underground and resemble tubers. It contains about 35 genera and 230 or so species, many of which are edible. Certain species are often found under particular trees, for example, *Tuber melanosporum* (the European truffle), which grows under oaks. They are thus thought to be mycorrhizal.

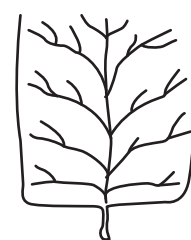
TRULLATE: A 2-dimensional shape; resembling a trowel blade, with 4 straight sides with the axis broadest below the middle and a length and breadth ratio of between 3:2 and 2:1.

TRUMPET CELL: An elongated cell, with most of them joined end to end to transport photosynthate or photosynthetic substances through the bodies of brown algae. The end walls are perforated with holes forming a sieve, similar to the arrangement of sieve elements in phloem.

TRUNCATABLE PRIME: A prime number n that remains a prime when digits are deleted from it one at a time. For example, 410256793 is a truncatable prime because each number created by the removal of the digit boldened produces a new prime: 41**0**256793, 41256**7**93, 412567**3**, 41**5**673, 456**7**3, 4**5**67, 4**6**7, 6**7**, 7. It is conjectured that there are infinitely many of these primes. If the digits from a prime can be deleted only from the right to leave a prime, then n is called a **right truncatable prime**. If they can be deleted only from the left to leave a prime, then n is a **left truncatable prime**.

TRUNCATION: 1. In botany, it refers to an abruptly transversed end as if cut off; for example, a truncated leaf is one in which the base is squared at the end, as if cut off, as shown in the diagram. 2. In mathematics, to truncate a number is to approximate a decimal number by dropping all decimal places past a certain point without rounding. 3. A truncated solid figure is one formed by cutting through a cone or prism with two planes or a geometric figure that has the apex or an end removed and replaced with a plane section, often parallel to the base. 4. Of a series, constituting a finite initial segment of an infinite series to which it approximates; in particular, a truncated decimal fraction is a finite initial segment of the terms of a non-terminating decimal. 5. In computer science, the term is used in reference to data types or variables, such as floating point numbers and strings. When information is truncated, it is ended abruptly at a certain spot. For example, if a program truncates a floating-point number 3.14159265 at four decimal places it becomes 3.14. This is not the same as rounding, as rounding would have 3.1416 as the answer. Truncation always rounds the number down. 6. Truncated crystal is a one that lacks the fully formed corners or faces that would be present in a simple form of the crystal.

TRUNCATION ERROR: In mathematics, it is the difference between a truncated value and the actual value, for example, the truncating error of reducing 99.987 to 99.98 is 0.007. In computing, truncation error is the discrepancy that arises from executing a finite number



Truncated leaf

of steps to approximate an infinite process. If not properly executed, truncation error can become serious mistakes in computer calculations involving very large mathematical operations.

TRUNK: 1. The main stem of a tree. 2. The body of a human or animal excluding the head and limbs. 3. The thorax of an insect. 4. A proboscis, especially the long prehensile proboscis of an elephant. 5. A main body, apart from tributaries or appendages, for example, the main stem of a blood vessel or nerve apart from the branches. 6. In computer science, it refers to a data line with large bandwidth capabilities that connects two locations.

TRUST-REGION METHOD: A class of approximate descent methods that replaces the step-length method computation by an estimation of the positive definiteness of the approximate Hessian. If this is found 'trustworthy', a step size of 1 is used, as in Newton's method; otherwise a search direction is generated, based on a measure of the trustworthiness itself updated through the calculation.

TRUTH: Deducibility of a proposition from the axioms of a theory in accordance with the rules of inference of that theory. Truth can be proved logically by theorems and formulae and logic paradigms and conforms and affirms reality and/or logic. However, there are truths that are not logical by nature; for example, a statement such as "It is true that I love my wife" cannot be proved logically, but it is truth.

TRUTH SET: A set of all the values that make a given logic or mathematical statement true when substituted in the statement.

TRUTH TABLE: 1. A table that summarises all possible outcomes of a logical operation. It is used to work out the truth or falsity of a compound statement in logic. 2. A table used to indicate the value of the output signal from a logic circuit or a device for every possible input. It is used in electronics and computing.

TRUTH-FUNCTION:

In logic, it is a connective such as conjunction, negation, disjunction and implication that determines the truth-value of a complex sentence solely in terms of the truth-values of the component sentences rather than their meanings.

TRUTH-VALUE: 1. In logic, either of the two possible values, true and false that may be assigned to a proposition or statement. 2. Any of the values that a semantic theory may be assigned to a statement.

TRUTH-VALUE GAP: In logic, the possibility in certain semantic systems of a statement being neither true nor false while not being determinately of any third truth value, so that the law of excluded middle fails. An example is a childless woman saying "all my children are asleep."

TRYMA: A drupelike nut such as a walnut or hickory nut with a fleshy exocarp that becomes tough and dry and eventually splits open.

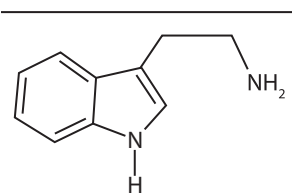
TRYPANOSOME: Any of various parasitic flagellate protozoans of the genus *Trypanosoma*, transmitted to the vertebrate bloodstream, lymph and spinal fluid by certain insects and often causing diseases

such as sleeping sickness and Chagas disease also known as American trypanosomiasis. Sleeping sickness is caused by *Trypanosoma brucei* and Chagas disease is caused by *Trypanosoma cruzi*.

TRYPANOSOMIASIS (SLEEPING SICKNESS): A disease caused by infestation by *Trypanosoma brucei rhodesiense* and *Trypanosoma brucei gambiense*, microscopic organisms that live as parasites in the blood and transmitted in the salivary glands of infected tsetse flies (*Glossina*). African sleeping sickness begins with a chancre at the site of the insect bite, an accelerated heartbeat, an enlargement of the spleen and rash and fever. Over the next few months the nervous system is attacked, with accompanying mood changes, sleepiness, lack of appetite, eventual coma and, frequently, death. Chagas disease also known as American trypanosomiasis, is a potentially life-threatening illness caused by the protozoan parasite, *Trypanosoma cruzi* (*T. cruzi*), which attacks children more frequently. It causes fever and damage to the spleen and nervous system, as well as to the liver and the heart muscles.

TRYPSIN: 1. An enzyme that digests proteins. It is secreted in an inactive form (trypsinogen) by the pancreas into the duodenum where it is acted on by an enzyme (enterokinase) produced in the duodenum. The active enzyme plays a role in the digestion of proteins in the anterior portion of the small intestine and also activates proteases in the pancreatic juice. 2. A proteolytic enzyme that cleaves peptide chains next to the basic amino acids arginine and lysine.

TRYPTAMINE: A naturally occurring alkaloid, having an indole ring system with a $-\text{CH}_2-\text{CH}_2-\text{NH}_2$ side chain in the 3-position of the nitrogen-containing ring. Its chemical formula is $\text{C}_{10}\text{H}_{12}\text{N}_2$.



Chemical Structure of Tryptamine

TRYPTIC PEPTIDE MAPPING:

The technique of generating a chromatographic profile, characteristic of the fragments resulting from trypsin enzyme cleavage of the protein.

TSETSEFLY: Any of about 21 species (genus *Glossina*, family Muscidae) of African blood sucking dipterans that are robust, sparsely bristled and usually larger than a housefly, found mostly in tropical Africa. It carries and transmits diseases to humans and animals. The two species that commonly transmit the protozoan parasites (trypanosomes) that cause human sleeping sickness are *Glossina palpalis* and *Glossina morsitans*.

TSUNAMI (TIDAL WAVE): A large seawave which is often generated by a underwater earthquake or volcanic eruption. It may be caused by underwater mudslide, earthquake, volcanic eruption or coastal landslide which generates waves over 10 m high, often travelling for hundreds of kilometres with speeds of up to 965 km/h.

TUB CONFORMATION: A conformation (of symmetry group) of an eight-membered ring in which the four atoms forming one pair of diametrically opposite bonds in the ring lie in one plane and all other ring atoms lie to one side of that plane. It is analogous to the boat conformation of cyclohexane.

TUBAL LIGATION: A permanent voluntary form of birth control (contraception) in which a woman's Fallopian tubes are surgically cut or blocked off to prevent pregnancy. It is usually performed using endoscopic surgery.

TUBAL PREGNANCY: A pregnancy that begins with implantation of the embryo in one of the fallopian tubes instead of in the uterus. Tubal pregnancy is usually caused by a defect in the tube that prevents the fertilised ovum from moving freely through it.

A	B	OUTPUT
1	0	0
0	1	0
0	0	0
1	1	1

Fig I: A Simple Truth Table

AND gate			OR gate		
INPUT A	INPUT B	OUTPUT	INPUT A	INPUT B	OUTPUT
0	0	0	0	0	0
1	0	0	1	0	1
0	1	0	0	1	1
1	1	1	1	1	1

Fig II:

Logic Gates Truth Tables showing AND Gate and OR Gate

TUBE: 1. A long hollow cylinder of metal, glass or rubber especially for holding or conveying liquids. 2. A hollow cylindrical organ that conveys liquids or gases around the body. 3. A narrow enclosed channel in a plant, for example, the organ in a germinating pollen grain that conveys the male gametes to the ovule or a roughly cylindrical fusion of the petals of a flower such as a daffodil.

TUBE NUCLEUS: One of the two nuclei produced when the haploid nucleus of a pollen grain divides by mitosis. The nucleus is believed to regulate growth of the pollen tube.

TUBE-WITHIN-A-TUBE SYSTEM: A type of body plan in which an animal has two openings, one for food and one for the elimination of waste and a specialised digestive system. Such animals are 10% more efficient at digesting and absorbing their food than animals with the sac-like body plan. The tube-within-a-tube plan allows specialisation of parts along the tube. The sac-like body plan has only one opening for both food intake and waste removal. Sac-like body plan animals do not have tissue specialisation or development of organs.

TUBER: A swollen underground stem or root of plants such as potato and yams characterised by buds or eyes usually adapted for storing food. It enables the plant to survive the dry season or winter and also a means of propagation. **Tuberoid** is a thickened root which resembles a tuber.

TUBERCLE: A small tuberlike swelling or projection.

TUBERCULATE (VERRUCOSE): Having a surface covered with small warty projections as, for example, the rind of ridge cucumbers. Pollen covered in wartlike structures (verrucae), for example, that of ivy (*Hedera helix*), is normally described as **verrucate**.

TUBERCULE: 1. A small rounded wartlike projecting part or outgrowth of the roots of some legumes. 2. A small rounded nodule or swelling, especially a mass of lymphocytes and epithelioid cells forming the characteristic lesion of tuberculosis. 3. A nodule or small eminence, especially one on a bone for attachment of a tendon.

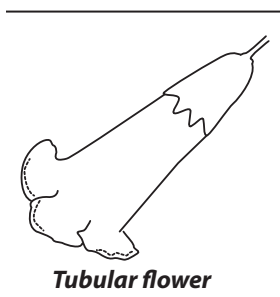
TUBERCULOSIS (TB): A wasting disease affecting various parts of the body's tissues such as kidneys, bones, lymph nodes, brain and especially the lungs. Pulmonary tuberculosis affects the lungs. It is caused by a bacterial microorganism, the tubercle bacillus, *Mycobacterium tuberculosis*. TB is transmitted from person to person, usually by inhaling bacteria-carrying air droplets. When a person sick with TB coughs, sneezes or speaks, small particles that carry two to three bacteria surrounded by a layer of moisture are released in the air, which may be inhaled by a healthy person and may develop into tuberculosis. Symptoms of TB include coughing, chest pain, shortness of breath, loss of appetite, weight loss, fever, chills and fatigue. Children and people with weakened immune systems are the most susceptible to TB. Half of all untreated TB cases are fatal.

TUBEROSITY: 1. A rounded protuberance, especially at a point on a bone where ligaments or muscles are attached. 2. The quality or condition of being tuberous.

TUBICOLOUS: Invertebrate animals that live in tubes that they have themselves constructed. For example, the polychaete fan worms build tubes of sand.

TUBULAR: Having the shape of a tube or cylinder. A tubular flower has a long, thin, straight-sided tube formed of united petals, often separating at the mouth into a flared shape.

TUBULAR REABSORPTION: A process in which waste products are reabsorbed incompletely so that they are mostly excreted in the urine, while useful products are reabsorbed completely so that they are mostly not excreted.



Tubular flower

TUBULAR SECRETION: The process in which ions and other waste products are transported into the distal tubules of the nephron. For example, H^+ and K^+ ions, are transported from peritubular capillaries into tubular lumen by diffusion or transcellular mediated transport. Some of the movements are coupled to reabsorption of Na^+ ions.

TUBULE: A minute elongated tube that is found in various structures or organs of the body. Examples are found in the kidney and testis.

TUBULIDENTATA: Order of mammals comprising the monospecific family Orycteropidae with the species *Orycteropus afer* (aardvark), an insect-eating mammal native to Africa.

TUBULIFLOUS: Of or referring to plants that have tubular corollas in the perfect flowers of a head, as occurs in certain composite plants. **Tubulous** refers to tubular flowers.

TUBULIN: One of several members of a small family of globular proteins. The most common members of the tubuline family are α -tubulin and β -tubulin, the proteins that make up microtubules.

TUFF: A rock made up of very small volcanic fragments compacted together. During an explosive volcanic eruption, the erupted material, called tephra, may be ejected into the air. Tephra consists of molten lava materials such as pumice, cinders and volcanic ash. When it cools and solidifies, it falls through the air after the eruption to form tuff. If it is composed mainly of glass, it is called vitric; if it is mainly crystal, it is classified as crystal chips; and lithic is if it is composed of the debris of pre-existing rocks.

TUFTED: Arranged in a dense cluster.

TUFTS: 1. A small bunch of grass, hair, feathers or fibres growing or held together at the base. 2. A small clump of trees or plants.

TULLGREN FUNNEL (BERLESE FUNNEL; BERLESE TRAP): In biology, it is an ordinary funnel into which a sample of soil or leaf litter containing living organisms such as insects, mites and other invertebrates are trapped. It works by placing a funnel above a collecting vessel and placing a light on top of the funnel to illuminate the sample that is placed in the collecting vessel. As living organisms in the sample move away from the source of light and heat, they end up falling into the collecting vessel where they are trapped and used for studies.

TUMBLER: 1. A flat-bottomed drinking glass without a handle or stem. 2. A part of a machine that moves or engages a gear. 3. A domestic pigeon that can perform backward somersaults in flight. 4. The part of a lock that must be engaged by a key in order to move the bolt. 5. A lever in a gunlock that forces the hammer forward when a trigger is pressed.

TUMBLR: A social networking platform that allows several different post formats, including quote posts, chat posts, video and photo posts as well as audio posts. It was founded in February 2007 and as in August 2018, has 115,000,000 active monthly users.

TURMERIC (*Curcuma longa*): The herbaceous, perennial plant in the ginger family, Zingiberaceae that grows to 1 metre tall. It is native to South Asia and cultivated in many countries such as parts of India, Nepal, Sri Lanka, Pakistan, Bangladesh, East and West Africa, Java, Peru, Caribbean Islands and Central America. It has a short stem with 7 to 12 leaves that alternate, arranged in two rows. The leaf blades are oblong in diameter and elliptic, narrowing at the tip. The plant produces bisexual flowers which are bilaterally symmetrical. The flowers are both gamosepalous (united sepals) and gamopetalous (united petals). It is cultivated for its cylindrical, many branched, yellow-orange, mildly aromatic rhizome called tumeric. The rhizome has a pungent, peppery, bitter flavour, used to process into a yellow powder, which is used in cooking and gives Indian curry its flavour and yellow colour. It is also used to flavour or colour some foods such as

some canned beverages, ice creams orange juice, yogurt, mustards, butter and cheese. It has been used medicinally to treat many diseases such as stomachache, liver problems, etc; has antioxidant and anti-inflammatory effects. It is used as a cosmetic; and also used by some people in parts of Asia to dye fabrics.

TUMID: Swollen or enlarged. **Tumescent** means somewhat swollen.

TUMORIGENESIS: The mechanism of tumour formation. It is a process by which normal cells are transformed into cancer cells.

TUMOUR: Overproduction of cells in a specific area of the body, often leading to a swelling or a lump or an uncontrolled growth or mass of body cells, which may be malignant and has no physiological function.

TUMOUR NECROSIS FACTOR (TNF): Either of two proteins that act as cytokines, eliciting numerous responses in virtually all types of cells by activating cellular signalling pathways and promoting the expression of gene. They are involved in many aspects of the body immune defence notably in their ability to destroy tumour cells. They assist in killing all infected cells by natural killer cells.

TUMOUR-INDUCING PRINCIPLE: A plasmid carried by the bacterium, *Agrobacterium tumefaciens*, which causes crown gall disease in plants. The plasmid is necessary for the transformation of normal host tissue into tumour tissue. Tissue removed from crown gall tumours can be grown in culture without the addition of auxin and cytokinin.

TUMOUR-SUPPRESSOR GENE (ANTI-ONCOGENE): Any of a class of genes that suppress the development of cancers in living cells. The products of tumour-suppressor genes are typically involved in monitoring replication of DNA and progress of the cell cycle and in promoting the cell of damaged DNA.

TUNABLE LASER: A high intensity source with narrow spectral bandwidth, with and without frequency doubling and/or Raman shifting. Its use may enable the spectral apparatus to be omitted. Tunable lasers may be either continuous (CW) or pulsed in nature.

TUNDRA: A terrestrial biome characterised by lack of trees and a permanently frozen subsoil lying between the ice cap and the timber line of North America, Europe and Asia.

TUNED CIRCUIT (LC CIRCUIT; RESONANT CIRCUIT): A circuit, which consists of an inductor with the symbol L and a capacitor, with the symbol C. When connected together, an electric current can alternate between them at the circuit's resonant frequency. LC circuits are used in tuning of electronic receivers such as a radio to a particular station, for voltage magnification, for current magnification and in induction heating.

TUNER CARD: An internal or external computer device that allows for connecting a cable television signal to a computer. The tuner card typically features a coaxial connection and some also include RCA (yellow, red and white) connections. Newer models also may feature HDMI connections. Internal tuner cards will install in a PCI slot, whereas external tuner cards plug in to a USB port. One of the more well-known brands of tuner cards is Hauppauge and their line of WinTV devices.

TUNGSTEN: A hard, heavy, lustrous, grey-white transition metallic element with the symbol W, atomic number 74, atomic weight 183.84, melting point 3,410 °C (3683 K), boiling point 5,900 °C (6173 K) and relative density of 19.3 (20 °C). It has the highest melting point and lowest vapour pressure of any metal. It is used to make the filaments in electric light bulbs and to make hard steels for cutting tools. It is extracted from wolframite, scheelite and other minerals.

TUNGSTEN CARBIDE: A black powder made by heating powdered tungsten metal with lamp black at 1600 °C (1873 K). It has a density of 15.8 g·cm⁻³ (solid), melting point 2,870 °C (3,143 K) and a boiling point of 6,000 °C (6,273 K). It is extremely

hard (9.5 on Mohs' scale) and is used in cutting tools such as dies, for making wear-resistant machine parts and abrasives.

TUNIC: 1. The loose membranous outer coating of a bulb or corm, for example, onion, tulip or crocus. 2. A coat or layer enveloping an organ or part.

TUNICA-CORPUS THEORY: A concept of the organisation and development of the apical meristem, in which the meristematic region is differentiated into an outer peripheral layer or layers, termed the tunica and an inner mass of cells, termed the corpus. The tunica is characterised by chiefly anticlinal divisions and the corpus mainly by periclinal divisions. The corpus gives rise to the interior part of the plant body and the tunica differentiates the outer layers including the epidermis.

TUNICATE: 1. An arrangement in sheathing, concentric layers, as the leaves of an onion bulb. 2. Various chordate marine animals such as salps, doliolids and pyrosomes, with some species commonly known as sea squirts, sea pork or sea tulips. They belong to the subphylum Tunicata or Urochordata and have cylindrical or globular body enclosed in a tough outer covering.

TUNING FORK: A metal-pronged fork that, when struck produces an almost pure tone of a predetermined frequency. It is used for tuning musical instruments and in experiments in acoustics.

TUNNEL: A passageway through a mountain, under a body of water or underground.

TUNNEL DIODE (ESAKI DIODE): A semiconductor diode which is based on tunnel effect. It consists of a highly doped p-n semiconductor junction, which short circuits with negative bias and has negative resistance over part of its range when forward biased. It is a useful device in many electronic fields due to its fast speed of operation. In the tunnel diode, the shape of the characteristic for current and voltage is affected sharply by the tunnel effect, with a large tunnelling current associated with tunnelling from the valence band to the conduction band. As the forward bias on the junction is increased the tunnel effect becomes less and less pronounced.

TUNNEL EFFECT: 1. An effect in which electrons are able to tunnel through a narrow potential barrier that would constitute a forbidden region if the electrons were treated as classical particles. The fact that there is a finite probability of an electron tunnelling from one classically allowed region occurs due to quantum mechanics. This concept is used in the tunnel diode. An example of a tunnelling process is alpha decay. 2. The ability of a particle to pass through a region of finite extent in which the particle's potential energy is greater than its total energy. This is a quantum-mechanical phenomenon which would be impossible according to classical mechanics.

TUPLE: A finite ordered list of elements; for example, a 5-tuple is an ordered set of five elements.

TUPPING: The mating between a male sheep and ewe for breeding.

TURBELLARIA: A class of free-living flatworms of the phylum Platyhelminthes comprising the planarians. They are found in wet soils, fresh water and marine environments. Their under surface is covered with cilia which are used for gliding over stones and weeds.

TURBID: Opaque and muddy as when particles and sediment are stirred up, for example; dense and cloudy or dark.

TURBIDIMETRIC TITRATION: Titration in which the end point is indicated by the developing turbidity of the titrated solution.

TURBIDITY: 1. The measure of relative clarity of a liquid. It is an optical characteristic of water and is an expression of the amount of light that is scattered by material in the water when a light is shined through the water sample. The higher the intensity of scattered

light, the higher the turbidity. It is used to estimate the cleanliness of water samples. Turbidity is measured with an instrument called Secchi disc. Some materials that cause water turbidity are clay, silt, finely divided inorganic and organic matter, algae, soluble coloured organic compounds and plankton and other microscopic organisms. Some human activities that disturb land, such as construction, can lead to high sediment levels entering water bodies during rain storms, due to storm water runoff and create turbid conditions. Turbidity can provide food and shelter for pathogens. If not removed, it can lead to waterborne diseases such as gastrointestinal diseases. 2. In light scattering, it is the apparent absorbance of the incident radiation due to scattering. For small particles, direct proportionality exists between turbidity and the Rayleigh ratio.

TURBIDITY CURRENT: Underwater current flowing swiftly downslope due to the weight of sediment it carries. This can also occur in fluids such as avalanches involving snow and rocks, pyroclastic volcanic flows, and volcanic lava flows.

TURBINE: A type of engine that uses steam, water, gas or air to spin a rotating shaft by pushing angled blades like a fan or a machine to produce rotational motion. Steam turbines are used to drive generators in power stations and ships' propellers; water turbines spin the generators in hydroelectric power plants; and gas turbines power most aircraft and drive machines in industry. Steam and water turbines provide some 95% of the world's electric power.

TURBO GENERATOR: A steam turbine driving an electric generator, used often in generating electricity in power stations.

TURBOJET: A jet engine in which a turbine-driven compressor draws in and compresses air, forcing it into a combustion chamber into which fuel is injected. Ignition causes the gases to expand and to rush first through the turbine and then through a nozzle at the rear. Forward thrust is generated as a reaction to the rearward momentum of the exhaust gases. The turbojet is the oldest kind of general-purpose jet engine. Frank Whittle in the United Kingdom and Hans von Ohain in Germany developed the concept independently into practical engines during the late 1930s.

TURBULENCE: A form of fluid flow in which the particles of the fluid move in irregular paths, resulting in an exchange of momentum from one portion of a fluid to another. Streamlining reduces turbulence of flow around an object such as an aircraft and reduces drag.

TURFGRASS: A spreading of stoloniferous grass, as opposed to a tufted ornamental grass. A turfgrass endures and typically requires, regular mowing.

TURGID: Distended or swollen by a build-up of fluid as found in a turgid bladder or turgid veins.

TURGOR: The condition in plant cells when its vacuole is distended with water, pushing the protoplasm against the cell wall. Pressure exerted by fluid in a cell presses the cell membrane against the cell wall. Turgor is what makes living plant tissue rigid. Loss of turgor, resulting from the loss of water from plant cells, causes flowers and leaves to wilt. Turgor plays a key role in the opening and closing of stomata in leaves.

TURGOR POTENTIAL (PRESSURE POTENTIAL): A component of water potential, represented by the symbol ψ_p . It is the pressure generated within a cell, being the net difference between the cell's osmotic potential and the water potential of the external environment. Pressure potentials may be negative, in which case they represent tensions. Tensions arise as a result of transpiration and are caused by resistance of the tissues to water flow.

TURGOR PRESSURE: In calculations of osmotic pressure, it is the hydrostatic pressure exerted by the contents of the cell against the cell wall.

TURNAROUND DOCUMENT: Output document produced by a computer with some extra data added to it, which is then input back into the computer.

TURNING POINT: In mathematics, a stationary point at which the first derivative of a function changes sign, so that, typically, its graph does not cross a tangent parallel to the axis of the independent variable (that is, horizontal in the usual two-dimensional Cartesian system). If the second derivative is strictly negative at a stationary point, it is a local maximum of the function; if it is strictly positive, it is a minimum.

TURNKEY SYSTEM: In computer science, a hardware and software system that is designed, supplied, built or installed fully that the user has only to switch on for it to start operation.

TURING MACHINE: A hypothetical computing device capable of storing information and responding to computational questions. It was invented in 1936 by Alan Turing (1912-1954), British mathematician, cryptanalyst, computer scientist and logician. It is an abstract computer and is used to define the concept of computability. The Turing machine mathematically models a machine that mechanically operates on a tape. On this tape are symbols, which the machine can read and write, one at a time, using a tape head. It is a sort of black box that carries out a calculation of some kind on an input number. If the calculation reaches a conclusion or halts, then an output number is returned. Otherwise, the machine theoretically carries on forever. A Turing machine consists of: (i) a tape divided into cells, one next to the other. Each cell contains a symbol from some finite alphabet; (ii) a head that can read and write symbols on the tape and move the tape left and right one (and only one) cell at a time; (iii) a state register that stores the state of the Turing machine; and (iv) a finite table, also called an action table or transition function, of instructions.

TURION: A winter bud covered with scale leaves and mucilage that is produced by certain aquatic plants such as frogbit.

TURKEY: A large bird with a bare, wattled head and neck. The tarsus is equipped with spurs and the tail feathers can be raised to form a vertical fan during courtship or aggressive displays. Turkeys belong to the family Phasianidae of the order Galliformes. They are sometimes placed in a separate family, Meleagrididae. The wild turkeys are classified as *Meleagris gallopavo* and the ocellated turkey, *Agriocharis ocellata*. The domestic turkey is descended from *Meleagris gallopavo*. Commonly black in colour with white freckles, it is bred for food and for its feathers, which are used as stuffing for cushions and pillows.

TURN OVER NUMBER: 1. The maximum number of molecules of substrate that can be converted to product per active site of a single enzyme molecule per unit time. 2. Also known as **turnover frequency** in catalytic processes, it is a value that indicates the amount of feed or substrate converted per a measured amount of catalyst.

TURN-AROUND FACT: In mathematics, an application of commutative properties of addition and multiplication, it is a multiplication or an addition fact in which the order of the addends or factors can be reversed but still arrive at the same answer. There are no turn-around facts for subtraction or division.

TURNER'S SYNDROME: A genetic disorder of women resulting from the absence of the second sex chromosome. It is characterised by a lack of ovaries and menstrual cycle. Although the affected women are sterile and lack secondary sexual characteristics, external genitals are present.

TURNPIKE THEOREMS: A set of economic theorems on optimal control problems, which characterise solutions (optimal paths) of optimal control problems and focus on the stationary optimal one and assert that an optimal or near optimal strategy always approaches the optimal growth rate and remains there.

TURNS RATIO: The ratio of the number of turns in the secondary winding (winding connected to the load) of a transformer to the number of turns in the primary winding (winding connected to an alternating current voltage source). Turns ratio determines the output product (voltage) at the secondary windings and determines if a transformer is either a step up or step down device.

TURNSTILE (GATEPOST): In logic, the symbol, $\vdash$ that denotes a binary relation and is used in different contexts, for example, to represent logical consequence when inserted between expressions to form a sequent or when prefixed to a single expression indicating that it is a theorem. For example, the expression $P \vdash Q$ means that Q is derivable from P in the system. Different variants of the symbol are often used to distinguish syntactic and semantic notions.

TURPENTINE: A type of oil obtained from certain trees such as conifers; or a brownish yellow sticky mixture of essential oil and resin that comes from the terebinth tree. It is used as solvents in mixing paints and varnish and in medicine. It is now mostly replaced by turpentine.

TURQUOISE: A mineral made up of a hydrated phosphate of aluminium and copper. It crystallises in the triclinic system and is generally blue in colour. It is used as a semiprecious stone. It is created by the action of surface waters on aluminium-rich rocks. Iran produces the best specimens.

TURTLE: 1. Any of approximately 300 species of the order Testudines (Chelonia) of the class Reptilia. They are tortoise-like reptiles that have bony shells overlaid with horny shields and found in most parts of the world. The world's largest turtle, the great leatherback sea turtle can reach a shell length of 2.4 centimetres and can weigh over 900 kg. 2. A small computer-controlled wheeled robot.

TUSsock: 1. A group of grass species in the poaceae family. They usually grow as singular plants in clumps, tufts, hummocks or bunches rather than forming sod or lawn in meadows, grasslands and prairies. Tussocks are perennial plants and most species live more than a season. 2. A small area of grass that is thicker or longer than the grass growing around it.

TUTORIAL: In computer technology, it is an interactive software program created as a learning tool to help users to learn new skills by using a step-by-step approach. For example, it assists users in learning how to use parts of a software product such as an office suite or any other application, operating system interface, programming tool, or video game.

TWEAK: To make minor improvements to a computer system, increasing the speed of the system or changing the system to perform the way you wish. A tweak can be either a hardware or software tweak. Examples of software tweak are removing system limitations and adding plug-ins or extensions to a computer's operating system.

TWEET: A short message sent through the free social networking microblogging service platform, Twitter. It may include text, keywords, mentions of specific users, links to websites and links to images or videos on a website. For example, you can tag the word "tech" in a tweet by typing "#tech" and have it easily found and sorted. All hashtags begin with a hash character, which is the symbol #.

TWEETER: A small loudspeaker capable of reproducing sounds of relatively high frequency, from around 2,000 Hz (2 kHz) to 20,000 Hz (20 kHz), generally considered to be the upper limit of human hearing. A tweeter is used in conjunction with a woofer, a loudspeaker used to reproduce low-frequency sounds, in high-fidelity equipment.

TWIG: A small shoot or branch from a tree.

TWILIGHT: A period of faint light that precedes sunrise and follows sunset. It is caused by the reflection of light from the upper layers of the atmosphere, which transmits sunlight to the observer for some time after sunset and before sunrise.

TWIN: 1. A compound crystal consisting of two mirror-image crystals that share a common plane. 2. One of two offspring produced from a single pregnancy or either of two individuals or animals born to the same mother at the same time. Fraternal or binovular twins arise from different egg cells and are equivalent to ordinary siblings, while monovular or identical twins are derived from the same egg cell and are genetically identical. 3. Somebody or something similar to or unusually closely associated with another.

TWIN PARADOX: A paradox that results from the special theory of relativity, which states that if one of a pair of twins remains on Earth while the other twin makes a journey to a distant star at close to the speed of light and subsequently returns to Earth, the twins will have aged differently. The twin remaining on Earth will have aged considerably more than the twin who has travelled to a star. This effect has been verified experimentally by measurements with atomic clocks.

TWIN PRIMES: Two prime numbers that differ by 2. For example, 11 and 13 are twin primes.

TWINNING: 1. The situation, in which two crystals of the same material form with orientations such that they are related to each other by a symmetry operation. This may either be a reflection in a plane or a rotation about an axis. 2. The bearing or reproduction of twins.

TWISTOR: A generalisation of the concept of a spinor that involves the use of complex numbers in an essential way. It is defined as a pair consisting of a spinor field and a complex conjugate spinor field satisfying the twistor equation. Twistor theory was first proposed by Roger Penrose in 1967, as a possible path to a theory of quantum gravity. The twistor approach is especially natural for solving the equations of motion of massless fields of arbitrary spin.

TWITTER: A social networking site that allows users to post a maximum of 140 character message about anything over the Internet, application or on phones. The message includes such things as what users are doing at a given moment, news items, links to websites or comments on events, etc. As of August 2018, its monthly users were 336,000,000.

TWO-BODY PROBLEM: The problem of finding the positions and velocities of two massive bodies that attract each other gravitationally, given their masses, positions and velocities at some initial time. It was first solved by Isaac Newton, who showed mathematically that the orbit of one body about another was an ellipse, a parabola or a hyperbola and that the centre of mass of the system moved with constant velocity.

TWO-DIMENSIONAL (2-DIMENSIONAL; 2D): A geometric figure such as a triangle or a square that has length and width but no depth. All two-dimensional figures can be drawn on a plane (flat surface).

TWO-DIMENSIONAL ANALYSIS: A technique used to separate mixtures of closely related molecules. One method involves two-dimensional paper chromatography, in which a chromatogram produced by a solvent running in one direction is then subjected to a second separation by placing the chromatogram at right angles in a second solvent. Another method involves separation by electrophoresis followed by paper chromatography.

TWO-HYBRID SCREENING (YEAST TWO-HYBRID SCREEN; Y2H): A molecular biology technique for detecting protein-protein interactions and protein-DNA interactions in yeast cells. It can be used in high throughput screening of libraries of genes or potential genes from any species to identify pairs of genes that encode interacting proteins.

TWO-PHOTON EXCITATION: Excitation resulting from successive or simultaneous absorption of two photons by an atom or molecular entity. This term is used for successive absorption only if some of the excitation energy of the first photon remains in the atom or molecular entity before absorption of the second photon. The simultaneous two-photon absorption can also be called **biphotonic excitation**.

TWO-PHOTON PHOTOCHROMISM: Photochromic process involving a two-photon process. The process might involve the simultaneous or the sequential absorption of two photons.

TWO-PHOTON PROCESS: A photophysical or photochemical event triggered by a two-photon excitation.

TWO-POINT CONTACT: In mathematics, it is the relation between two curves, surfaces, etc., at a point at which they touch and have a common tangent; a tacpoint.

TWO-RANKED: In botany, it refers to occurring in two vertical rows on exactly opposite sides of an axis so as to form two ranks, for example, leaves on a branch.

TWO-SAMPLE PROBLEM: In statistics, any problem requiring a test to be applied to two independent samples.

TWO-SIDED: In mathematics, of a limit, equal to both the one-sided limits from above and from below, as the independent variable approaches a given value.

TWO-STEP WORD PROBLEMS: Word problems that require the use of two mathematical operations or steps to determine the solution, for example, two different operations (like multiplication and addition) or two of the same operation (like subtraction and subtraction).

TWO-TAILED TEST (NON-DIRECTIONAL HYPOTHESIS): In statistics, it is the standard test of significance to determine if there is a relationship between variables in either direction (whether the given sample is greater or less than a particular range of values). In other words, it is a hypothesis that has no direction. For a significance level of 0.05, half (0.025) is used to test the statistical significance in one direction and another half for testing statistical significance in the other direction. If the testing sample lies in the null hypothesis, then the null hypothesis is rejected and the alternative hypothesis is accepted. If, for example, the null hypothesis states that the mean is equal to 10, the alternative hypothesis is that the mean is less than or greater than 10. The region of rejection then consists of a range of numbers that are less than or greater than 10.

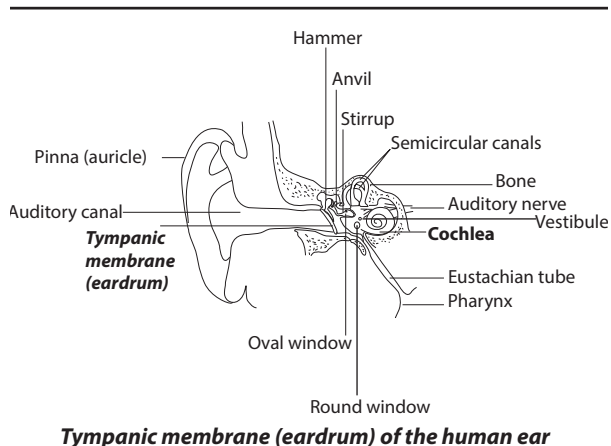
TWO-WAY TABLE: A table used to illustrate two variables when handling data.

TYCHONOFF SPACE: A completely regular T_1 -space. Tychonoff spaces can be characterised as the topological spaces that are homeomorphic to a subspace of some cube.

TYCHONOFF'S THEOREM: The theorem, which states that the (infinite) Cartesian product of compact topological spaces is compact with respect to the product topology. It is named after Andrey Nikolayevich Tikhonov, 1906-1993), Russian mathematician.

TYLOSE: Parenchymatous cells that associate with xylem vessels or tracheids in proximity. The parenchyma cells attach to tracheids through pits occurring in cell walls of the tracheids.

TYMPANIC MEMBRANE (EARDRUM): A thin membrane separating the outer and middle ear in tetrapod vertebrates. It vibrates in response to sound waves and transmits these vibrations via the ear ossicles of the middle ear to the inner ear.



TYNDALL EFFECT: The scattering of light by minute particles in its path, such as dust in the air or the phenomenon observed when dust particles collide randomly in a beam of sunlight. This accounts, for example, for the blue appearance of tobacco smoke.

TYPE: In logic, it is a class of expressions or of the entities they represent, that can all enter into the same syntactic relations. In Russell's theory of types, the type of a function, including a predicate, is determined recursively by the types of its arguments and values.

TYPE I ERROR: In statistics, the error that rejects the null hypothesis when it is in fact true and instead accepts the alternative hypothesis, despite it being attributed to chance. In other words, it is the error of detecting an effect that is not present.

TYPE II ERROR: In statistics, the error that occurs when one accepts a null hypothesis that is in fact false. The error rejects the alternative hypothesis, even though it does not occur due to chance. In other words, it is the error of failing to detect an effect that is present.

TYPE A AND B METALS: The classification of metal ions according to the stability of their complexes for a given ligand. Type A metal cations include the ions of group 1 (Li^+ to Cs^+), the ions of group 2 (Be^{2+} to Ba^{2+}) and ions of lighter transition metals in high oxidation states, for example, cobaltic ion (Co^{3+}), titanate ion (Ti^{4+}), ferric ion (Fe^{3+}). The type B metal cations are those of heavier transition metals in lower oxidation states, for example, silver ion (Ag^+), cuprous ion (Cu^+) and nickelous ion (Ni^{2+}). Certain ligands tend to form more stable complexes with type A metals; others form more stable complexes with type B. For example, halide anions tend to complex with type A metals.

TYPE SPECIMEN: The specimen used for naming and describing a species or sub-species. A specimen or in some cases a group of specimens of an organism to which the scientific name of that organism is formally attached. A type is an example which serves to anchor or centralise the defining features of that particular taxon.

TYPEFACE: A set of characters such as letters, numbers, punctuation marks and symbols of the same design. Examples include Arial, Helvetica, Times New Roman and Tahoma. There are many typefaces and most computers have a few installed. Because typefaces are vector-based (not bitmaps), typefaces can be scaled very large and still look sharp. A **font** is a specific size and style of a typeface, for example, Tahoma is a typeface, while Tahoma 10 pt bold is a font.

TYPHOID: A serious and sometimes fatal bacterial infection of the digestive system, caused by ingesting food or water contaminated with the bacillus *Salmonella typhi*. It causes fever, severe abdominal pain and sometimes intestinal bleeding.

TYPHOON: A violent, revolving tropical storm (cyclone), which is a hurricane in the western Pacific and Indian Oceans. It is a mature tropical cyclone that develops in the north-western part of the Pacific Ocean between latitudes 180° and 100°E . Japan Meteorological Agency classifies a tropical cyclone with maximum sustained winds reaching wind speeds of 64 knots or 119 km/h as a **typhoon** or **hurricane**. It is the highest category on its scale. It is an intense circular storm that originates over warm tropical oceans and is characterised by low atmospheric pressure, high winds and heavy rain. It draws energy from the sea surface and maintains its strength as long as it remains over warm water. In extreme cases, wind speeds may exceed 240 km/hour and gusts may exceed 320 km/hour. Accompanying these strong winds are torrential rains and a devastating phenomenon known as the **storm surge**, an elevation of the sea surface that can reach 6 metres above normal levels. A **tropical depression** is the lowest category that the Japan Meteorological Agency uses for a tropical system that has wind speeds not exceeding 33 knots or 61 km/h. A tropical depression with sustained wind speeds exceeding 34 knots or 63 km/h is upgraded to a **tropical storm**. If the storm intensifies further and reaches sustained wind speeds of 48 knots or 89 km/h, then it is classified as a **severe tropical storm**.

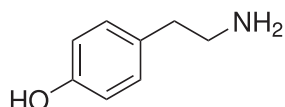
TYPHUS

TYPHUS: An infectious disease caused by one of the bacteria in the family Rickettsiae and marked by fever, great weakness and appearance of purple spot on the body.

TYPICAL VARIANT: The variant of a species to which belongs the type specimen of that species.

TYPOLOGICAL SPECIES CONCEPT: A concept developed by Aristotle defining a species as individuals possessing characteristics that distinguish them from other species. Other concepts, especially the biological species concept has now supplanted the typological species concept.

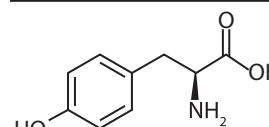
TYRAMINE: A biological active amine, derived from tyrosine, whose effects are similar to that of adrenaline. It occurs naturally in body tissues and can also be found in ergot, mistletoe, putrefied animal tissue and cheese. It causes increased heart activity and raises blood pressure in the body.



**Chemical Structure
of Tyramine**

TYROXINE

TYROSINE: An amino acid with the chemical formula $C_9H_{11}NO_3$. It is obtained from the hydrolysis of proteins such as casein, which is a precursor of epinephrine, melanin and thyroxine.



**Chemical Structure
of Tyrosine**

TYROSINE KINASE: Any of a large family of proteins that catalyse the phosphorylation of a tyrosine residue of a protein by adenosine triphosphate (ATP). They are components of numerous signalling pathways inside cells, notably ones regulating cell growth and differentiation.

TYROSINE KINASE RECEPTOR: A receptor protein in the plasma membrane that responds to the binding of a signal molecule by catalyzing the transfer of phosphate groups from adenosine triphosphate (ATP) to tyrosines on the cytoplasmic side of the receptor. The phosphorylated tyrosines activate other signal-transduction proteins within the cell.

TYROXINE: The main hormone secreted by the thyroid gland. It stimulates metabolism and is very important for normal growth.



UART (UNIVERSAL ASYNCHRONOUS RECEIVER-TRANSMITTER): A type of integrated circuit used for serial communications over a computer or peripheral device serial port that converts the bytes into serial bits and transmits those bits through an asynchronous transmission, stripping out the start and stop bits for each character.

UBIQUINONE (COENZYME Q; COENZYME Q10; CoQ10): Any of a group of compounds derived from quinine within the mitochondria that serves as electron carrier in the electron transport chain reactions of cellular respiration. It also stabilises cell membranes and acts as an antioxidant.

UBIQUITIN: A small 76-amino-acid protein, found in the cytoplasm and nucleus of all prokaryotes and eukaryotes that tag protein destined for degradation by proteasomes. It forms a covalent bond with lysine residues in an adenosine triphosphate (ATP)-dependent reaction called ubiquitination. It is important in both the normal life in the cell and in a cell's response to stress and it is regarded as a heat-shock protein. Failure of the ubiquitin system may cause the development of cancerous cells.

UBIQUITOUS: Widespread, occurring in a wide variety of habitats, especially at the same time; omnipresent.

UBIQUITOUS COMPUTING: In computer science, computing becomes proactive with a network of small inexpensive processing devices in any location and in any format.

UBUNTU: A popular free and open source operating system that is noted for its ease of installation and use. It was developed by Linux in 2004. Packaged with a variety of open source applications, including Firefox and Apache OpenOffice, Ubuntu also comes pre-installed on laptops, desktops and servers. Ubuntu is an African word meaning "humanity to others" that embodies the belief in the bond of sharing connecting all humanities

UDDER: The mammary gland found in animals such as sheep, cow and goats in the form of a bag, containing two or more milk-secreting glands, each with its own teat.

UDP (USER DATAGRAM PROTOCOL): An alternative protocol to the TCP/IP that runs on top of IP and is sometimes referred to as UDP/IP. Unlike TCP/IP, UDP does not divide each transmission into packets, which allows for a faster transmission. However, UDP does not provide error checking.

UID (UNIQUE IDENTIFIER): In computer science, it is a numeric or alphanumeric string, such as a product code, registration code and national identification number that is exclusive to that particular entity within a given system.

ULCER: A slow-healing sore of skin marked by tissue disintegration or on the surface of a mucous membrane, especially the membrane lining the stomach or other parts of the digestive tract.

ULLAGE: The extra volume of gas above the liquid propellant held in sealed tanks. This space allows for thermal expansion of the propellant or the build-up of gases released by the propellant.

ULNA: The larger and longer of the two bones in the lower forearm of vertebrates. It lies medially, away from the thumb side. It articulates with carpals at the wrist and with the radius and humerus at the elbow.

ULNAR ARTERY: The larger of the two terminal branches of the brachial trunk. It is the main blood vessel, which supplies oxygenated blood to the muscles of the forearm and wrist and hand. It arises from the brachial artery and terminates in the superficial palmar arch, which joins with the superficial branch of the radial artery. It is palpable on the anterior and medial aspect of the wrist.

ULNAR NERVE: A nerve that arises in the axilla from the medial cord of the brachial plexus, descends through the medial part of the upper arm to the back of the medial epicondyle and passes between the heads of the flexor *Carpi ulnaris* to enter the forearm.

ULTIMATE: In botany, it refers to the final section or division of a structure.

ULTIMATE CAPACITY (STOICHIOMETRIC CAPACITY): The theoretical maximum capacity of a solvent containing a given concentration of extractant for a solute under any conditions.

ULTIMATE CAUSATION: The hypothetical evolutionary explanation for the existence of a certain pattern of animal behaviour. The ultimate causes of social behaviours include their evolutionary or historical origins and the selective processes that have shaped their past and current functions. In the case of an isolated herd of animal, the development of a better defence against predators that results in increased survival of individuals remaining in groups would be an ultimate cause for the tendency to reunite with the herd.

ULTRA HIGH FREQUENCY (UHF): A range of radio waves of very short wavelength with frequencies from 300 MHz to 3 GHz used, for example, for television broadcasting. This corresponds to a wavelength range of 1 m to 0.1m.

ULTRACENTRIFUGE: A high or low speed centrifuge, used for quantitative measurement of sedimentation velocity or sedimentation equilibrium and also to separate macromolecules, such as proteins or nucleic acids from solutions. They are driven by electricity and can operate at speeds of up to 60,000 rpm.

ULTRADIAN RHYTHM: A biological cycle that repeats in less than a day. For example, the body temperature of hibernating animals often shows periodic transient increase corresponding to brief periods of arousal. Some of the other ultradian cycles of the body are hormonal release, heart rate, sleep, urination, bowel activity, nostril dilation and appetite.

ULTRAFILTRATION: 1. A filtration process by which substances in solution are separated on the basis of their molecular size. Particles of colloidal size are retained by a filter medium while solvent and accompanying low-molecular-weight solutes are allowed to pass through. Its uses include: (i) separating colloid from suspending medium; (ii) separating particles of one size from particles of another size; (iii) determining the distribution of particle sizes in colloidal systems by the use of filters of graded pore size. 2. A filtration process that uses a porous membrane to isolate and remove particles such as viruses and bacteria. The process is used in the pharmaceutical industry and for water purification.

ULTRAMICROSCOPE: A type of microscope that uses the Tyndall effect to show the presence of particles that cannot be seen with a normal optical microscope. Colloidal particles, smoke particles, etc., are suspended in a liquid or gas in a cell with a black background and illuminated by an intense cone of light that enters the cell from the side and has its apex in the field of view. The particles then produce diffraction-ring systems, appearing as bright specks on the dark background.

ULTRAPOWERS: The ultrapower of a set with itself.

ULTRAPRODUCT: In mathematics, it is the quotient of a Cartesian product of an infinite family of sets $\{A_i : i \in I\}$ with respect to the equivalence relation induced by an ultrafilter U on $I : \{a_i\} = \{b_i\}$ if $\{i \in I : a_i = b_i\}$ belongs to U . If the sets $A_i = A$ are identical this is